FRG Analysis of LRMMs

Bernardo Atolin

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1 Introduction

Physical models endowned with long-ranged interactions have in recent years reached the spotlight of condensed-matter and statistical physics, first, as a consequence of the rich and diversified properties that those models exhibit at criticality, and second, because of recent development of experimental techniques with trapped ions, multimode cavities, and polar molecules, among others; which have allowed physicists to engineer theoretical long- and short-ranged models in experimental settings with calibrated level of precision; for example through atoms stored in a Penning Trap, supercold atoms coupled to photons in a cavity, among others. The meaningful progress, that this field of study has made in the last 20 to 50 years, leads to the hope, that the combined theoretical and experimental efford in the area can eventually be used for the fabrication of materials with exotic properties, for industrial application in the design of new technologies and components.[1—12, 42, 44, 63, 97]

On the theoretical side, breakthrough progress has in this connection in the last two to three decades emerged. The method of Renormalization Group, pioneered by Wilson and subsequently developed further by Wetterich, has enabled for the reliable and systematic study of critical phenomena, both qualitatively and quantitatively. A significant count of papers, particularly in the areas of statistical and condensed matter physics, has successfully utilized this method for the accurate characterization and quantification of continuous second order phase transitions.[33—42, 44, 59—65, 94—97]

An existing gap in literature persists nevertheless, which concerns the abscence of quantitative insight in the region of cross-over of the long-range model, and in particular in two-dimensional systems. One avers namely the following facts:

- According to predictions in the area of conformal field theory, it is theorized that two-dimensional systems are endowned of infinite conformal symmetries, as a result of properties of holomorphic functions. As a consequence of those symmetries, one expects that two-dimensional systems tunned to specific parameters must dispose over multicritical fixed points characterized by classes of virasoro minimal models indexed by integer indicies.

- On the other side of the coin, long-ranged models are endowned of a cross-over point, at which the model transitions from the long-ranged to a short-ranged universality class. According to Sac's argument (which this article will discuss in more details), the cross-over is completely affixated by the anomalous dimension of the corresponding short-ranged model to which the long-ranged model is expected to converge to in the infrared.

- The combination of the two aforementioned facts leads to the hypothesis that long-ranged models may be tunnable to specific parameters, for which the long-ranged model collapses (in the infrared) onto multicritical fixed points characterised by the universality classes of the corresponding virasoro primaries predicted by solving the corresponding conformal field theory. One expects therefore, exclusively in two dimensions, that by tunning the strength of the long-ranged interaction to particular quantities, new criticality families should emerge corresponding to the respective families of virasoro minimal models predicted by the conformal field theory.

A series of recent publications have done significant progress in investigating the the aforementioned hypothesis, from the point of view of conformal field theory. The original formulation in [88, 89] appears to consist of employing deformations of virasoro minimal models by means of introducing a long-ranged gaussian free-field, in the style of [73, 98—101]. The goal is to obtain a dual description for the cross-over.

Given the existing gap in literature concerning functional renormalization group approach to the long range minimal models in the style of [88, 89]; as well as the recent success of the functional renormalization group approach for predicting both qualitative and quantitative physical phenomena at criticaliy, including recent, meaningful contributions from [94—96]; this thesis proposes a functional renormalization group approach to the long range minimal models of [88, 89], in order to extract meaningful quantitative and qualitative information from the renormalization group the main hypothesis flow.[73, 88—96, 98—103]

This thesis is thus organized as follows:

- Part I provides a state-of-art description of the method of functional renormalization group; qualitative and quantitative insights; as well as methods of (both numerical and semi-analytical) computation. In this section, often one falls back to the simplest case of a scalar ϕ^4 theory, in order to instructively provide example-calculations and example-results, in order to elucidate the implementation of the method.

- Part IV describes in detail, the main elements of conformal field theories that this thesis will rely on; and more broadly, the elements that [88, 89] relies on for formulating its theoretical framework.

- Finally, Part V employs the insights obtained in subsequent Part I and Part IV, in order to derive analytical and theoretical results concerning the long-range minimal models proposed in [88, 89].

2 A Guide to Citations

In the following, a broad overview of the main pieces of literature that this thesis relies on is provided. The work leading up to this paper can be roughly summarized as follows:

- [108] shows a theoretical phase transition in one dimension, when long range interactions are accounted for. In subsequent years, most remarkably [107—110] study the long-range model in more details and derive the main phenomenological phenomena in this regard. Noteworthy contributionsalso include [105, 106]. [110] additionally predicts the long-short-range crossover at $s^\star$; which is the parameter where the long range universality class transitions to the short-range universality class.

- On the other side of the spectrum, [39—41] proposes the renormalization group equations to describe coupling renormalizations, continuous phase transitions at criticality; and in theory provide a non-perturbative description of field-theoretical phenomena. [43] employs the newly-developed formalism to describe critical phenomena; and [33] relies on this framework to distinguish relevant, irrelevant, and marginal parameters in effective field theories. [34—38] formulate the renormalization group flow equations in terms of the effective action; and from this point on this newly-developed framework is implemented in order to derive a multitude of new physical phenomena. Here, the LPA and LPA' methods which are today widely used in Functional Renormalization Group formalism are for the first time developed. [45] then proposes the Litim regulator which is today widely implemented in LPA(') problems. The method is now widely implemented by many authors in order to predict phenomenological properties of universality of phase transitions; such as [17, 42, 46, 47, 94—96]. Today, the field of functional renormalization group is widely well-established, with mention-worthy publications and/or textbooks including 16, 18, 19, 41, 42, 46].

- Around simultaneously, topics of conformal field theories begin to be studied in more detail. [73, 79, 98—101] made great contributions to the classification and study of the so-called virasoro minimal

models; based on the works of [83, 84, 87]. [82] showed the equivalence between unitary, scale-invariant theories and conformal theories in two dimensoins; and [80] introduced for the first time the concept of operator product expansions. Today, the field is well-established, with a number of textbooks in this area, including [69—72, 74—77, 85, 86].

- [93] points to a number of inconsistencies between the currently-accepted nature of long-range interactions, and predictions from the point of view of conformal field theory. [90] proposes a modified long-range theory in the style of CFT-deformations studied by [98, 99], with the goal of fixing those inconsistencies, and with the goal of making numerical predictions for the theory near $s \lesssim s^{\star}$. The concept of dualities gains some propulsion. A series of new contributions are made in [88, 91, 92], and [102, 103] extends the description of [90] to the study of models in one dimensions, among others.

Part I

The Functional Renormalization Group Method

3 Functional Methods in Field Theories

Functional methods play a central role in the modern formulation of field theories, providing an alternative and often more flexible framework than canonical approaches. A key advantage is that the physics of a system can be systematically encoded in functional objects, such as the partition function and its associated generating functionals. In particular, the generating functional of connected correlation functions and the effective action — which are related by a Legendre transform — serve as central quantities from which correlation and vertex functions can be obtained via functional differentiation. This provides an unified framework for the computation of physical observables, for systematically introducing symmetries and gauges to a theory, and for developing approximation schemes. Moreover, this formalism establishes a direct connection between quantum field theory and statistical physics, by interpreting fields as statistical degrees of freedom.[13—15, 25]

This section briefly summarizes the main elements concerning functional methods used in this work.[a] Let the theory be described by the lagrangian density $\mathcal{L}[\phi]$ for a continuous field $\phi(x)$. The statistical weight of a given field configuration is given by the Boltzmann factor, Eq. (3.1), where $\mathcal{S}[\phi]$ is the bare action.

$$P[\phi] = exp\left(-\mathcal{S}[\phi]\right) \;; \qquad \mathcal{S}[\phi] = \int \mathcal{L}[\phi]\, d^d x \tag{3.1}$$

One defines the integration measure so that it satisfies the normalization condition, Eq. (3.2).[b]

$$1 = \int D\mu[\phi] = \int D\phi \cdot P[\phi] = \int D\phi \cdot exp\left(-\mathcal{S}[\phi]\right) \tag{3.2}$$

One defines the Partition Function, Eq. (3.3), where $j(x)$ is the external source.

$$\mathcal{Z}[j] = \int D\mu[\phi]\, exp\left(\int j(x)\,\phi(x)\, d^d x\right) \tag{3.3}$$

[a] *See* Appendix Manipulation of Functional Objects for additional details on the manipulation of functional derivatives.
[b] This normalization is not necessary. It is chosen in this thesis, without loss of generality, for simplicity.

Correlation functions are generated by taking functional derivatives with respect to $\mathcal{Z}$, Eq. (3.4).

$$\left(\frac{\delta^n \mathcal{Z}[j]}{\delta j(x_1)\,\delta j(x_2)\ldots}\right)_{j=0} = \int \phi(x_1)\,\phi(x_2)\ldots D\mu[\phi] = \langle \phi(x_1)\,\phi(x_2)\ldots\rangle \tag{3.4}$$

Useful functionals are furthermore the free energy $\mathcal{W}[j]$ and its correpsonding Legendre Transform $\Gamma[\phi]$, often labelled as the effective action or the free energy, Eq. (3.5).

$$\mathcal{W}[j] = log(\mathcal{Z}[j]) \;; \qquad \Gamma[\phi] + \mathcal{W}[j] = \int j(x)\,\phi(x)\,d^d x \tag{3.5}$$

Additionally, the following properties follow directly from the transform, Eq. (3.6).

$$\frac{\delta \mathcal{W}[j]}{\delta j(x)} = \phi(x) \;; \qquad \frac{\delta \Gamma[\phi]}{\delta \phi(x)} = j(x) \tag{3.6}$$

By taking functional derivatives with respect to the free energy, one extracts connected correlation functions, Eq. (3.7). Here the "connected" correlation function is meant to be distinguished from the correlation function, in that it removes the trivial contributions introduced by disconnected parts; that is to say, it accounts only for geniune correlations inctroduced by interactions and fluctuations. Given the normalization condition Eq. (3.2), $\mathcal{Z}[0] = 1$ for the particular purposes of this paper.[14—16, 21, 23, 24, 29, 30]

$$\left(\frac{\delta^n \mathcal{W}[j]}{\delta j(x_1)\,\delta j(x_2)\ldots}\right)_{j=0} = \frac{1}{\mathcal{Z}[0]}\int \phi(x_1)\,\phi(x_2)\ldots D\mu[\phi] = \langle \phi(x_1)\,\phi(x_2)\ldots\rangle_c \tag{3.7}$$

The effective action plays an additional important role. One considers Eq. (3.8).

$$\Gamma^{(2)}(x,y) = \left(\frac{\partial^2 \Gamma[\phi]}{\partial\phi(x)\,\partial\phi(y)}\right) \tag{3.8}$$

The correlation function can be calculated by taking the inverse of Eq. (3.8), as demonstrated in Eq. (3.9).[c] In other words, be the effective action known, correlation functions can be calculated and the physics of the theory is completely specified. The terminology "effective action" arises here, because Γ describes the bare action embedded by interactions and quantum fluctuations, such that the effective action defines the full inverse-propagator. The effective action can be understood as the quantum-corrected action; or the inverse propagator dressed by quantum effects. It is the generating functional of 1-particle-irreducible diagrams.[13, 14, 24, 30]

$$\delta^d(x-z) = \int G(x,y)\,\Gamma^{(2)}(y,z)\,d^d y \tag{3.9}$$

4 A Motivation for Functional Renormalization Group

4.1 Perturbative Regime

Renormalization emerges naturally in field theories, even at the perturbative level, as a consequence of scale dependancy, UV divergencies, and coarse-graining. As an initial motivation, one thus considers, at first, a

[c] In more details,

perturbative regime. Such a regime typically arises when interactions are weak and act as small corrections around a stationary, reference field configuration ϕ_*; that is to say, Eq. (4.1).[16—19, 24, 29, 30]

$$\frac{\delta S\left[\phi_*\right]}{\delta\phi} = 0 \tag{4.1}$$

In this case, the weak interactions suggest a perturbative expansion in the vicinity of the stationary point. This is justified for theories whose properties are mostly captured by the reference field configuration, and fluctuations play a lesser role. Under this approximation, the assumptions given in Eq. (4.2) below are justified, and fluctuations can correspondingly be expanded around ϕ_*.[15, 18, 19]

$$P\left[\phi_*\right] = max\left(P\left[\phi\right]\right) \ ; \qquad\qquad Z\left[j=0\right] = e^{-\mathcal{S}\left[\phi_*\right]} \times \text{corrections} \tag{4.2}$$

Let a scalar ϕ^4 theory be specified by the bare action, Eq. (4.3).

$$S\left[\phi\right] = \int d^d x \left(\partial_\mu\phi\partial^\mu\phi + m^2\phi^2\right) + \mathcal{O}\left(\phi^4\right) \tag{4.3}$$

One considers Eq. (4.4).

$$e^{-\Gamma\left[\phi\right]} = \int D\chi exp\left(-\mathcal{S}\left[\phi+\chi\right] + \int j\left(x\right)\chi\left(x\right)d^d x\right) \tag{4.4}$$

By taylor-expanding the action term in Eq. (4.4) to the leading order around ϕ_*, one finds Eq. (4.5),[d] where $G\left(p\right)$ is the propagator in momentum space. At this approximation level, the bare and effective actions are the same, and the propagator corresponds to the free propagator of a gaussian free-field.

$$\Gamma\left[\phi_*\right] = S\left[\phi_*\right] \ ; \qquad\qquad G\left(p\right) = \frac{1}{\left(p^2+m^2\right)} \tag{4.5}$$

The next-order correction is obtained from Eq. (4.6),[e] by expanding the bare action to the next order in the taylor expansion. The integral in Eq. (4.6) is ultraviolet divergent. Formally, one regulates those divergencies by introducing a cut-off Λ to the integration space.

$$\Gamma_1\left[\phi\right] = TrLog\left(\frac{\partial^2\mathcal{S}}{\partial\phi\left(x\right)\partial\phi\left(y\right)}\right)_{\phi_*} = \int^{p^2=\Lambda^2}\frac{d^4p}{\left(2\pi\right)^4}\frac{\lambda}{p^2+m^2} = \frac{\lambda}{8\pi^2}\left\{\frac{\Lambda^2}{2} - m^2 log\left(\frac{\Lambda^2+m^2}{m^2}\right)\right\} \tag{4.6}$$

4.2 Regularization

At this point, it is worth taking a pause to understand the origination of those divergencies, and the justification for the process of regulatization.[13—18, 23—25, 29, 30, 46]

- **Locality as the Origin of Divergencies**: the loop diagrams in aforementioned theory can be

$$\delta^d\left(x-z\right) = \int d^d y\frac{\partial\phi\left(y\right)}{\partial j\left(z\right)}\frac{\partial j\left(x\right)}{\partial\phi\left(y\right)} = \int d^d y\left(\frac{\partial^2\mathcal{W}\left[j\right]}{\partial j\left(x\right)\partial j\left(y\right)}\right)\left(\frac{\partial^2\Gamma\left[\phi\right]}{\partial\phi\left(y\right)\partial\phi\left(z\right)}\right) = \int G\left(x,y\right)\Gamma^{(2)}\left(y,z\right)d^d y$$

[d] *See* Appendix Perturbative Calculations for a detailed derivation.
[e] *See* Appendix Computation of n-Point Vertex Functions and Next Order Correction to the Propagator.

understood as virtual interactions occuring at arbitrarily short or long wavelengths. The locality of the theory leads to integrations over an unbounded momentum space. Because there are no other terms otherwise suppressing those contributions, virtual processes at arbitrarily large energies contribute to the loop integral. The locality of the interaction leads to the absence of a preferred length scale for the interactions. The evaluation of the loop integral is understood, that virtual interactions are not measurable, and such uncertainty is therefore encoded by considering the averaged contribution over all possible virtual processes. The absence of the length scale, in turn, leads to unbounded virtual processes, and, by extension, to divergencies.

- **Justification of the Regularization Step**: The introduction of a cut-off is justified by the fact that those local interactions are ill-defined. By introducing a cut-off, one specifies a length scale $a \sim \Lambda^{-1}$, such that fields evaluated withing the scale smeared off. A resolution scale to the theory, above which no virtual processes occur. Formally, one takes the limit $\Lambda \to \infty$, so that local interactions where two fields are evaluated at the same spacetime point are removed.

- **Independence of Regularization Choice**: Although the limit $\Lambda \to \infty$ is formally taken, in practice, no expression with Λ survives after renormalization and regularization. This is a mathematical artifact to handle divergencies. Furthermore, the regularization scheme is not unique. Different regularization schemes exist and lead to different solutions to the integral, Eq. (4.6). Crucially, although the regularization scheme changes, physically-measurable quantities must be the same, independent of regularization scheme. The introduction of the cut-off (and, subsequently, of a renormalization scale μ) are mathematical artifacts to handle divergences, and they disappear when physical quantities are measured.

4.3 Renormalization

The next formal step consists of renormalization. Divergent contributions to the two-point correction $\Gamma_1^{(2)}$are local and can be absorbed into a redefinition of the bare mass term in the lagrangian. One defines a renormalized mass $m(\mu)$, and one re-defines the bare mass, Eq. (4.7).[23, 24, 29, 30]

$$m_0^2 = m^2(\mu) + m_{CT}^2(\mu) \tag{4.7}$$

The counterterm mass m_{CT}^2 is chosen to explicitly remove the dependance on the divergent cut-off Λ, Eq. (4.8).[f]

$$m_{CT}^2(\mu) + m_{loop}^2(\mu) = \frac{\lambda m(\mu)^2}{16\pi^2} log\left(\frac{m(\mu)^2}{\mu^2}\right) \tag{4.8}$$

Here, the introduced renormalization scale μ is completely arbitrary, and all observable quantities are independent of Λ and μ, by construction, and as justified in more details above. Furthermore, in principle, the regularization and renormalization procedure may introduce momentum-dependant terms to the definition of the physical mass and coupling strength (this is the case, e.g., in QED). In those cases, although the

[f] In more details,

$$m_{CT}^2(\mu) + m_{loop}^2(\mu) = -\frac{\lambda}{16\pi^2}\left(\Lambda^2 - m(\mu)^2 log\left(\frac{\Lambda^2}{\mu^2}\right)\right) + \frac{\lambda}{16\pi^2}\left(\Lambda^2 - m(\mu)^2 log\left(\frac{\Lambda^2}{m^2}\right)\right) = \frac{\lambda m(\mu)^2}{16\pi^2} log\left(\frac{m(\mu)^2}{\mu^2}\right)$$

physical quantities continue to be independent of Λ and μ, a momentum dependency emerges, and the physical quantities run as a function of the momentum, so that experiments conducted at different energy levels lead to different observations of physical mass and coupling strength.[18, 21, 22]

$$m_{phys}^2 = m_{phys}^2(p) \ ; \qquad \lambda_{phys}^2 = \lambda_{phys}^2(p) \tag{4.9}$$

The physical mass (up to the first loop correction, as higher-order loops introduce further complications to the pole condition) corresponds to the pole of the Green's function, Eq. (4.10). To emphasize, the actual physical mass (observed by expermient) does not change as function of the scale μ. The running occurs, rather, on the renormalized mass parameter that appears in the lagrangian.[23, 24, 29]

$$m_{phys}^2 = m(\mu)^2 + \frac{\lambda m(\mu)^2}{16\pi^2} log\left(\frac{m(\mu)^2}{\mu^2}\right) \tag{4.10}$$

The beta function measures the running of the coupling with respect to the scale, and it is defined by Eq. (4.11).[g, h]

$$\beta_{m^2} = \frac{\mu}{m^2}\frac{dm_{phys}^2}{d\mu} = \frac{\lambda}{8\pi^2} \tag{4.11}$$

A similar analysis is carried out, in order to obtain the beta function corresponding to the coupling parameter λ. In this case, the 4-point vertex function must be evaluated. The beta function is found as Eq. (4.12).[i]

$$\beta(\lambda) = \mu\frac{\partial\lambda}{\partial\mu} = \frac{3\lambda^2}{8\pi^2} \tag{4.12}$$

5 Functional Renormalization Group

5.1 Qualitative Description

The perturbative method introduces interactions, essentially by "freezing-out" all of the interactions onto a mean-field configuration, and perturbatively introducing fluctuations back, via functional taylor expansion around the gaussian reference theory. This procedure leads to the shortcomings discussed above. Because the perturbative method treats all wavelengths at equal footing, it fails to give accurate results for theories (or regions of a theory) where scale takes precedence over perturbative order. In those cases, perturbation theory leads to uncontrolled approximations that, even when many orders of the taylor expansion are calculated, still yield incorrect results. In that connection, the motivation for the FRG arises as an alternative way of dealing with those divergencies and interactions, in a more systematic way. Furthermore, because the

[g] In more details, from Eq .(4.10), one finds

$$\frac{\mu}{m^2}\frac{dm_{phys}^2}{d\mu} = \frac{\mu}{m^2}\frac{dm^2}{d\mu} + \frac{\lambda}{16\pi^2}\left\{\frac{\mu}{m^2}\frac{dm^2}{d\mu}\left(1 + log\left(\frac{m(\mu)^2}{\mu^2}\right)\right) - 2\right\} = \beta_{m^2} + \frac{\lambda}{16\pi^2}\left\{\beta_{m^2}\left(1 + log\left(\frac{m(\mu)^2}{\mu^2}\right)\right) - 2\right\} = 0$$

[h] Here, one implicitly assumes that β_{m^2} is at least (and at most) order $\mathcal{O}(\lambda^1)$. This is justified by a proof by contradiction: assume that β_{m^2} is of order $\mathcal{O}(\lambda^0)$, such that the formula will read $\beta_{m^2} = \frac{A(\mu)\lambda}{16\pi^2}$ for some constant $A(\mu)$ of at least order $\mathcal{O}(\lambda^0)$. This leads β_{m^2} to be of order $\mathcal{O}(\lambda^1)$, contradiction.

[i] *See* Appendix 4-Point Vertex Correction for a detailed analysis.

FRG equation is (in principle)[j] exact, this method can in principle be used to extract key properties of the system, even for strong-interacting theories, or for theories close to a critical phase transitions — where the perturbative regime otherwise fails. On the one hand, the hope is to implement FRG methods in order to derive new physics that the perturbation method alone cannot predict. On the other hand, FRG provides a new perspective over known results. Although no exact, analytical solution for the RG flow equation is believed to exist, qualitative analysis of the RG flow equation is sufficient to predict non-trivial behaviour near criticality; and computational methods have shown themselves as reliable and efficient tools, for the identification of critical points and calculation of critical coefficients.[14—17, 20, 32, 42—46, 54, 55]

To give a qualitative description of the RG method, one recognizes that different microscopic theories lead to similar macroscopic behaviour. The RG approach consists of building an effective macroscopic theory, by freezing out the short-distance nodes that don't crucially affect the macroscopic physics. Physical systems described by different microscopic theories, but which are macroscopically identical, will then build a trajectory in the RG flow. Additionally, a "critical surface" exists in the RG flow, corresponding to models at criticality. Models at criticality converge to points in the domain of the critical surface, whereas models away from criticality describe RG trajectories moving away from this critical surface. All systems converging to a particular point of the critical survace are said to belong to the same "universality class", since they have the same long-range physics. Points near the critical surface also build universality classes, corresponding to systems close to criticality, whose physics is described by power-law behaviours. Those systems will "looks" critical, so long as the correlation length remains high.[14, 15, 18—20, 25, 26, 39, 40, 44, 46, 97]

The FRG approach consists, roughly, of defining a family of running effective actions $\Gamma_k[\phi]$, $0 < k < \Lambda$, for some cut-off $\Lambda \to \infty$, such that $\Gamma_k[\phi]$ corresponds to a physical theory where all fluctuations at low-frequency nodes $k' < k$ are freezed-out. One creates an effective field theory that is valid at a scale k, and which suppresses fluctuations at scales lower than k. At $k = \Lambda$, all of the fluctuations are suppressed, and one obtains the mean field solution, $\Gamma_{k=\Lambda}[\phi] = \mathcal{S}[\phi]$. At $k = 0$, no suppression occurs, and one obtains the actual physical solution, with interactions, and without any screening effects, $\Gamma_{k=0}[\phi] = \Gamma[\phi]$. The renormalization group equation describes the evolution of the effective physical theory, starting from a scale $\Gamma \propto a^{-1}$ down to the exact theory. Here, k plays the role of an IR cut-off; and the RG flow, correlation functions, and the existence of fixed points, are all contained in Γ_k.[16—18,35—38, 44, 46]

In order to implement this scheme, one decouples the slow modes from the partition function by endowing such modes with a large mass, whilst keeping the quick modes vastly unchanged. This is motivated by the inverse relation between compton wavelength and mass; such that large masses lead to a reduction in the compton wavelength corresponding to the node, and, by extension, to the distance at which fluctuations meaningfully contribute to the physics of the system. Therefore, one introduces a k-dependant mass term to the action, which, at a given scale k, kills-off all of the low-frequency modes, while retaining the high-frequency modes almost unaltered. The motivation behind introducing this "regulator", that is, the k-dependant mass-term, comes from the fact that the correlation length is inversely proportional to the mass, $\xi^{-1} \propto m$. By increasing the mass, the correlation length gets suppresed, and the effect of quantum fluctuations accordingly. This regulator be infinitely large at Λ (in practice, $\approx \Lambda^2$), all fluctuations are suppresed at the mean-field limit, whereas it should be zero at $k = 0$, so that the actual physical theory is recovered. In the intermediary $0 < k < \Lambda$, rapid modes $k' > k$ remain largely unaffected, whereas slow modes are freezed-out $k' < k$. Qualitatively, the renormalization flow consists of systematically excuding or including physics that matter or don't at a particular scale. The choice of the regulator itself is in principle arbitrary,

[j] In practice, one relies on truncations to solve the FRG Equation. The equation is exact as a functional identity only.

so long as these qualitative properties are satisfied. The choice of the regulator will affect the renormalization flow, however, the physics of the system, that is to say, the quantities that can actually be measured and the respective universality class that the flow specifies, remains the same, independent of the regulator. The momentum-dependant mass term acts as a cutoff, and builds a family of models.[16—19, 35—38, 44, 46]

As a final remark, qualitatively, one notices already that this method, by construction, avoids both IR and UV divergencies completely. Namely, the freezing of the slow modes automatically prevents IR divergencies, whereas the fact that, by construction, the cut-off is defined to automatically satisfy $\Gamma_{k=\Lambda}[\phi] = \Gamma_{MF}[\phi] = \mathcal{S}[\phi]$, leads to no emergence of UV divergencies, as, at a scale k, all the high-frequency modes $p >> k$ have already been integrated out, and the integral is only contributed to by a shell in the vicinity of $p \approx k$.[16, 18, 32]

5.2 Formulation of the Method

One considers now the renormalization flow. The effective action under the mean field theory is given by $\Gamma_{MF}[\phi] = \mathcal{S}[\phi]$, and one successively integrates-out the momentum nodes, until the effective action given by the theory is reached. One defines a scale $0 < k < \Lambda$ for some cut-off Λ, and one defines a family of running effective actions $\Gamma_k[\phi]$, such that $\Gamma_{k=\Lambda}[\phi] = \mathcal{S}[\phi]$ and $\Gamma_{k=0}[\phi] = \Gamma[\phi]$. Aforementioned procedure consists of introducing a momentum-dependent mass-term to the hamiltonian, Eq. (5.1), where $R\left(\nabla^2\right)$ is the so-called "regulator". This is referred to as a "mass-term", because it is squared in the field. The regulator introduces the required running of the mass term with respect to the scale k. Furthermore, the regulator must satistfy the properties that $R_{k=0} = 0$ and $R_{k=\Lambda} \approx \Lambda^2$. This leads to the total suppression of modes at $k = \Lambda$, and no suppression at $k = 0$. Additionally, one would require that for $0 < k < \Lambda$, $R_k\left(z^2\right) \approx 0$, $z^2 >> k^2$ and $R_k\left(z^2\right) >> 1$, $z^2 << k^2$; such that all low-frequency nodes are freezed-out, and high-frequency nodes remain unchanged, respectively in relation to the scale.

$$\Delta\mathcal{S}_k[\phi] = \frac{1}{2}\int \phi(x) R_k\left(\nabla^2\right)\phi(x)\, d^d x = \frac{1}{2}\int \phi(p) R_k\left(p^2\right)\phi(p)\, d^d p \tag{5.1}$$

5.3 Wetterich Equation

From this point, one re-write Eq. (3.5), the legendre transform, as Eq. (5.2).

$$\Gamma_k[\phi] + \mathcal{W}_k[j] = \int j(x)\phi(x)\, d^d x - \frac{1}{2}\int \phi(x) R_k\left(\nabla^2\right)\phi(x)\, d^d x \tag{5.2}$$

The source is written as Eq. (5.3).

$$j(x) = \frac{\partial\Gamma_k}{\partial\phi} + \int R_k\left(\nabla^2\right)\phi(x)\, d^d x \tag{5.3}$$

Correspondingly, one re-writes Eq. (4.4) as Eq. (5.4).

$$e^{-\Gamma_k[\phi]} = \int D\chi exp\left(-S[\phi] - \Delta S_k[\phi] + \int \frac{\partial\Gamma_k}{\partial\phi}\chi(x)\, d^d x\right) \tag{5.4}$$

By taking functional derivatives with respect to Eq. (5.2), one finds Eq. (5.5).

$$\frac{\partial^2\Gamma_k[\phi]}{\partial\phi(y)\,\partial\phi(z)} = \frac{\partial j(x)}{\partial\phi(y)} - R_k \tag{5.5}$$

The exact propagator is slightly modified in this new formalism, Eq. (5.6).[k]

$$\delta^d(x-z) = \int G(x,y)\left(\frac{\partial^2\Gamma_k[\phi]}{\partial\phi(y)\,\partial\phi(z)} + R_k\right)d^dy \tag{5.6}$$

The goal is to describe the variation of the effective action as a function of scale. Let the RG-time $t = log\left(\frac{k}{\Lambda}\right)$ be defined, such that at $t=0$, $k=\Lambda$ and at $t\to\infty$, $k\to 0$. By taking the derivative of Eq. (5.4)[l] with respect to the RG-time and simplifying, one obtains Eq. (5.7), where, on the last line, one also writes down the momentum-space representation and in terms of the trace. This is the Wetterich Equation.

$$\partial_t\Gamma_t[\phi] = Tr\left(\partial_t R_t\left(\Gamma_t^{(2)} + R_t\right)^{-1}\right) \tag{5.7}$$

Finally, one re-casts Eq. (5.7) into Eq. (5.8)[m] for convention.

$$\partial_k\Gamma_k[\phi] = \tilde{\partial}_k Trlog\left(\Gamma_k^{(2)} + R_k\right) \tag{5.8}$$

5.4 Discussion

One emphasizes a few important elements concerning the Wetterich equation:[17—19, 32, 42, 45—47, 51—53]

- First of all, the effective action, up to one-loop, is given by $\Gamma_k^{(1L)}[\phi] = Trlog\left(S_k^{(2)}[\phi] + R_k\right)$. Comparing with Eq. (5.8) above, one notices that the Wetterich equation describes the "exact loop". Furthermore, one notices that the "exact propagator" can be derived from Eq. (5.6), and it takes the form $G_k[\phi] = \left(\Gamma_k^{(2)}[\phi] + R_k\right)^{-1}$. These two elements lead to a very important distinction between the perturbative and the FRG methods. In the FRG approach, assuming that Γ_k would have been computed, all of the information is automatically encoded in the 1-loop contribution — there is only one integral to be evaluated. For the perturbative approach, on the other hand, the expansion is truncated at l-loops, and l integrals must be accordingly computed.

- If $S[\phi]$ and the functional measure $d\mu[\phi]$ respect some group symmetry, and a regulator can be found so that ΔS_k also respects the symmetry, then Γ_k will also respect this symmetry. This is not the case, for example, for gauge symmetries. In this case, the gauge is broken and only recovered at $k\to 0$. Furthermore, the composition law of RG transformations will continue to be satisfied even if

[k] In more details,

$$\delta^d(x-z) = \int d^dy\frac{\partial\phi(y)}{\partial j(z)}\frac{\partial j(x)}{\partial\phi(y)} = \int d^dy\left(\frac{\partial^2\mathcal{W}_k[j]}{\partial j(x)\,\partial j(y)}\right)\left(\frac{\partial^2\Gamma_k[\phi]}{\partial\phi(y)\,\partial\phi(z)} + R_k\right) = \int G(x,y)\left(\frac{\partial^2\Gamma_k[\phi]}{\partial\phi(y)\,\partial\phi(z)} + R_k\right)d^dy$$

[l] In more details,

$$\partial_t\Gamma_t[\phi]\,e^{-\Gamma_t[\phi]} = \int D\chi\left(\int\phi(x)\,\partial_t R_t\left(\nabla^2\right)\phi(x)\,d^dx\right)exp\left(-S[\phi] - \Delta S_t[\phi] + \int\frac{\partial\Gamma_t}{\partial\phi}\chi(x)\,d^dx\right)$$
$$\partial_t\Gamma_t[\phi] = \int\partial_t R_t\left(\nabla^2\right)\langle\phi(x)\,\phi(x)\rangle\,d^dx = \int\partial_t R_t\left(p^2\right)\langle\phi(p)\,\phi(-p)\rangle\,d^dp = Tr\left(\partial_t R_t\left(\Gamma_t^{(2)} + R_t\right)^{-1}\right)$$

[m] In more details,

$$\partial_k\Gamma_k[\phi] = Tr\left(\partial_k R_k\left(\Gamma_k^{(2)} + R_k\right)^{-1}\right) = \frac{\partial R_k}{\partial k}\frac{\partial}{\partial R_k}Trlog\left(\Gamma_k^{(2)} + R_k\right) = \tilde{\partial}_k Trlog\left(\Gamma_k^{(2)} + R_k\right)$$

the Wetterich equation is truncated, or an Ansatz is used to solve to "guess" the expression for Γ_k. This is obvious, since, in those cases, the Wetterich equation continues to be exact on the solution-space of the Ansatz, even if the Ansatz itself is not an exact representation of the physical system at question.

- The automatic regularization of the RG approach in the UV and IR limits are made explicit from the Wetterich Equation. The regulator, by construction, ensures the IR regulatization. Meanwhile, its presence on the trace on the left-handed side of Eq. (5.8) protects UV regularization, since it vanishes for $p > k$. This ensures that, in the Wetterich equation, only the momenta in the shell $q \approx k$ will contribute to the RG flow at a given scale; and both UV and IR ranges come automatically regularized. In the framework of this method, The RG flow is calculated first, and only afterwards the renormalized quantities show up. The perturbation expansion can then be retrieved by truncating the renormalization group equation, and the renormalization "comes as bonus".

- k acts as an IR regulator, and the pole at $k \to 0$ is removed. No phase transition is thus observed in the regularized theory, i.e., $k > 0$. On the other hand, precursors to phase transitions can be observed in the limit. Furthermore, the fact that the pole at $k = 0$ is regularized, means that Γ_k is a regular function of ϕ, even if Γ itself is not. This guarantees that Γ_k can be expanded in orders of derivatives of the field. Truncation methods discussed in subsequent sections explicitly rely on this fact.

- An odd aspect of this equation, is that the computation of a physical quantity, the effective action, crucially depends on the regulator R_k, which is arbitrary, except at the end-points of the RG trajectory, $k = 0$, $k = \Lambda$. This is understood in the sense that the RG trajectory of Γ_k connecting the bare action to the effective action does not matter, so long as the end-points are preserved. Furthermore, by working with dimensionless and renormalized quantities, the RG equation can be made explicitly independent of k and Λ. This is important, as it guarantees that the composition rule for RG transformations is satisfied, and the self-similarity property of RG transformations is valid, even after the expansion series is truncated. This property allows for the identification of fixed points and the computation of critical exponents associated with the theory. Although the exact theory should be independent of the choice R_k, in practice, the introduction of approximations might lead to the emergence of undesired dependencies on R_k. In principle, so long the choice of regulator is such that this dependance is small, this must not be a problem.

- The Wetterich Equation is highly non-linear, and approximation methods are generally needed to solve it. Truncation methods, on the other hand, are not necessarily required. In subsequent sections, computational implementations will be shown, where the RG flow can be integrated numerically, and those integrations are not truncated solutions. On the other hand, to the extend that truncation methods are concerned, one notices that those methods are an improvement from perturbative solutions, only insofar as they resum the diagrams in a more efficient manner. Furthermore, solutions to the Wetterich Equation consist of projecting the equation onto an Ansatz: one solves the exact Wetterich Equation, in a restricted functional space. The quality of the result depends only on the choice of that space. The RG flow itself is thus perfectly analytical, up to the choice of Ansatz. All of the properties concerning RG flow, that the Wetterich Equation would otherwise predict, are retained; and approximations arise only insofar as the choice on Ansatz is concerned.

- The Wetterich Equation leads furthermore to the decoupling of massive nodes: suppose a theory with some mass M, $k << M$. Those nodes add nearly no contribution to the flow; such that the main

contributions arise in the range $M < k < \Lambda$. When $k << M$, it is not possible to assert whether the underlying theory that led to this effective theory involves or massive particle or not, since, at this scale, the massive nodes are nearly decoupled, and the high- and low-energy sectors of the theory are decoupled from each other. This fact leads to universality as a consequence, namely because different theories, with different high-energy sectors, result in the same low-energy theories.

6 Method of Green Functions and Vertex Expansion

6.1 Description of the Method on a ϕ^4 Scalar Theory.

In this portion, the methodology is briefly discussed and main results for a simple ϕ^4 scalar theory are stated out for comparison with the perturbative method.[n] Given the Wetterich Equation, Eq. (5.7), one begins by explicitly computing the RG flow for the n-point Vertex functions, by taking functional derivatives against both sides in the Wetterich Equation; example Eq. (6.1).[38, 45, 49]

$$\begin{aligned}\partial_k \Gamma_k^{(2)} = &\int \left(\partial_k R_k\left(q\right)\right) G_k\left(q\right) \Gamma_k^{(3)}\left(q,p\right) G_k\left(p\right) \Gamma_k^{(3)}\left(q,r\right) G_k\left(r\right) dqdpdr \\ &- \int \left(\partial_k R_k\left(q\right)\right) G_k\left(q\right) \Gamma_k^{(4)}\left(q,p,p_1\right) G_k\left(p\right) dqdp \end{aligned} \tag{6.1}$$

As a final step, an Ansatz for $\Gamma_k$as well as a regulator R_k is typically chosen. Typical choices include, for example, a Litim regulator and an Ansatz specified by Eq. (6.2) below.

$$\Gamma_k\left[\phi\right] = \frac{1}{2}\phi^2 p^2 + \frac{1}{2}\phi^2 m^2 - \frac{1}{4!}\phi^4 \lambda \; ; \qquad R_k\left(q\right) = \left(q^2 - k^2\right)\Theta\left(q^2 - k^2\right) \tag{6.2}$$

Eq. (6.1) solves explicitly to Eq. (6.3).

$$\partial_k m_k^2 = \frac{\lambda_k}{16\pi^2}\frac{k^5}{\left(k^2 + m_k^2\right)^2} \approx \frac{\lambda_k k}{16\pi^2} \; ; \qquad \partial_k \lambda_k = \frac{\lambda_k^2}{16\pi^2}\frac{k^5}{\left(k^2 + m_k^2\right)^3} \approx \frac{1}{k}\frac{3\lambda_k^2}{16\pi^2} \tag{6.3}$$

Differently from the result obtained through steps of regularization and renormalization in perturbation theory, notice that in this case the renormalization scale has a mass dimension $[k] = \left[m^1\right]$. One defines the renormalization time $t = log\left(k/\Lambda\right)$, and Eq. (6.3) reduces to Eq. (6.4), such that the choice $\frac{\mu}{m^2} \to \frac{1}{k^2}$ makes the equivalence of both methods explicit.

$$\frac{1}{k^2}\partial_t m_t^2 = \frac{\lambda_t}{16\pi^2} \; ; \qquad \partial_t \lambda_t = \frac{3\lambda_t^2}{16\pi^2} \tag{6.4}$$

6.2 Discussion

At this point, one emphasizes important elements with respect to the method of vertex expansion:[17, 18, 42, 44, 46]

- The method creates a hirearchy of flows equations for the Green functions $G^{(n)}$. In other words, one creates an infinite tower, where $\partial_t \Gamma_k^{(n)}$ can be expressed in terms of $\Gamma_k^{(n+1)}$ and $\Gamma_k^{(n+2)}$. This equation in

[n] *see* Appendix Wetterich Vertex Expansion Algebra, for detailed algebraic manipulations concerning this method.

principle cannot be solved analytically. In practice, one truncates the tower, thus closing this hirearchy, so that only the first n terms contribute to the flow equations (in the case discussed above, $\Gamma_k^{(2)}$ and $\Gamma_k^{(4)}$). The RG flow is expressed by a finite number of flow equations. These equations are solved self-consistently.

- although the FRG method is technically non-perturbative in formulation, the solution above is actually perturbative. The calculation implicitly assumes here that the theory could be solved by perturbative expansion, because one proposes that the effective action could be expressed in terms of coefficients $\Gamma_k^{(2)} = p^2 + m_k^2$, $\Gamma_k^{(4)} = \lambda_k$. One implicitly chosees a background field ϕ_* to expand Γ_k around when this particular choice for the coeffients is made. The key distinction between this method and the method with perturbation consists of the following, that $\Gamma_k^{(n)}$ is not evaluated at the bare action S_*, but rather at some flowing effective action Γ_k.

- In effect, the flow re-summs infinitely many diagrams in the perturbation order, so that the propagator already comes fully dressed. This can be seen from the term $\left(\Gamma_k^{(2)} + R_k\right)$ in the denominator, which corresponds to a geometric progression expansion in powers of λ. The solution continues to be perturbative, but the regulator re-organizes the contributions, so that scale is given preference, and at each step of the flow, the contributions $p \approx k$ are integrated out. This leads to a perturbative solution that is IR-consistent. The regularization and renormalization come as bonus.

- To emphasize, the solution is still perturbative. The FRG, in the approximation scheme formulated in the foregoing, does not solve the exact effective action. Instead, it projects the exact RG flow onto the chosen solution (Ansatz). The answer is exact up to truncation choice. The theory is an improvement with respect to perturbation, because it resums infinite diagrams, and therefore accurately predicts perturbative phenomena in the IR scale, as well as strong coupling, which typical perturbation theory is conventionally unable to predict without modifications. In other words, this method predicts local fluctuations in the field; it correctly gives the shape of the effective action; and correctly qualifies the location and nature of its minima (i.e., spontaneous symmetry breaking). This is because the phase transition is IR-divergent at $\xi \to \infty$, $m \to 0$, and the resummation accomplished by the FRG exactly cures those IR divergencies. So long as one's interest is constrained to long-distance physics, the truncation in which only the lowest powers of Γ_k are maintained actually correctly describes the physical phenomena, even when the perturbation theory itself fails. This is the key distinction here.

- The truncation itself comes with a number of undesirable effects. The truncation choice does not accurately predict anomalous dimensions, for example the momentum dependance of verices respectively the running wavefunction renormalization. These higher-order effects can in principle still be predicted by changing the choice of truncation.

- The Ansatz choices for $\Gamma_k^{(2)}$ and $\Gamma_k^{(4)}$ are not random guesses. They are entirely based on the assumption that the effective action can be expanded in a power series up to the first few terms. The term $\Gamma_k^{(3)} = 0$ is conversely imposed by the $\mathbb{Z}^2$ symmetry. One observes furthermore that the perturbation expansion can be recovered from the FRG method, by implementing the trucation up to $\Gamma_k^{(2)}$ and $\Gamma_k^{(4)}$, and explicitly replacing the running parameters with the bare parameters.

7 Fixed Points and Criticality

7.1 Qualitative Overview

There are two length scales in statistical systems, the first corresponds to the lattice spacing $a \sim \Lambda^{-1}$ (or correspondingly the compton wavelength, for quantum systems), and the second is the correlation length $\xi \sim m^{-1}$. At criticality, the correlation length diverges, $\xi \to \infty$, and the physics of the system is accordingy dominated by ξ, whereas the dependance on a disappears. This divergence on the correlation length gives origin to the concepts of scale invariance and universality, which correspondingly lead to power-law behaviours for universal quantities in scale-invariant systems. This makes sense qualitatively: because if the correlation length goes to infinity, quantum fluctuations are expected to have the same significance in all scales, that the theory is symmetric under scale transformations. (The connection to conformal invariance will be discussed at later sections). Universality, accordingly, corresponds to the notion that different microscopic systems have similar macroscopic properties, since in the macroscopic scale, collective behaviours are statistically averaged-out. To given an example, $1mol$ of gas has $\mathcal{O}\left(10^{24}\right)$ microscopic degrees of freedom. On the other hand, only a few set quantities must be defined in order to accurately describe the system in the macroscopic scale (such as temperature, volume, pressure, under others). In this connection, the existence of fixed points in the RG flow, i.e., points where the RG flow freezes and couplings stop running, become directly connected to the notion of criticality: those fixed points are symmetric under scale transformations; which means that the correlation length diverges; which corresponds to a second-order phase transition (and, sometimes, to spontaneous symmetry breaking). Those fixed points furthermore define the notion of universality: theories whose respective RG-flows lead to the same fixed point in the RG phase-space correspond to the same universality class, because their "look the same" at the fixed point. Those notions alone are furthermore sufficient to justify the power-law behaviour or thermodynamic quantities near criticality.[15, 20, 25—27, 31, 32, 39—42, 97]

7.2 Fixed Points of a Scalar ϕ^4 Theory

From the dimensionless flows for $\tilde{m_t}^2$ and $\tilde{\lambda}_t$ obtained previously, Eq. (7.1), a fixed point exists only solong as $\partial_t \tilde{m_t}^2 = 0$ and $\partial_t \tilde{\lambda}_t = 0$ simultaneously, that is, when $\tilde{m_t}^2 = \tilde{\lambda}_t = 0$. This is the only fixed point of a ϕ^4 theory in 4 dimensions, and it is called a gaussian fixed point, because at this fixed point the interactions vanish and the system behaves like a free theory. In other words, although the "bare theory" had interactions, as the scale approaches the gaussian fixed point, these interactions are washed-out by the RG flow and one obtains a free theory at all scales. The mean field treatment becomes exact. The existence of a gaussian fixed point indicates furthermore that the theory is marginally stable, i.e., it corresponds to the statement that perturbative treatment of the theory is well-behaved and leads to "good results".[17, 18]

$$\partial_t \tilde{m_t}^2 = -2\tilde{m_t}^2 + \frac{\tilde{\lambda}_t}{16\pi^2}\frac{1}{\left(1+\tilde{m_t}^2\right)^2} \; ; \qquad \partial_t \tilde{\lambda}_t = \frac{3\tilde{\lambda_t}^2}{16\pi^2} \tag{7.1}$$

The existence of a fixed point naturally leads to power-law scaling behavious near criticality. Looking at the scaleless RG-flow equations, Eq. (7.1), we can explicitly solve the differential equation corresponding to λ_k, Eq. (7.2), where Λ is some initial condition in the RG flow.[42, 46]

$$\tilde{\lambda}_t \sim \frac{16}{3\pi^2}\frac{1}{log\left(k/\Lambda\right)} \tag{7.2}$$

Similarly, the mass equation can be approximately calculated to Eq. (7.3).[o]

$$m_k^2 = k^2 \tilde{m_t}^2 \sim \frac{k^2}{log\,(k/\Lambda)} \tag{7.3}$$

The correlation length has the scaling relation given by Eq. (7.4), such that the critical exponent is $\nu = -\frac{1}{2}$ corresponding to the scaling between the correlation length and the distance to criticality.

$$\xi \sim m_{phys}^{-1} \sim \left(m_{k,*}^2\right)^{-\frac{1}{2}} \sim \left(\frac{k_*^2}{log\,(k_*/\Lambda)}\right)^{-\frac{1}{2}} \sim \frac{k_*}{\left(log\,(k_*/\Lambda)\right)^{\frac{1}{2}}} \tag{7.4}$$

7.3 Additional Remarks

Here one emphasizes a couple of important points:[13, 17, 18, 32, 34—38, 56, 58, 62]

1. The scaling behaviour of thermodynamical quantities near criticality arises seamlessly: near criticality the RG flow equations become linearized, which automatically leads to power laws. In fact, the existence of fixed points in the RG flow itself, is sufficient to explain phase transitions and and critical exponents respectively the scaling behaviour of universal quantities near criticality. This is one of the great successes of the FRG method. By the mere qualitative/semi-quantitative analysis of the FRG equations, one predicts criticality and phase transitions in systems, even if an exact analytical solution for the RG flow cannot be formally written down. Furthermore, the fact that one projects the RG equation into an Ansatz whilst preserving the RG composition law, implies that even when approximation techniques are carried-out, important properties of the system with respect to the RG flow are still in principle accurately captured. The RG flow is not spoiled by the Ansatz.

2. diffferent from mean field theory that predicts exclusively power-law scalings, the introduction of fluctuations leads the FRG approach to additionally predict logarithmic corrections to the scaling, at least for the particular theory currently investigated.

7.4 Running of Couplings

See Appendix section 11.8 for a more generic discussion concerning this section. One notices from Eqs. (7.2, 7.3) that in this particular case, both the couplings m_k^2 and λ_k attain an anomalous scaling. Namely, contrary to the expectation that $m_k^2 \sim k^2$ and $\lambda_k \sim k^0$ due to scaling arguments, one in reality observes the additional emergence of a logarithmic dependence on the scale, that is to say, Eq. (7.5).

$$m_k^2 \sim \frac{k^2}{log\,(k/\Lambda)}\ ; \qquad \lambda_k \sim \frac{1}{log\,(k/\Lambda)} \tag{7.5}$$

[o] In more details,

$$\frac{d\tilde{m_t}^2}{dlogk} = -2\tilde{m_t}^2 + \frac{\lambda_t}{16\pi^2}\frac{1}{\left(1+\tilde{m_t}^2\right)^2} \approx -2\tilde{m_t}^2 + \frac{1}{3}\frac{1}{log\,(k/\Lambda)} \Longrightarrow \frac{d}{dt}\left(e^{2t}\tilde{m_t}^2\right) = \frac{1}{3}\frac{e^{2t}}{t} \Longrightarrow m_k^2 = k^2\tilde{m_t}^2 \sim \frac{k^2}{log\,(k/\Lambda)}\ ,$$

where one substitutes $\tilde{\lambda_t}$, uses the gaussian fixed point condition, $\tilde{m_t} \to 0$, and remembers the RG-time definition, $t = log\,(k)$. In the last step, one approximates the solution to the differential equation by assuming that e^{2t} varies much quicker and is the dominant term, such that $\frac{1}{t}$ can be approximately treated as a constant. This is valid in this particular analysis, since the criticality is found in the IR limit $k \to 0$, such that t varies slowly in comparison to e^{2t}.

One remarks that those runnings of the couplings with respect to the scale occur on the actual, physical and observable couplings, and not on the dimensionless quantities. Therefore, these correspond to real, physically-observable scale-dependancies induced to the couplings. Literature normally illustrates and characterize tihs behaviour by referring to it as a "running of the coupling constants", that is justified because the ocuplings are "screened-out by virtual interactions". In other words, one understands that the microscopic interactions characterized by the coupling λ_k essentially vanish or get "averaged-out" in the macroscopic, mean-field limit; such that the effective theory in the IR limit becomes independent of λ_k.[29, 30, 42, 46]

This particular analysis is carried out specifically in $d = 4$ here. In Appendix A Remark on Dimensional Analysis for example, a similar analysis is carried out in $d = 2$ and shows that in that case m_k^2 does not run, and λ_k runs as Eq. (7.6), such that the IR limit is actually characterized by a relevant λ_k parameter.

$$\lambda_k \sim \frac{k^2}{log\left(\Lambda/k\right)} \tag{7.6}$$

8 Local Potential Approximation (LPA)

8.1 Method Description of a scalar ϕ^4 Theory

Once again the detailed algebraic manipulations are subordinated to an Appendix.[p] In essence, one introduces the Ansatz for the running effective action, Eq. (8.1), where one separates the contributions into a momentum-dependant kinetic term and a local potential $U_k\left[\phi\right]$. One additionally makes the $\mathbb{Z}^2$ symmetry explicit by setting $\rho = \frac{1}{2}\phi^2$.[16—18, 42, 44][q]

$$\Gamma_k\left[\phi\right] = \int d^dx \left\{\frac{1}{2}\partial_\mu\phi\partial^\mu\phi + U_k\left(\rho\right)\right\} \; ; \qquad R_k\left(q\right) = \left(q^2 - k^2\right)\Theta\left(q^2 - k^2\right) \tag{8.1}$$

One additionally defines the dimensionless quantities, Eq. (8.2), similar to the method of vertices.[r]

$$m = k\tilde{m} \; ; \quad \rho = k^{d-2}\tilde{\rho} \; ; \quad g_4 = k^{-d+4}\tilde{g}_4 \; ; \quad g_{2n} = k^{d-n(d-2)}\tilde{g}_{2n} \; ; \quad \partial_\rho = \left(\partial_\rho\tilde{\rho}\right)\partial_{\tilde{\rho}} = k^{2-d}\partial_{\tilde{\rho}} \; ; \tag{8.2}$$

Finally, one defines the RG-time, Eq. (8.3).

$$\partial_k = \left(\partial_k t\right)\partial_t = k^{-1}\partial_t \tag{8.3}$$

The Wetterich Equation ultimately takes the form[s] Eq. (8.4), where the notation using the tilde, $(\tilde{\ldots})$, indicates that the quantities are dimensionless, as specified in Eq. (8.2) above. furthermore, the term A_d is a normalization constant corresponding to the integral over a d-dimensional ball of radius k. In 4 dimensions, for example, one obtains $A_4 = \frac{1}{32\pi^2}$.

$$\partial_t\tilde{U}\left(\tilde{\rho}\right) = -d\cdot\tilde{U}\left(\tilde{\rho}\right) + \left(d-2\right)\tilde{\rho}\tilde{U}_t'\left(\tilde{\rho}\right) + A_d\cdot\frac{1}{1 + \tilde{U}'\left(\tilde{\rho}\right) + 2\tilde{\rho}\tilde{U}''\left(\tilde{\rho}\right)} \tag{8.4}$$

[p] *See* Appendix Derivation of the LPA-Wetterich for a ϕ^4 Scalar Theory.

[q] In principle it is not a requirement that the system be $\mathbb{Z}^2$-symmetric, although it is commonly accepted to be the case at least for the vast majority of physical systems. Furthermore, the equations can in similar way be solved even when this symmetry is not imposed, only that the algebraic steps become more complicated.

[r] *See* Appendix 11.8 for a more detailed discussion of those dimensionless quantities.

[s] *See See* Appendix Derivation of the LPA-Wetterich for a ϕ^4 Scalar Theory.

8.2 Discussion

First of all, one would like to notice some of the key elements of this formulation:[17—20, 34—38, 48—53, 59—65]

1. The formulation of the LPA relies on the regularization of the pole of the theory at $k = 0$ by means of R_k; since this regularization ensures that Γ_k is a regular function of ϕ (even if Γ itself is not), such that an expansion of Γ_k in orders of the field and its derivatives exists to begin with. Furthermore, the LPA scheme can be seen as an RG-improved Landau Theory, in the sense that one introduces a model with as minimal structure as possible, and the running of the couplings arises from FRG alone. Another point to note, is that this choice for the Ansatz is very natural, since at $k \to \Lambda$ the Ansatz corresponds to the effective action in the mean field formulation; such that this method could informally be thought of a mean-field/perturbative solution projected onto the RG flow.

2. The effective action is local. Non-local operators are ignored; and the relevant physics of the theory is primarily dominted and described by the effective potential U_k, whereas the momentum dependance on vertices is assumed to be sub-leading and is neglected; as well as couplings between momenta and the field. This assumption namely explicitly misses the prediction of the anomalous dimension η, because by construction one sets a fixed wavefunction renormalization. This means that from start the LPA already fails to capture the running of couplings and masses with the momentum.

3. The LPA is motivated by the desire of capturing IR physics, with minimal complexity added. It assumes that the theory is well-described by an effective potential with perturbative dominance, and it in those regimes therefore correctly predicts the a number of qualitative observations, including: (i) convexity of the effective potential; (ii) the emergence of IR phenomena such as phase transitions, symmetry breakings, and scaling laws; and (iii) it is therefore well-suited for (in those systems) investigating universality and identifying universality classes. The LPA in principle even allows the computation of critical coefficients, although with limitations insofar as accuracy is concerned. The LPA therefore captures the IR physics that is otherwise not captured by perturbative method, without the need of resuming graphs.

4. In the mathematical side, this formulation has the advantage that it reduces the Wetterich Equation to a simple differential equation on the flow of the effective potential, thus reducing the non-linearity of the original equation to a more tacktable formulation. The solution can furthermore be systematically improved by including extra truncation terms.[t] On the negative side, those truncations inherit unphysical dependancies on the regulator. Furthermore, as one demonstrates in later sections, at the lower critical dimension $d = 2$ the method fails completely,[u] because the assumption that the tower of differential equations can be truncated is in this particular case not justified.

5. Those shortcomings of the LPA lead to a number of issues. At the lower critical dimension, the LPA for example falsely predicts the spontaneous symmetry breaking of the XY model.[v] On the other hand, it correctly predicts the phase transition in the ising model; the emergence of critical exponents

[t] Although one makes the observation here, that insofar as the study of universality classes is concerned, meaningful qualitative information about the system is primarily described by leading terms; the so-called "relevant coefficients".

[u] A forewarning to this statement must be given: it is not exactly accurate. Specifically, the truncated short range ising model fails in $d = 2$, which is a fact that will be more explicitly demonstrated in Section Connection to Mermin-Wagner. As an example, long-ranged models, which are also investigated in this paper, can attain critical dimension $d_{CRIT} < 2$ depending on the choice of parameters; in which case truncation methods are well-justified.

[v] This statement is in fact not entirely accurate; although it is often (falsely) claimed in the literature. In Section Connection to Mermin-Wagner, the author demonstrates that although this spontaneous symmetry breaking is naively predicted by the

(although only qualitatively, and quantitative estimation can generally be untrustworthy); and it (by construction!) fails to correctly predict physics in systems that are dominated by momentum terms, due to the assumption of subleading momentum dependancy.[w] Gauge theories are another challange (although ultimately resolvable), since the LPA breaks the gauge symmetry, leading to modified Ward identities. In summary: the LPA fails primarily in the lower critical dimension, and/or when the dominant fluctuations are introduced by the gradient fluctuations as opposed to field fluctuations.

6. On the bright side, the implementation of semi-analytical, computational methods — as discussed in this paper in Section Analytical and Computational Methods in FRG — allows for the explicit numerical solution to the RG-flow for the potential; correspondingly the prediction of phase transitions as well as critical coefficients, without in principle the requirement of imposing any form of truncation at all.

9 The Tower Equations in the LPA

9.1 The Tower Equations

From Eq. (8.4) above, one obtains an infinite tower of differential equations corresponding to each of the couplings, by taking derivatives with respect to $\tilde{\rho}$ on both sides of Eq. (8.4). In Eqs. (9.1, 9.2, 9.3) below, the first three equations in this tower are stated-out. One makes the following observations:[17, 18, 32, 38, 42, 44, 46, 56—58, 62—65]

1. In practice, this method of building-up a tower of equations is a truncation method. Namely, one must choose a trunctation n such that by definition one sets $\tilde{U}_t^{(n)}(\tilde{\rho}) = 0$ for all t and $\tilde{\rho}$, in order to obtained a closed system of differential equations. This method is therefore in its essence perturbative; and in fact up to truncation choice it recovers the same results as those obtained by previously by vertex expansion.

2. On one side, this method can be used in order to extract qualitative results for the system. On the other side, and as already discussed in the foregoing, the assertation of quantitative results is accomplished to better aptitude by instead explicitly solving the RG flow equation for the exact potential, as this paper discusses in more details in Appendix Analytical and Computational Methods in FRG.

3. This method provides a quantitative view of the qualitative observation we made earlier: we create an infinite tower tower of differential equations representing the RG flow, and we eventually truncate this tower at some order. The equations are afterwards self-consistently solved. Furthermore, the composition law of RG transformations will continue to be satisfied even in the truncated solution, since the Wetterich equation continues to be exact on the solution-space of the Ansatz, even if the Ansatz itself is not an exact representation of the physical system at question. This is because we are projecting the exact RG flow into the solution-space of the particular Ansatz.

LPA in $d = 2$, in contradiction to the Mermin-Wagner Theorem, a self-consistent treatment of the flow equations makes it explicit that such a phase transition would not be physical, namely because it requires negative wavefunctions to occur; such that the LPA correctly predicts the absence of spontaneous symmetry breaking in $d = 2$. In the same section, the author furthermore extends an "FRG-like" Mermin-Wagner argument to generic systems, not exclusively those endowned by short-range interactions. As far as the author is aware, both those results have not been demonstrated before.

[w] This particular shortcoming can at least partially be resolved by explicitly introducing a wavefunction renormalization to the Ansatz, which is called the LPA-prime method, and will be introduced in subsequent sections.

$$\partial_t \tilde{U}_t^{(1)}(\tilde{\rho}) = -2\tilde{U}_t^{(1)}(\tilde{\rho}) + (d-2)\,\tilde{\rho}\tilde{U}_t^{(2)}(\tilde{\rho}) - A_d \cdot \frac{3\tilde{U}_t^{(2)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(3)}(\tilde{\rho})}{\left(1+\tilde{U}_t^{(1)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(2)}(\tilde{\rho})\right)^2} \tag{9.1}$$

$$\begin{aligned}\partial_t \tilde{U}_t^{(2)}(\tilde{\rho}) &= (d-4)\,\tilde{U}_t^{(2)}(\tilde{\rho}) + (d-2)\,\tilde{\rho}\tilde{U}_t^{(3)}(\tilde{\rho}) \\ &\quad - A_d \cdot \frac{5\tilde{U}_t^{(3)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(4)}(\tilde{\rho})}{\left(1+\tilde{U}_t^{(1)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(2)}(\tilde{\rho})\right)^2} + 2A_d \cdot \frac{\left(3\tilde{U}_t^{(2)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(3)}(\tilde{\rho})\right)^2}{\left(1+\tilde{U}_t^{(1)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(2)}(\tilde{\rho})\right)^3}\end{aligned} \tag{9.2}$$

$$\begin{aligned}\partial_t \tilde{U}_t^{(3)}(\tilde{\rho}) &= (2d-6)\,\tilde{U}_t^{(3)}(\tilde{\rho}) + (d-2)\,\tilde{\rho}\tilde{U}_t^{(4)}(\tilde{\rho}) - A_d \frac{7\tilde{U}_t^{(4)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(5)}(\tilde{\rho})}{\left(1+\tilde{U}_t^{(1)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(2)}(\tilde{\rho})\right)^2} + \\ &\quad + 6A_d \frac{\left(5\tilde{U}_t^{(3)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(4)}(\tilde{\rho})\right)\left(3\tilde{U}_t^{(2)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(3)}(\tilde{\rho})\right)}{\left(1+\tilde{U}_t^{(1)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(2)}(\tilde{\rho})\right)^3} - 6A_d \frac{\left(3\tilde{U}_t^{(2)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(3)}(\tilde{\rho})\right)^3}{\left(1+\tilde{U}_t^{(1)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(2)}(\tilde{\rho})\right)^4}\end{aligned} \tag{9.3}$$

9.2 Connection to Renormalizability

Although Eqs. (9.1, 9.2, 9.3) may appear overwhelming at first glance, one infers very interesting qualitative properties from them, which are outlined here:[18, 25, 33, 42, 62—65, 66—68]

- First of all, as already discussed, the RG flow can be linearized in the IR limit. This means that in Eqs. (9.1, 9.2, 9.3) above, the linear terms correspond to the the dominant contributions to the RG flow, Eqs. (9.4), where one easily obtains the leading contribution to $\partial_t \tilde{U}^{(n)}(\tilde{\rho})$ by induction.

$$\begin{cases} \partial_t \tilde{U}^{(1)}(\tilde{\rho}) & \sim -2\tilde{U}_t^{(1)}(\tilde{\rho}) \\ \partial_t \tilde{U}^{(2)}(\tilde{\rho}) & \sim (d-4)\,\tilde{U}_t^{(2)}(\tilde{\rho}) \\ \partial_t \tilde{U}^{(3)}(\tilde{\rho}) & \sim (2d-6)\,\tilde{U}_t^{(3)}(\tilde{\rho}) \\ \partial_t \tilde{U}^{(n)}(\tilde{\rho}) & \sim (n(d-2)-2)\,\tilde{U}_t^{(n)}(\tilde{\rho}) \end{cases} \tag{9.4}$$

- This observation alone is sufficient to justify renormalizability and non-renormalizability from the point of view of FRG: on the one hand, the term $\partial_t \tilde{U}^{(1)}(\tilde{\rho})$ always corresponds to the RG flow for the "mass" term, and it has a scaling relation (to the first order and in the IR limit) $\left[m_k^2\right] = k^2$. On the other hand, the $\partial_t \tilde{U}^{(n)}(\tilde{\rho})$ always corresponds to the $2n^{\text{th}}$ order coupling $g_{\tilde{2n}}$ pegged to a polynomial term $\tilde{\phi}^{2n}$ on the field; and its scaling relation is dominated and specified by the term $(n(d-2)-2)\,\tilde{U}_t^{(n)}(\tilde{\rho})$; such that, assuming $d \neq n$, the scaling relation for the coupling, in the IR limit and to its dominant order, will read as Eq. (9.5).[x]

$$g_{\tilde{2n}} \sim k^{(n(d-2)-2)} \sim m_k^{\frac{n}{2}(d-2)-1} \tag{9.5}$$

- When $(n(d-2)-2) < 0$, the coupling strength corresponding to $g_{\tilde{2n}}$ increases as it approaches the IR limit $k \to 0$, and its contribution plays a role in describing an effective field theory. Conversely, when $(n(d-2)-2) > 0$, the coupling strenght $g_{\tilde{2n}}$ gets exponentially suppressed as it approaches the IR limit and plays no role in describing the (IR) theory; in other words, $g_{\tilde{2n}}$ is suppressed near the IR

[x] In more details,

$$\frac{\partial g_{\tilde{2n}}}{\partial log\,(k)} \sim (n(d-2)-2)\,g_{\tilde{2n}} + \ldots \Longrightarrow g_{\tilde{2n}} \sim k^{(n(d-2)-2)} \sim m_k^{\frac{n}{2}(d-2)-1}$$

fixed point, such that it is an irrelevant degree of freedom in the theory; i.e., the theory in the IR leads to the same results, whether or not this coupling is explicitly introduced to the theory.

- The coupling at $n = d$ is special, since the linear contribution in $\partial_t \tilde{U}^{(n=d)}(\tilde{\rho})$ is killed off. For example, in the case of $d = 4$, this corresponds to the interaction $g_{\tilde{2n}} = \tilde{\lambda}$. In this case, Eq. (9.2), that is to say, Eq. (6.4), takes the form Eq. (9.6); and the coupling scales logarithmically with the scale instead of scaling as a power law, which is the result derived previously. In this case, the IR limit suppresses the coupling $\tilde{\lambda} \to 0$ logarithmically.[y]

$$\tilde{\lambda} \sim \frac{1}{log\left(\Lambda/k\right)} \tag{9.6}$$

- This means that, by just looking at this tower of differential equations, Eqs. (9.1, 9.2, 9.3), one effordlessly tells that in (even) d dimensions, with $\mathbb{Z}^2$ symmetry, in addition to the mass term m_k^2, one obtains (at most) $n(d) = \frac{2}{d-2}$ renormalizable couplings that need to be tuned to describe the theory. This means that all models with $d > 4$ are IR-triviall in the FRG sense;[z] namely because $n(d = 4) = 1$ and $n(d > 4) < 1$. This can be seen directly from the tower of differential equations: in a $d > 4$ theory, the closest to a renormalizable coupling is Eq. (9.7). However, for $d > 4$, this leads to $\tilde{g}_4 \sim k^C$ for $C > 0$, such that the coefficient is suppressed at IR. $d = 4$ is thus defined as the upper critical dimension, above which fluctuations become irrelevant.

$$\partial_t \tilde{U}^{(2)}(\tilde{\rho}) \sim (d-4)\,\tilde{U}_t^{(2)}(\tilde{\rho}) \tag{9.7}$$

- The identical analysis can be carried out the other way around: the lower critical dimension $d = 2$. One finds that the leading order contribution to the nth coupling is given by Eq. (9.8). This means that the wave function has no scaling dimension, $[\rho] = \left[k^0\right]$, whereas all coefficients scale as $[g_n] = \left[k^2\right]$. This leads to the conclusion that the truncation of the power series is in this case not justified, because all of the infinite many couplings are relevant to the theory, and they do not get suppressed by the renormalization flow. This statement gives an initial insight towards the Mermin-Wagner theorem,[aa] that at the lower critical dimension no spontaneous symmetry breaking of a continous symmetry at finite temperature can occur, because the contributions due to the fluctuations are not suppressed by the RG flow.

$$\partial_t \tilde{U}^{(3)}(\tilde{\rho}) \sim \left(n(d-2) - 2\right)\tilde{U}_t^{(n)}(\tilde{\rho}) = -2\tilde{U}_t^{(n)}(\tilde{\rho}) \tag{9.8}$$

- Qualitatively, the RG flow equations provides an incredible justification, why the microscopic degrees of freedom and infinitely many couplings lose importance as the flow reaches the macroscopic scale, so that the physics of the system can be (almost) entirely described but just a few macroscopic degrees

[y] In more details,

$$\partial_t \tilde{\lambda} = \frac{9}{16\pi^2} \cdot \frac{\tilde{\lambda}^2}{\left(1+\tilde{m}^2\right)^3} \approx \frac{9\tilde{\lambda}^2}{16\pi^2} \Longrightarrow \tilde{\lambda} \sim \frac{1}{log\left(\Lambda/k\right)}$$

[z] That is to mean: such theories are characterized exclusively by a gaussian fixed point corresponding to the free-theory.

[aa] Here, one must emphasize that this is a strong insight, although it falls short of being a full proof. The analysis of the FRG flow namely shows that there is some "special" lower dimension, the critical lower dimension, which is endowned of special properties. In this dimension namely all couplings become relevant. Fundamentally, one observes that this statement agrees with the Mermin-Wagner picture, because Mermin-Wagner argues, in essence, that SSB is prevented in $d \leq 2$ due to the existence of IR fluctuations that can be arbitrarily generated with little energy cost. The FRG analysis agrees with this picture,

of freedom: one considers the RG phase-space. As those microscopic degrees of freedom move through the RG flow direction, they tend to degenerate into hypersurfaces of lower dimension, so that the different degrees of freedom become degenerate and vanish quickly with scale. An analogy emerges in literature, is to imagine the RG hypersurface as a landscape, and the multiple RG flows of the microscopic degrees of freedom as small rivers that eventually flow into or merge onto a larger river; which corresponds to those macroscopic degrees of freedom. In this analogy, the explicit solution of the RG flow equations shows that the the increase of scale leads to the degeneration of those microscopic degrees of freedom onto a submanifold of the RG hypersufrace. As the scale increases, the microscopic degrees (and eventually even the macroscopic ones, at criticality) "observe" the correlation length tending to infinity, $\xi \to \infty$, such that they fall into a gaussian fixed point with respect to their eigendirection, so that the macroscopic theory "sees" a theory in which those degrees of freedom vanish. This analysis furthermore justifies the reason that the inclusion of non-renormalizable terms in field theories is useless in the context of studying universal properties of the system, in consideration of the following, that they are exactly those degrees of freedom that die out at larger scales, i.e., those "small rivers that flow into the big river", in aforegiven analogy.

- The bare action $\mathcal{S}[\phi]$ might depend only on a few powers of ϕ. On the other hand, $\mathcal{S}[\phi]$ is not a physically measurable quantity; but rather the effective action $\Gamma[\phi]$; and the effective action does include arbitrary n-point correlation functions; which in turn in principle would require the tunning of invinitely many degrees of freedom. The FRG equations appear in principle not to make any distinction between renormalizable and non-renormalizable couplings in that sense. On the other hand, the question of renormalizability is resolved under the FRG framework, in that (in a small RG-time) all of the degrees of freedom on the RG hypersurface collapse into a submanifold whose dimension correspond to the number of renormalizable couplings in the theory. Those "non-renormalizable" degrees of freedom accordingly correspond to the degenerate degrees of freedom that eventually fall into this submanifold in the RG hypersurface. Once the RG flow collpses into this submanifold, the flow behaves as if it is driven by a finite number of couplings that predict the flow in this submanifold (the "big river", in the analogy). Once the flow collpses into this submanifold, it cannot be inverted back to the main manifold, as there are infinitely many non-unique ways to describe the reversal of the flow. Once the flow is initialized at a particular scale μ, it is not possible to invert the flow direction, "look back" and recover lower scales. Once the RG flow collapses onto the submanifold, the degrees of freedom become degenerate and cannot be traced back (i.e., once the "rivers" collapse into the "big river", the water loses memory from where it came from, so the trajectory cannot be reversed back). This is exactly the property that characterizes universality: once the RG flow reaches this this submanifold, universal quantities of systems belonging to the same universality class behave identically, independent of the specifics of the microscopic theory that describes it; exactly because at the limit of the scale, they collapse to the same submanifold (in the parameter-space of couplings) that describe the theory.

- In this FRG sense, renormalizability is therefore interpreted as the ability to "trace back" the RG flow along one or some of its particular eigendirections. The couplings that have always belonged to the submanifold corresponding to the fixed point are renormalizable, because their trajectories can be perfectly traced back, whereas the couplings that fall into this submanifold only near the fixed point

are non-renormalizable, because they are absorved by the renormalizable couplings. The existence of this submanifold near fixed points, into which irrelevant eigendirections in the RG flow collpase in the IR limit, is what ensures that only a finite number of parameters need to be tuned to ensure predictive power of the theory. The theory is perturbatively renormalizable, when there exists only a finite number of couplings in the basin of this fixed point. Non-renormalizable couplings are irrelevant, because they are absorbed or suppressed by the flow. The perturbative method is therefore "good" for computing universal quantities and "bad" for computing non-universal quantities, because it accurately characterizes the submanifold near the fixed point, whereas it by nature of its own formulation fails to capture effects outside this submanifold. The FRG method in principle describes the whole flow exactly, however, the implementation of truncations and approximation techniques in practice puts it back to the perturbative regime, so that non-universal information is lost. In this sense, the theory is renormalizable and predictive when there is a finite number of couplings that need to be tuned to perfectly describe the whole theory; and the theory is said to be non-renormalizable when the RG flow does not collapse to a submanifold in the IR.

9.3 Application to the Scalar ϕ^4 Theory

9.3.1 Correspondence to Vertex Method

One assumes a generic ϕ^n theory whose potential is given by $\tilde{U}(\tilde{\rho}) = \tilde{m}^2\tilde{\rho} + \frac{\tilde{g}_4}{6}\tilde{\rho}^2 + \frac{\tilde{g}_6}{30}\tilde{\rho}^3 + \ldots$. One truncates the tower at $\tilde{g}_6 = 0$. Eqs. (9.1, 9.2, 9.3) reduce to Eqs. (9.9, 9.10, 9.11). Importantly, one keeps track of the correct multiplicative factors: $3\tilde{U}^{(2)}(\tilde{\rho}) = \tilde{g}_4$ and $5\tilde{U}^{(3)}(\tilde{\rho}) = \tilde{g}_6$.

$$\partial_t \tilde{m}^2 = -2\tilde{m}^2 - \frac{A_d \tilde{g}_4}{(1+\tilde{m}^2)^2} \tag{9.9}$$

$$\partial_t \tilde{g}_4 = (d-4)\,\tilde{g}_4 + \frac{6A_d \tilde{g}_4{}^2}{(1+\tilde{m}^2)^3} - \frac{3A_d \tilde{g}_6}{(1+\tilde{m}^2)^2} \tag{9.10}$$

$$\partial_t \tilde{g}_6 = (2d-6)\,\tilde{g}_6 - \frac{30A_d \tilde{g}_4{}^3}{(1+\tilde{m}^2)^4} + \frac{30A_d \tilde{g}_4 \tilde{g}_6}{(1+\tilde{m}^2)^3} \tag{9.11}$$

For $d = 4$, one considers $A_d = \frac{1}{32\pi^2}$. One sees without difficulty that the results correpsond, up to choice of truncation and notation, to the results obtained by the vertex method. The method is furthermore an improvement from the perturbative method in the same way as previously discussed in section section 11.6 Wetterich Vertex Expansion Algebra.

9.3.2 Universality class of ϕ^4 Theory in $d = 4$

One supposes then that an additional order ϕ^6 is added to the truncation, that is, $\tilde{g}_6 \neq 0$. Two possible regimes exist:

- if $\tilde{g}_6$ is initially not being supressed, $\tilde{g}_6 >> 1$, then the corresponding flow equation is, up to leading order, dominated by Eq. (9.12). This means that, along the RG flow, $\tilde{g}_6$ will scale as $\tilde{g}_6 \sim k^2$, such that towards the IR region, $\tilde{g}_6$ will be exponentially suppressed. In other words, the regime where $\tilde{g}_6$ is not yet suppressed is short-lived, and $\tilde{g}_6$ quickly degenerates onto a suppressed regime along the RG

because it shows that indeed, when $d \leq 2$, all of the couplings, that is, loosely speaking, all of the available interaction chains, become available and can contribute arbitrarily in the IR limit.

flow towards the IR; because in this non-supressed regime, $\tilde{g_6}$ is suppressed exponentially along the RG flow towards the infrared.

$$\partial_t \tilde{g_6} \sim 2\tilde{g_6} + \dots \tag{9.12}$$

- Now $\tilde{g_6}$is already suppressed, that one treats it as if $\tilde{g_6} \approx 0$. In that case (also using $\tilde{m}^2 \to 0$ near the gaussian fixed point), the flows corresponding to $\tilde{m}^2$ and $\tilde{g_4}$ reduce back to the ϕ^4 theory, since the third term in Eq. (9.10) tends to zero, Eq. (9.13), such that the scaling corresponding to $\tilde{g_4}$ is $\tilde{g_4} = Ct^{-1}$ where t is the RG time.[ab]

$$\frac{3A_d\tilde{g_6}}{(1+\tilde{m}^2)^2} \sim 0 \tag{9.13}$$

$$\frac{30A_d\tilde{g_4}\tilde{g_6}}{(1+\tilde{m}^2)^3} \sim 0 \tag{9.14}$$

- In this regime, one can furthermore solve Eq. (9.11) analytically, by taking Eq. (9.14) and combining Eq. (9.2) with Eq. (9.3). One obtains Eq. (9.15).[ac]

$$\tilde{g_6} = -\frac{5}{3}\tilde{g_4}^2 \ , \qquad \tilde{g_6} \sim t^{-2} \tag{9.15}$$

This result tells a very interesting story: at first, $\tilde{g_6}$ is large and it is its independent degree of freedom. By nature of scale, the RG flow exponentially suppresses this coupling, so that the degree of freedom is short-lived and eventually degenerates onto a submanifold. Once this submanifold is reached, $\tilde{g_6}$ degenerates or gets "absorved" onto the $\tilde{g_4}$ degree of freedom. This is exactly the "river picture" discussed above. First, the IR theory is fully specified without the need of introducing $\tilde{g_6}$ coefficient. Near the gaussian fixed point, the interactions corresponding to $\tilde{g_6}$ are radiatively generated by $\tilde{g_4}$, and those radiative perturbations get suppressed along the RG flow at the square of the suppression rate of $\tilde{g_4}$. It is easy to see that this analysis can be arbitrarily extended: suppose one truncates the RG flow now at $\tilde{g_{10}} = 0$ and $\tilde{g_8} \neq 0$. The flow of $\tilde{g_8}$ will contain terms of $\tilde{m}^2$, $\tilde{g_4}$ and $\tilde{g_6}$, in adition to a linear term $+4\tilde{g_8}$. The linear term will once again justify the suppression described in the first bullet-point above. In the suppressed regime, the linear term doesn't contribute, and $\tilde{g_8}$ will be parametrized by $\tilde{m}^2$, $\tilde{g_4}$ and $\tilde{g_6}$. But furthermore, now $\tilde{g_6}$ is no longer a degree of freedom and it instead drops to a degeneracy $\tilde{g_6} \sim \tilde{g_4}^2$, such that the $\tilde{g_8} \sim \tilde{g_4}^2 \sim t^{-4}$. By induction, all $\tilde{g_{2n}}$ $(n > 2)$ are in a similar manner suppressed. This result justifies the qualitative analysis concerning renormalizability that was previously carried-out.[18, 66, 67, 68]

9.3.3 Theϕ^4 Scalar Theory in $d > 4$

In the previous section, the universality of ϕ^4 in $d = 4$ was investigated. Namely, it was demonstrated that the choice of truncating or not at $\tilde{g_6} = 0$ is irrelevant up to universality, because all ϕ^{2n} theories reduce to ϕ^4 in the IR. In fact, one notices that the analysis easily carries-over to all $d > 4$. Namely, the only change for $d > 4$ is that $\tilde{g_4}$ moves from being a marginal coefficient to becoming an irrelevant coefficient, which

[ab] This corresponds namely to the result already derived in Eq. (7.2).

[ac] In more details,

$$\partial_t \tilde{g_6} \approx -\frac{5\tilde{g_4}^3}{8\pi^2} = \frac{2\times 3\times 5}{2\times 3\times 8\pi^2}\tilde{g_4}^2\tilde{g_4} = \partial_t\left(-\frac{5}{3}\tilde{g_4}^2\right) \Longrightarrow \tilde{g_6} = -\frac{5}{3}\tilde{g_4}^2 \text{ and } \tilde{g_6} \sim t^{-2}$$

means that the analysis concerning the coefficients $\tilde{g_{2n}}$ $(n > 2)$ additionally carries over to $n > 1$ as well, such that the only fixed point at $d > 4$ is the trivial point, and all ϕ^n theories behave as free theories in the IR.[16—18, 32]

9.3.4 The ϕ^4 Scalar Theory in $d = 3$

The analysis is easily carried out to three dimensions. For $O(1)$ symmetries, this model corresponds to the ising universality class, describing ferromagnets, liquid-gas phase transitions, among others. For generic $O(N)$, this model also describes the qualitative behaviour of the superfluid transition in He^4, superconducting films, among others.[31, 32, 97]

The case of $d = 3$ leads to $\tilde{g_4}$ becoming a relevant parameter and $\tilde{g_6}$ becoming a marginal parameter, by simple analysis of Eqs. (9.10, 9.11). This means that in principle one cannot truncate the theory at $\tilde{g_6} = 0$, as this might leave-out important physics.

One chooses $A_3 = \frac{1}{6\pi^2}$. If the fixed point is at $\tilde{g_6} = 0$, one easily finds Eq. (9.16) by setting Eq. (9.17), solving for the flow of $\tilde{g_4}$ and replacing it on the flow of $\tilde{m}^2$.

$$\tilde{m}^2 = -\frac{1}{3}\ ; \qquad \tilde{g_4} \approx \pi \left(\frac{2}{3}\right)^5 \tag{9.16}$$

$$\partial_t \tilde{g_4} = 0\ ; \qquad \partial_t \tilde{m}^2 = 0 \tag{9.17}$$

The trivial gaussian point is also obviously a solution. If otherwise $\tilde{g_6} \neq 0$ at the fixed point , a quadratic must be solved, solve the quadratic, yielding Eq. (9.18).

$$\tilde{m}^2 = \frac{\left(1+\tilde{m}^2\right)}{12 g_6} \times \left(1 \pm \sqrt{1 + \frac{2\tilde{g_6}}{\pi^4 \left(1+\tilde{m}^2\right)^5}}\right)\ ; \quad \tilde{g_4} = \frac{\pi^2 \left(1+\tilde{m}^2\right)^3}{2 g_6} \times \left(1 \pm \sqrt{1 + \frac{2\tilde{g_6}}{\pi^4 \left(1+\tilde{m}^2\right)^5}}\right) \tag{9.18}$$

Since both $\tilde{m}^2$ and $\tilde{g_4}$ have been now affixed, one can in the third differential equation consider the $\tilde{m}^2$ and $\tilde{g_4}$ dependencies as independent variables with respect to $\tilde{g_6}$ in the vicinity of the fixed point, in order to get an approximate solution for the flow of $\tilde{g_6}$ near the fixed point. Let this step be clarified to more precision:

- The construction expressly fixes the values of $\tilde{m}^2$ and $\tilde{g_4}$ to constants, and the parameter $\tilde{g_6}$ is accordingly parametrized as $\tilde{g_6} \equiv \tilde{g_6}\left(\tilde{m}^2, \tilde{g_4}, t\right)$.
- Now, given such fixed parameters $\tilde{m_*}^2, \tilde{g}_{4,*}$ that affix some $\tilde{g}_{6,*}$, it is required that one finds whether such a set of parameters exists, for which the RG flow $\partial_t \tilde{g}_{6,*}$ vanishes. Ths is more precisely betoned by re-writing Eq. (9.11) as Eq. (9.19), where some constants A and B are defined for ease.

$$\partial_t \tilde{g}_{6,*}\left(\tilde{m_*}^2, \tilde{g}_{4,*}, t\right) = \frac{\tilde{g}_{6,*}\left(\tilde{m_*}^2, \tilde{g}_{4,*}, t\right)\tilde{g}_{4,*}}{5\pi^2} \frac{1}{\left(1+\tilde{m_*}^2\right)^3} - \frac{5\tilde{g}_{4,*}^3}{\pi^2} \frac{1}{\left(1+\tilde{m_*}^2\right)^4} = A\tilde{g}_{6,*} + B \tag{9.19}$$

This differential equation can be solved explicitly, such that one obtains a dependency $\tilde{g}_{6,*} \sim k^{-1}$, such that $\tilde{g}_{6,*} \to 0$ as $k \to 0$, that is, in the IR limit. This analysis shows that the only case where a fixed point can be found requires $\tilde{g_6} = 0$. $\tilde{g_6}$ therefore goes from being a marginal to becoming a useless degree of

freedom, and the truncation step prpoosed initially is justified, so that the solution in the 2-level truncation, $\tilde{m}^2 = -\frac{1}{3}$ and $\tilde{g_4} \approx \pi \left(\frac{2}{3}\right)^5$, was correct from start.

This analysis shows furthermore a number of different things:

- first, one finds two fixed points for this theory; one the trivial gaussian fixed point; the other the so-called fisher-willson fixed point. In addition to that, one demonstrates that $\tilde{g_6}$ is an an irrelevant coupling in the vicinity of those fixed points.[ad]

- The emergence of a (dimensionless) negative squared "mass" might appear surprising here. It is however not. Since those are simply the dimensionless parameters (or, after scaling, the dimensionful effective parameters going into the bare action), they don't actually correspond to actual, physically-measurable quantities. The dimensionless negative "mass" simply qualifies the nature of this spontaneous symmetry beaking. One considers namely the minima of the potential and its respective condition on the wavefunction $\tilde{\rho_*}$, Eqs. (9.20, 9.21). $\tilde{\rho_*}$ must be a strictly-positive quantity, from the definition of the wavefunction. The coefficients $\tilde{g_4}$, $\tilde{g_6}$ are also built to be stricly positive by definition.

$$\tilde{U}\left(\tilde{\rho}\right) = \tilde{m}^2\tilde{\rho} + \frac{\tilde{g_4}}{6}\tilde{\rho}^2 + \frac{\tilde{g_6}}{30}\tilde{\rho}^3 \tag{9.20}$$

$$\rho_{*,\pm} = \frac{5}{2\tilde{g_6}}\left(-\frac{\tilde{g_4}}{3} \pm \sqrt{\left(\frac{\tilde{g_4}}{3}\right)^2 - \frac{4}{5}\tilde{g_6}\tilde{m}^2}\right) > 0 \tag{9.21}$$

- For $\tilde{m}^2 > 0$ one is forced to choose the condition Eq. (9.22) in order to enforce the condition that $\tilde{\rho_*}$ must be strictly positive. But this leads to a single valid solution at $\tilde{m}^2 = 0$, and the potential is then an upward-bending parabola with its minima at zero. This is the trivial gaussian point and adds no further insight to the analysis.

$$\Delta = \left(\left(\frac{\tilde{g_4}}{3}\right)^2 - \frac{4}{5}\tilde{g_6}m^2\right) = 0 \tag{9.22}$$

- On the other hand, one picks $\tilde{m}^2 < 0$, $\Delta \neq 0$ and thus correctly enforces $\rho_{*,+} > 0$ (one still fails to corrently enforce $\rho_{*,-} > 0$ nevertheless). By remembering that $\rho = \frac{1}{2}\phi^2$, one obtains to two solutions at $\phi_{*,\pm} = \pm\sqrt{\rho_{*,+}}$. This corresponds to spontaneous symmetry breaking (SBB),and it leads to the famous "mexican hat" potential. In other words, $\tilde{m}^2 < 0$ leads to a maxima in the potential at $\phi = 0$. This is an unstable configuration that spontaneously collapses, either to $\phi_{*,+}$ or to $\phi_{*,-}$; most crucially, without any preference whether to pick the positive or the negative vacuum. It is then said that the field acquires a non-zero vacuum expectation value as a result of this broken symmetry.

- To briefly expand this discussion, here, one considers the SBB of a discrete, $\mathbb{Z}^2$ symmetry: the original system was symmetric under $\phi \to -\phi$, however, the introduction of the vacuum expectation value breaks this symmetry, because $\phi_{*,+} = -\phi_{*,-}$. The analysis carried-out above can be easily carried-over to $O\left(N\right)$ symmetries. In this case, a continous symmetry is broken and the system degenerates to $O\left(N-1\right)$ symmetry. Goldstone theorem then states that for every continuous symmetry that is spontaneously broken, each broken symmetry generator introduces the creation of massless excitations, which are called goldstone bosons.[24, 25, 29, 30]

[ad] At the start, one could only claim that $\tilde{g_6}$ was a marginal coupling.

9.3.5 Connection to Multicritical Points

In the discussion above, one finds $\tilde{g}_6 = 0$ for a ϕ^4 theory in $d = 3$, such that the theory disposes of only two fixed points, even at 3–level truncation (namely the GFF and the WF points). By a similar consideration, if one were to progressively introduce additional couplings $\tilde{g}_{2n}$, $n > 3$, those couplings would all satisfy $\tilde{g}_{2n} = 0$ at the fixed point. In other words, up to the choice of Ansatz and regulator, in section 9.3.4 above one derives the exact fixed points of the theory. The introduction of new couplings does not (in this case) change the number nor the nature of the fixed points (although it may change the critical coefficients, as it will be seen in section §10).

This is not generally the case. In general, a theory with n relevant coefficients is expected to have at most n critical points (although not necessarily all of them). This is justfied by the fact that all marginal and irrelevant coefficients flow to zero in the IR, such that their fixed point value will certainly be $\tilde{g}_{2n} = 0$. On the other hand, long-range theories, such as the ones discussed in section §27, are expected to have $n \geq 3$ relevant couplings — depending on the tuning of the long-range interaction — such that one expects theories with at least three or more critical points. Such class of theories is said to dispose over multicrtical fixed points; that different tunning of the fixed point on the coupling-parameter soace will lead to different criticality classes characterized by different critical coefficients.[31, 32, 33, 97]

10 Analytical and Computational Methods in FRG

10.1 Numerical Integration of the Potential

10.1.1 Discussion

As an alternative to the perturbative, truncation methods discussed previously in sections 6, 8, and 9, one may choose instead to solve the differential equation corresponding to the potential explicitly, Eq. (10.1). This method is of great advantage, because in practice it leads to a non-truncated solution to the flow equations. In this sense, the method is "exact", up to the Ansatz choice for the LPA (*see* the next paragraph for a more precise characterizatoin of this sentence). On the negative side, due to the highly non-linear nature of this equation, numerical and computational solutions must be implemented. An additional challenge is solving Eq. (10.1) consists on the fact that two initial conditions must be affixed, and which are, however, in principle not known. This challenge is however tackled by implementation, as shown in the subsequent paragraphs to come.

$$\partial_t \tilde{U}(\tilde{\rho}) = -d \cdot \tilde{U}(\tilde{\rho}) + (d-2)\,\tilde{\rho}\tilde{U}_t'(\tilde{\rho}) + A_d \cdot \frac{1}{1 + \tilde{U}'(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}''(\tilde{\rho})} \tag{10.1}$$

The great advantage of numerically integrating the differential equation for the potential is that this method is "exact", up to the Ansatz choice for the LPA. The precise meaning of "exact" in this sentence is clarified as follows:[17, 18, 36—38, 42, 44]

- The Ansatz choice for the LPA itself is not exact; namely it fails completely in capturing sub-leading dependencies on momenta. The author refers back to discussions from section §8:

  > "*The effective action is local. Non-local operators are ignored; and the relevant physics of the theory is primarily dominted and described by the effective potential U_k, whereas the momentum dependance on vertices is assumed to be sub-leading and is neglected; as well as*

couplings between momenta and the field ... from start the LPA already fails to capture the running of couplings and masses with the momentum. ... [the method furthermore may] inherit unphysical dependancies on the regulator."

- On the other hand, although previus methods discussed in sections 6, 8, and 9 relied on truncations (that is to say, they were expressely perturbative methods); the choice of instead realizing and explicit, numerical integrationof the differential equation, Eq. (10.1), leads to an expressly non-perturbative solution. The non-perturbativeness of the solution is what the author clarifies to mean "exact" in this particular context.

- Furthermore, it must be emphasized that additional inexactness emerges from numerical instabilities; as well as the particular choice of regulator that allows for a closed differential equation for the potential.

10.1.2 Algorithmic Implementation

In practice, one implements an integration algorithm requiring the following steps:[59—65], [ae]

- One sets $\partial_t \tilde{U}(\tilde{\rho}) = 0$ such that the Eq. (10.1) reduces to an ordinary differential equation in $\tilde{\rho}$ describing the $\tilde{U}$ as a function of $\tilde{\rho}$ exactly at the fixed point of the RG flow. Furthermore, one initializes the algorithm at $\tilde{\rho} = 0$, at each step integrating outwards towards $\tilde{\rho} \to \infty$.

- The initial condition for the convexity of the potential is specified by a free-float parameter σ, namely by requiring $\tilde{U}'_t(\tilde{\rho} = 0) = \sigma$. The initial condition for the potential is self-consistently fixed so as to satisfy Eq. (10.1) at the initial condition $\tilde{\rho} = 0$, namely, one affixes Eq. (10.2).

$$\tilde{U}(\tilde{\rho} = 0) = \frac{A_d}{d} \frac{1}{1+\sigma} \tag{10.2}$$

- In face of foregoing constraints, one obtains a family of solutions $\tilde{U}_\sigma(\tilde{\rho})$ indexed by the free parameter σ. Additionally, one must require a "reality" condition, namely that the (dimensionless) wavefunction $\tilde{\rho}$ must be finite for all finite $\tilde{U}$ and vice-versa. This additional condition strongly constrais the solution-space of the index-parameter σ, because in practice, due to the emergence of $\tilde{\rho}$ on the denominator of Eq. (10.1), the equation diverges for the vast majority of parameter-choices for σ.

- Consequently, one lets the numerical integration of Eq. (10.1) run as a function of $\tilde{\rho}$ from 0 towars ∞, with execution terminated at some $\tilde{\rho}_\infty^{(\sigma)}$ for which the solution is divergent; and one plots $\tilde{\rho}_\infty^{(\sigma)}$ as a function of σ. This plot is hereinafter referred to as a "spike plot". The values of σ for which $\tilde{\rho}_\infty^{(\sigma)}$ spikes correspond to values for which the computer simulation executed for a larger range of $\tilde{\rho}$, without that the dimensionless wavefunction diverged. These correspond therefore to potential candidates for the fixed points of the potential.

- In practice, the algorithm may lead to false fixed points, namely because the differential equation is never integrated from 0 to ∞, rather, from 0 to some threshold ρ_{MAX}, such that the algorithm may capture slowly-diverging initial conditions σ as if they were real fixed points. Once a potential candidate σ is identified, additional investigation must therefore be carried out in order to ascertain whether this solution corresponds to a real fixed point.

[ae] *See* Appendix **??** for an implementation of this algorithm.

- The anomalous dimension is introduced self-consistently. Namely, given $\tilde{\rho}$ and $\tilde{U}$ and its derivatives on the previous step, the anomalous dimension is completely specified from Eq. (10.3); up to a choice of initial condition which in principle can be set to $\eta = 0$ without loss of generality.

$$\eta = \frac{2A_d \left(\tilde{U}'''(\tilde{\rho})\right)^2}{\left(1 + \tilde{U}'(\tilde{\rho}) + 2\tilde{\rho}U''(\tilde{\rho})\right)^4} \times \left(1 - \frac{\eta}{d+2}\right) \tag{10.3}$$

10.2 Critical Coefficients by Method of Stability Matrix

One would like to calculate the critical coefficients corresponding to a particular phaes transition. For illustrative purposes, one considers a truncation where $\tilde{g_6} = 0$, such that Eq. (9.11) is always zero. One solves Eqs. (9.9, 9.10) explicitly and finds the following fixed point conditions (in addition to the gaussian fixed point), Eq. (10.4).[15, 16, 20, 32, 36, 37]

$$\tilde{m}^2 = \frac{d-4}{16-d} \; ; \qquad \tilde{\lambda} = \frac{-1}{6A_d}\left(\frac{12}{16-d}\right)^3 (d-4) \tag{10.4}$$

The critical coefficients can be obtained from the eigendirections fo the RG flow. In order to identify the eigendirections of the RG flow, one considers linear perturbations in the vicinity of the fixed point $\left(\tilde{m}_*^2, \tilde{\lambda}_*\right)$ specified by Eq. (10.5) below.

$$\tilde{m_*}^2 \to \tilde{m_*}^2 + \delta\tilde{m}^2 \; ; \qquad \tilde{\lambda}_* \to \tilde{\lambda}_* + \delta\tilde{\lambda} \tag{10.5}$$

Alternatively, one defines the beta function for simplicity, and the stability matrix D is defined as the jacobian of the beta function, Eq. (10.6).

$$\beta_{\tilde{m}^2} = \partial_t \tilde{m}^2 \; ; \qquad \beta_{\tilde{\lambda}} = \partial_t \tilde{\lambda} \; ; \qquad D = \begin{pmatrix} \frac{\beta_{\tilde{m}^2}}{\partial \tilde{m}^2} & \frac{\beta_{\tilde{\lambda}}}{\partial \tilde{m}^2} \\ \frac{\beta_{\tilde{m}^2}}{\partial \tilde{\lambda}} & \frac{\beta_{\tilde{\lambda}}}{\partial \tilde{\lambda}} \end{pmatrix} \tag{10.6}$$

The linear perturbations specified in Eq. (10.5) above can instead be treated as perturbatoins to the coefficient-space by the stability matrix, Eq. (10.7), such that by explicitly finding the eigenvalues and eigenvectors of the stability matrix D, one obtains the required RG eigendirections and correspondingly the critical coefficients.

$$\vec{g} \to \vec{g} + \vec{\varepsilon} D \tag{10.7}$$

Given Eqs. (9.9, 9.10) and the fixed point condition, Eq. (10.4), one finds Eq. (10.8) corresponding to the stability matrix evaluated at the critical point.

$$D_* = \begin{pmatrix} -\frac{(d+2)}{3} & -A_d\left(\frac{16-d}{12}\right)^2 \\ -\left(\frac{12(d-4)}{16-d}\right)^2 \frac{1}{2A_d} & -(d-4) \end{pmatrix} \tag{10.8}$$

One obtains the eigenvalues of the RG flow by solving the eigenproblem corresponding to D_*, that is to say, by solving the quadratic, Eq. (10.9).

$$6\lambda^2 + 4\,(2d-5)\,\lambda + (d-4)\,(16-d) = 0 \tag{10.9}$$

In this praticular case, one chooses the truncation $\tilde{g_6} = 0$ and solves the eigenproblem on a 2×2 matrix in order to obtain the critical coefficients. If the truncation were instead to have occurred at a later truncation point, say, $\tilde{g_{2n}} = 0$; in that case one would obtain a $n \times n$ matrix and would in principle need to implement computational techniques in order to diagonalize the matrix (unless $n < 5$ in which case a closed-form analytical solution for the polynomial equation in principle exists). Improved solutions are thus obtained by introducing additional additional couplings.

One emphasizes that even in the case where the additional introduced couplings are irrelevant in the FRG sense, these insertions nevertheless lead to improved estimations to the leading eigenvalue of the stability matrix. Furthermore, one notices that an $n \times n$ matrix will lead to a total of n eigenvalues. The leading eigenvalues are those corresponding to the relevant directions of the RG flow, and they are used for the computation of critical coefficients, as shown in subsection below. The remaining eigenvalues will correspond to the maginal and irrelevant directions; and they correspond to sub-leading corrections to the scaling; which are also discussed in more details in subsection below.[15, 20, 36, 37]

10.3 Connection to Critical Coefficients

As already discussed, near fixed points, universal macroscopic quantities expect to be embossed by power-law behaviours due to the collapse of the RG flow onto a finite number of degrees of freedom. Those power law behavours are governed by critical coefficients, which in turn arise from the eigenvalues corresponding ot the surviving eigendirections of the RG flow near criticality.[20, 21, 25—27]

Solving the quadratic, Eq. (10.9), leads to two roots. The dominant root is often labelled y_t, whereas the other root, commonly labelled as ω, is conventionally denoted as the "leading irrelevant exponent", an it controls sub-leading corrections to scaling. In fact, one can in principle obtain a hirearchy of sub-leading irrelevant exponents $\omega_1 < \omega_2 < \ldots < \omega_n$ corresponding to each subsequent truncation order of the parameter-space of the RG flow. The sub-leading corrections to scale then enter, for example, as Eq. (10.10). One additionally observes that $\omega \to 0$ corresponds to a marginal direction that decays more slowly towards the IR fixed point, whereas large ω corresponds to highly-irrelevant eigendirections that decay and vanish quickly as it approaches the fixed point.[18, 32, 42, 44]

$$\xi \sim t^{-\nu}\left(1 + a_1 t^{-\omega_1 \nu} + a_2 t^{-\omega_2 \nu} + \ldots\right) \tag{10.10}$$

The eigenvalue y_h on the other hand correpsonds to the $\mathbb{Z}^2$–asymmetric operator $\sim h\phi$, that it is completely fixed by dimensional analysis, Eq. (10.11).

$$y_h = \frac{d+2-\eta}{2} \tag{10.11}$$

In order to derive the connection between critical exponents and RG eigendirections, one considers that near fixed points the free energy density is a homogenous function of t and h, satsifying Eq. (10.12), where y_t and y_h are the eigenvalues corresponding to the two relevant eigendirections of the RG flow. In the ising model, one conventionally interprets t as the reduced temperature parameter and h as the external magnetic field; such that y_h corresponds to the eigendirection respectively to the ϕ^3 operator. b on the other hand is the RG scaling factor.[22, 25—27, 42, 44]

d	y_t	$y_h = \frac{d+2-\eta}{2}$	ω	$\nu = \frac{1}{y_t}$	$\beta = \frac{d-y_h}{y_t}$	$\gamma = \frac{2y_h-d}{y_t}$	$\delta = \frac{y_h}{d-y_h}$	$\alpha = 2 - \frac{d}{y_t}$
2	1.83	2	2.52	0.55	0	1.09	∞	0.91
3	1.84	2.5	1.18	0.54	0.27	1.09	5	0.37
4	2	3	0	$\frac{1}{2}$	$\frac{1}{2}$	1	3	0

Table 1: The eigenvalues y_t and y_h of the stability matrix D_* in $d = 2, 3, 4$ dimensions, along with the corresponding critical coefficients.

$$f(t,h) = b^{-d} f\left(tb^{y_t}, hb^{y_h}\right) \tag{10.12}$$

One sets $b = t^{-\frac{1}{y_t}}$ and the following relations are satisfied, such that the critical exponents ν, α, β, and γ correspond respectively to the scaling laws for the correlation length, specific heat, magnetization, and succeptibility.[31, 32]

$$f(t,h) = t^{-\frac{d}{y_t}} f\left(1, hb^{\frac{y_h}{y_t}}\right) \;;\quad \xi \sim b \sim t^{-\frac{1}{y_t}} \;;\quad C \sim \partial_t^2 g \sim t^{2-\frac{d}{y_t}} \;;\quad M \sim \frac{\partial f}{\partial h} \sim t^{\frac{d-y_h}{y_t}} \;;\quad \chi \sim \frac{\partial M}{\partial h} \sim t^{\frac{d-2y_h}{y_t}} \tag{10.13}$$

The critical exponent δ correpsonding ot the critical isotherm on the other hand correpsonds to M at the critical temperature, Eq. (10.14).

$$M(h) = b^{-d+y_h} M\left(0, hb^{y_h}\right) = h^{\frac{-d+y_h}{y_h}} M(0,1) \tag{10.14}$$

This particular analysis considered the case where only two eigendirections in the RG flow and their respective eigenvalues y_t and y_h are relevant. In systems with additional relevant eigendirections, one obtains additional RG-eigenvalues and, correspondingly, additional critical coefficients, which in turn regulate the behaviour of other thermodynamic variables in the vicinity of multi-critical fixed points.[18, 31, 32, 44, 97]

10.4 Semi-Analytical Criticality Analysis on a scalar ϕ^4 Theory with Method of Stability Matrix

10.4.1 Results and discussion

By explicitly solving Eq. (10.9) and using the relations for the critical exponents derived in the previous sub-section, the corresponding eigenvalues and critical exponents for the scalar ϕ^4 theory are given in table 1 below. These results are to be compared with those presented in table 2, corresponding to the best-known values existing in current literature, and which are based on experimental techniques and/or computational simulations. One notices, furthermore, that the results derived analytically in this paper, that is, those in table 1, correspond to the LPA estimation, that is, with anomalous dimension $\eta = 0$. This paper will repeat the analysis with $\eta \neq 0$ in the subsequent section.

One draws the following observations in regards to the results obtained above:

1. The method fails completely in $d = 2$. This follows from the fact that all truncation levels, that is, all couplings are relevant in $d \leq 2$, such that in this case the truncation of the coupling hirearchy at $n = 2$, that is, by considering only $\tilde{m}^2$ and $\tilde{g}_4$ as relevant coefficients, is invalid. The failure of the truncation

d	y_t	$y_h = \frac{d+2-\eta}{2}$	ω	$\nu = \frac{1}{y_t}$	$\beta = \frac{d-y_h}{y_t}$	$\gamma = \frac{2y_h-d}{y_t}$	$\delta = \frac{y_h}{d-y_h}$	$\alpha = 2 - \frac{d}{y_t}$
2	1	$\frac{15}{8}$	2	1	$\frac{1}{8}$	$\frac{7}{4}$	15	0
3	1.59	2.48	0.83	0.63	0.33	1.24	4.79	0.11
4	2	2	0	$\frac{1}{2}$	$\frac{1}{2}$	1	3	0

Table 2: The eigenvalues and eigenvectors correpsonding to table 1, as predicted by experiments and computer simulation.

for the lower critical dimension has already been discussed in mode details in section 9.2. As it will be demonstrated in more detail in section §29, the introduction of an anomalous dimension has, in the essence, the effect of reducing the lower critical dimension from $d = 2$ to $d = 2 - \eta$, such that, at least in principle, the critical coefficients can be predicted with greater accuracy, even for $d = 2$.

2. In $d = 4$ on the other hand, the results obtained by the FRG approach at this truncation level, $n = 2$, are *exact*. This is justified by the fact that the only fixed point at $d = 4$ is the gaussian fixed point. In fact, one finds that for all d, the critical coefficients at the gaussian fixed point are exact and correspond to those in the $d = 4$ theory (up to rescaling by an effective dimension). One then classifies the gaussian fixed point universality class; and (as shown in section §27) one finds that even long-ranged ising models may fall onto the short-range fixed-point universality class (up to rescaling my an effective dimension), depending on the choice of parameters that define the model.

3. In particular, the exactness of results at the gaussian fixed point stamms from the fact that all $\tilde{g}_{2n}$ are equal to zero at the gaussian fixed point, and furthermore (specifically for $d = 4$) all of their flows are irrelevant or at least marginal, such that the stability matrix, even when truncated at the 2×2 level, comes automatically diagonalized and with the entry $(D_*)_{11}$ corresponding to the leading relevant eigenvalue. Furthermore, because the anomalous dimension is proportional to $\eta \sim (\tilde{g}_4)^2$, it is exactly zero at the gaussian fixed point and therefore does not contribute to the critical coefficients.

4. On the other hand, for $d = 3$ the results come in a variety of accuracies and inaccuracies. Most critical coefficients are underestimated by factors between 10% and 20%; whereas δ is overestimated by roughly 5% and α by over 300%. The main source leading to those inaccuracies arises due to the fact that the LPA Ansatz assumes zero anomalous dimension, $\eta = 0$. In subsequent sections, the author shows that the introduction of an anomalous dimension leads to significantly improved results. Another source of error arises from the fact that at $d = 3$ the coefficient $\tilde{g}_6$ moves from irrelevant to marginal, such that in principle the 2–level truncation of the stability matrix becomes inaccurate, and at least the 3rd level to the truncation should be introduced instead.

10.4.2 Stability Matrix Results with Non-Zero Anomalous Dimension

The results obtained in table 1 can be significantly improved by the introduction of an anomalous dimension. Namely, as one argues in section §29 that the introcution of an anomalous dimension has the effect of lowering the lower critical dimension. One expects, from this fact, that for $d = 2$ some of the couplings will become marginal and/or irrelevant; so that (at least in principle) truncation methods should become accurate (although not necessarily a two-level truncation); whereas for $d = 3$ the results should receive improvemets.

10.5 Computational Implementation of Stability Matrix

10.5.1 Discussion

From section 10.1, one in practice finds the fixed point directly from the differential equation correpsonding to the local potential. Correspondingly, the method of Stability Matrix must be implemented computationally, so that the eigendirections of the RG flow are found *not* from the beta function, but rather by applying linearized perturbations to the potential directly in the vicinity of the fixed point. One considers an infinitesimal perturbation around the RG fixed point of the potential, Eq. (10.15), where by definition the perturbation is linear on the RG flow.[59—32]

$$\tilde{U}\left(\tilde{\rho}\right) = \tilde{U_*}\left(\tilde{\rho}\right) + \varepsilon\tilde{v}\left(\tilde{\rho}\right) \ ; \qquad \partial_t\tilde{v}\left(\tilde{\rho}\right) = y \cdot \tilde{v}\left(\tilde{\rho}\right) \tag{10.15}$$

By substituting the perturbation onto the RG flow for the potential, Eq. (10.1), and expanding the denominator in Eq. (10.1) by taylor expansion so that a linear order in ε emerges, one obtains Eq. (10.16).

$$\left(y+d\right)\tilde{v}\left(\tilde{\rho}\right) - \left(d-2\right)\tilde{\rho}\tilde{v}'\left(\tilde{\rho}\right) = -A_d \cdot \frac{\tilde{v}'\left(\tilde{\rho}\right) + 2\tilde{\rho}\tilde{v}''\left(\tilde{\rho}\right)}{\left(1 + \tilde{U_*}'\left(\tilde{\rho}\right) + 2\tilde{\rho}\tilde{U_*}''\left(\tilde{\rho}\right)\right)^2} \tag{10.16}$$

As a qualitative analysis, one observes that at the asymptotic limit $\tilde{\rho} \to \infty$, the right-handed side of Eq. (10.16) vanishes, such that one obtains the asymptotic solution, Eq. (10.17). This corresponds to the expectation of power-law behaviour near criticality.

$$\tilde{v}\left(\tilde{\rho}\right) \sim \tilde{\rho}^{\frac{(y+d)}{(d-2)}} \tag{10.17}$$

Furthermore, one must impose initial conditions for $\tilde{v}$. By setting $\tilde{v}\left(0\right) = 1$, which is an allowed normalization condition, $\tilde{v}'\left(0\right)$ is automatically fixed by the Eq. (10.16). Namely, one obtains the initial condition specified by Eq. (10.18).

$$\tilde{v}'\left(0\right) = -\frac{1}{A_d}\left(y+d\right)\left(1 + \tilde{U_*}'\left(0\right)\right)^2 \tag{10.18}$$

10.5.2 Algorithmic Implementation

The integration algorithm must therefore constitute of the following steps:[59—32], [af]

- Starting from the implementation in section 10.1, one finds the family σ which solves the potential $\tilde{U}\left(\tilde{\rho}\right)$ at the fixed point in question. One additionally initializes the system at $\tilde{\rho} = \varepsilon \to 0$, such that the eigenvalue can be writte nas Eq. (10.19) below, and the initial conditions $\tilde{v}\left(0\right) = 1$ and Eq. (10.18) can be perturbed by an infinitesimal shift to the wavefunction's initial condition $0 \to \varepsilon$, namely by means of Eq. (10.20).

[af] *See* Appendix **??** for an implementation of this algorithm.

$$\tilde{v}\left(\varepsilon\right) = \tilde{v}_{*} + \varepsilon \tilde{v}_{*}^{(1)} + \frac{\varepsilon^{2}}{2} \tilde{v}_{*}^{(2)} + \mathcal{O}\left(\varepsilon^{2}\right) \tag{10.19}$$

$$\tilde{v}''\left(\tilde{\rho}\right) = \frac{-1}{2\tilde{\rho}} \left\{ \frac{\left(1 + \tilde{U}_{*}{}'\left(\tilde{\rho}\right) + 2\tilde{\rho}\tilde{U}_{*}{}''\left(\tilde{\rho}\right)\right)^{2}}{A_{d}} \left(\left(y + d\right)\tilde{v}\left(\tilde{\rho}\right) - \left(d - 2\right)\tilde{\rho}\tilde{v}'\left(\tilde{\rho}\right)\right) - \tilde{v}'\left(\tilde{\rho}\right) \right\} \tag{10.20}$$

- One extends the shooting method proposed in section 10.1 to this problem. One proposes different eigenvalues y and numerically integrates $\tilde{v}\left(\tilde{\rho}\right)$ up to the threshold $\tilde{\rho}_{\infty}^{(\sigma)}$ obtained in section 10.1 for the particular family of solutions $\tilde{U}\left(\tilde{\rho}\right)$ indexed by σ. In other words, given σ obtained in section 10.1, one now integrates both $\tilde{U}\left(\tilde{\rho}\right)$ and $\tilde{v}\left(\tilde{\rho}\right)$ simultaneously, constrained by the initial condition specified by σ for $\tilde{U}\left(\tilde{\rho}\right)$, specified by $\tilde{v}\left(0\right) = 1$ and by Eqs. (10.18), and additionally ranged over possible eigenvalues y.
- One obtains a family of asymptotic solutions $\tilde{v}_{\infty}^{(\sigma,y)} \equiv \tilde{v}^{(y)}\left(\tilde{\rho}_{\infty}^{(\sigma)}\right)$ indexed by the eigenvalue y, correpsonding to the asymptotic behaviour of the eigenvector $\tilde{v}$ corresponding to the eigenvalue y, for a given critical point characterized by σ. Using additionally Eq. (10.17), one finds that real, physical solutions require $\tilde{v}_{\infty}^{(\sigma,y)} = 0$. By plotting $\tilde{v}_{\infty}^{(\sigma,y)}$ as function of y, the intersections of $\tilde{v}_{\infty}^{(\sigma,y)}$ at zero therefore correspond to the eigenvalues of the RG flow. Negative eigenvalues will correpsond to relevant RG-eigendirections, and positive eigenvalues will correpsond to irrelevant RG-eigendirections.

11 Appendicies in Concern of Functional Renormalization Group Methods

11.1 Manipulation of Functional Objects

Functionals are loosely defined as functions of functions. One considers the functional spaces $\mathcal{X}$, $\mathcal{Y}$, and let $\phi \in \mathcal{X}$ be a function in this space. An operator-valued functional is defined by the map Eq. (11.1).

$$F : \mathcal{X} \longrightarrow \mathcal{Y} \; ; \qquad \phi \longmapsto F\left[\phi\right] \tag{11.1}$$

If F instead depends on a collection of linearly independent functions, $\{\phi_i\}$, then functional derivatives satisfy the property Eq. (11.2).

$$\frac{\delta F\left[\phi_i\right]}{\delta \phi_i} = \sum_{j} \frac{\delta F\left[\phi_i\right]}{\delta \phi_j} \frac{\delta \phi_j}{\delta \phi_i} \; ; \qquad \frac{\delta \phi_j}{\delta \phi_i} = \delta_{ij} \tag{11.2}$$

If instead of indexation by discrete indicies i one considers a continuous index $x \in \mathbb{R}$, $\phi \overset{\text{def.}}{=\!=} \phi\left(x\right)$, then the property in Eq. (11.2) generalizes to Eq. (11.3).

$$\frac{\delta F\left[\phi\right]\left(p\right)}{\delta \phi\left(q\right)} = \int \frac{\delta F\left[\phi\right]\left(p\right)}{\delta \phi\left(\xi\right)} \frac{\delta \phi\left(\xi\right)}{\delta \phi\left(q\right)} d\xi \; ; \qquad \frac{\delta}{\delta \phi\left(p\right)} = \int \frac{\delta \phi\left(\xi\right)}{\delta \phi\left(p\right)} \frac{\delta}{\delta \phi\left(\xi\right)} d\xi \; ; \qquad \frac{\delta \phi\left(p\right)}{\delta \phi\left(q\right)} = \delta\left(p - q\right) \tag{11.3}$$

The chain rule is given by Eq. (11.4).

$$\frac{\delta F\left[G\left[\phi\right]\right](p)}{\delta\phi\left(q\right)}=\int\frac{\delta F\left[G\left[\phi\right]\right](p)}{\delta G\left[\phi\right](\xi)}\frac{\delta G\left[\phi\right](\xi)}{\delta\phi\left(q\right)}d\xi \tag{11.4}$$

The chain rule can be arbitrarily extended, Eq. (11.5).

$$\frac{\delta F\left[G\left[H\left[\phi\right]\right]\right](p)}{\delta\phi\left(q\right)}=\int\frac{\delta H\left[\phi\right](\xi)}{\delta\phi\left(q\right)}\frac{\delta G\left[\phi\right](\zeta)}{\delta H\left[\phi\right](\xi)}\frac{\delta F\left[H\left[\phi\right]\right](p)}{\delta G\left[\phi\right](\zeta)}d\xi d\zeta \tag{11.5}$$

The following property also holds, Eq. (11.6).

$$\frac{\delta F\left[\phi\right](p)}{\delta F\left[\phi\right](q)}=\int\frac{\delta F\left[\phi\right](p)}{\delta\phi\left(\xi\right)}\frac{\delta\phi\left(\xi\right)}{\delta F\left[\phi\right](q)}d\xi=\frac{\delta\phi\left(p\right)}{\delta\phi\left(q\right)}=\delta\left(p-q\right) \tag{11.6}$$

The functional taylor expansion, Eq. (11.7)

$$\begin{aligned}F\left[\phi+\chi\right]&=F\left[\phi\right]+\frac{\delta F\left[\phi\right]}{\delta\phi\left(x\right)}\chi+\frac{1}{2}\chi\frac{\delta^2F\left[\phi\right]}{\delta\phi\left(x\right)\delta\phi\left(y\right)}\chi+\ldots\\&=F\left[\phi\right]+\int\frac{\delta F\left[\phi\right]}{\delta\phi\left(x\right)}\chi\left(x\right)dx+\frac{1}{2}\int\chi\left(x\right)\frac{\delta^2F\left[\phi\right]}{\delta\phi\left(x\right)\delta\phi\left(y\right)}\chi\left(y\right)dxdy+\ldots\end{aligned} \tag{11.7}$$

The Gaussian Integral generalizes readily to Eq. (11.8), where the trace consists of projecting the operator to a basis.

$$log\left(\mathcal{I}\right)=log\int D\phi\cdot exp\left(-\frac{1}{2}\phi\left(x\right)F\left(x,y\right)\phi\left(y\right)\right)=-\frac{1}{2}Trlog\left(F\left[\phi\right](x,y)\right)\ ;\quad TrF\left[\phi\right]=\int F\left[\phi\right](x,x)\,dx \tag{11.8}$$

Consider a convolution, Eq. (11.9).

$$\begin{aligned}\tilde{\phi}^2\left(p-q\right)&=\int e^{ipx-iqy}\phi^2\left(x\right)\delta\left(x-y\right)dxdy=\int e^{i(p-q)x}\phi^2\left(x\right)dx\\&=\int e^{i(p-q+p_1+p_2)x}\phi\left(p_1\right)\phi\left(p_2\right)dxdp_1dp_2=\int\delta\left(p-q+p_1+p_2\right)\phi\left(p_1\right)\phi\left(p_2\right)dp_1dp_2\end{aligned} \tag{11.9}$$

Functional derivatives acting on the convolution reads Eq. (11.10).

$$\frac{\delta\tilde{\phi}^2\left(p\right)}{\delta\phi\left(q\right)}=2\int\delta\left(p+q+p_1\right)\phi\left(p_1\right)dp_1 \tag{11.10}$$

11.2 Perturbative Calculations

One considers the action term, Eq. (11.11)

$$S\left[\phi\right]=\int d^dx\left\{\partial_\mu\phi\partial^\mu\phi+m^2\phi^2+\mathcal{O}\left(\phi^4\right)\right\} \tag{11.11}$$

First we compute the identity given by Eq. (11.12).

$$e^{-\Gamma[\phi]} = exp \int D\phi exp \left(-\mathcal{S}\left[\phi\right] + \int j\left(x\right)\left(\phi\left(x\right) - \varphi\left(x\right)\right) d^d x \right) = \int D\chi exp \left(-\mathcal{S}\left[\phi + \chi\right] + \int j\left(x\right)\chi\left(x\right) d^d x \right) \tag{11.12}$$

Consider the functional taylor expansion around the fluctuation field, Eq. (11.13).

$$\begin{aligned} \mathcal{S}\left[\phi + \chi\right] &= \mathcal{S}\left[\phi\right] + \frac{\partial \mathcal{S}\left[\phi\right]}{\partial \phi\left(x\right)}\chi + \frac{1}{2}\chi \frac{\partial^2 \mathcal{S}\left[\phi\right]}{\partial \phi\left(x\right)\partial \phi\left(y\right)}\chi + \ldots \\ &= \mathcal{S}\left[\phi\right] + \int \frac{\partial \mathcal{S}\left[\phi\right]}{\partial \phi\left(x\right)}\chi\left(x\right) d^d x + \frac{1}{2}\int \chi\left(x\right) \frac{\partial^2 \mathcal{S}\left[\phi\right]}{\partial \phi\left(x\right)\partial \phi\left(y\right)}\chi\left(y\right) d^d x d^d y + \ldots \end{aligned} \tag{11.13}$$

The first order gives Eq. (11.14).

$$j\left(x\right) = \left(\frac{\partial \Gamma\left[\phi\right]}{\partial \phi\left(x\right)} \right)_{\phi_*} = 0 \tag{11.14}$$

We obtain Eq. (11.15).

$$e^{-\Gamma[\phi]} = \int D\chi exp \left(-\mathcal{S}\left[\phi + \chi\right] + \int j\left(x\right)\chi\left(x\right) d^d x \right) = exp\left(-\mathcal{S}\left[\phi_*\right]\right) \int D\chi \tag{11.15}$$

That is, Eq. (11.16).

$$\Gamma\left[\phi_*\right] = S\left[\phi_*\right] \tag{11.16}$$

We two-point vertex function in the mean field for a ϕ^4 theory is calculated explicitly via Eq. (11.17).

$$\Gamma^{(2)}_{MF}\left(x, y\right) = \frac{\partial^2 \mathcal{S}}{\partial \phi\left(x\right)\partial \phi\left(y\right)} = \left(-\partial^2 + m^2 + \frac{\lambda}{2}\phi^2\left(x\right) \right) \delta\left(x - y\right) \tag{11.17}$$

The fourier transform can be calculated from Eq. (11.17) by making use of the convolution Eq. (11.9), Eq. (11.18).

$$\mathcal{S}^{(2)}\left(p - q\right) = \int dx dy \left(-\partial^2 + m + \frac{\lambda}{2}\phi^2\left(x\right) \right) \delta\left(x - y\right) e^{ipx - iqy} = \left(p - q\right)^2 + m^2 + \frac{\lambda}{2}\tilde{\phi}^2\left(p - q\right) \tag{11.18}$$

One notices that functional derivatives of $\Gamma^{(2)}$ can be easily calculated with Eq. (11.19).

$$\begin{aligned} \tilde{\phi}^2\left(p - q\right) &= \int e^{ipx - iqy}\phi^2\left(x\right)\delta\left(x - y\right) dx dy = \int e^{i(p-q)x}\phi^2\left(x\right) dx \\ &= \int e^{i(p-q+p_1+p_2)x}\phi\left(p_1\right)\phi\left(p_2\right) dx dp_1 dp_2 = \int \delta\left(p - q + p_1 + p_2\right)\phi\left(p_1\right)\phi\left(p_2\right) dp_1 dp_2 \end{aligned} \tag{11.19}$$

The correlation function is computed formally from Eq. (11.18), Eq. (11.20), where the terms denoted $(\ldots)$ in $\left(-\partial^2 + m^2 + \ldots\right)$ refer to the background field, which is eventually set to zero, $\phi_* = 0$.

$$\int \frac{\partial^2 \mathcal{S}[\phi]}{\partial\phi(x)\,\partial\phi(y)} G(y-z)\, d^d y = \int \left(-\partial^2 + m^2 + \ldots\right)_{\tilde{\phi}=0} \delta(x-y)\, G(y-z)\, d^d y = \left(-\partial^2 + m^2\right) G(x-z) = \delta(x-z)$$
$$\Longrightarrow \int \frac{d^d p}{(2\pi)^d} \left(p^2 + m^2\right) G(p)\, e^{ip(x-z)} = \delta(x-z) = \int \frac{d^d p}{(2\pi)^d} e^{ip(x-z)} \Longrightarrow G(p) = \frac{1}{(p^2+m^2)} \tag{11.20}$$

Or simply in momentum space, Eq. (11.21).

$$\mathcal{S}^{(2)}(p)\, G^{-1}(p) = \left(p^2 + m^2 + \frac{\lambda}{2}\tilde{\phi}^2(p)\right)_{\tilde{\phi}=0} G^{-1}(p) \Longrightarrow G(p) = \frac{1}{(p^2+m^2)} \tag{11.21}$$

11.3 Computation of n-Point Vertex Functions

At this point, let us calculate a few identities that will be used to evaluate results in the subsequent sections. For a ϕ^4 Theory, the n-Point Vertex Functions up to fourth order are easily calculated from Eq. (11.19), Eqs. (11.22, 11.23, 11.24).

$$\Gamma^{(2)}(p,q) = \frac{\lambda}{2} \int dr_r dr_2 \delta(-p-q+r_1+r_2)\, \phi(r_1)\, \phi(r_2) \tag{11.22}$$

$$\Gamma^{(3)}(p,q,p_1) = \frac{\delta\Gamma^{(2)}(p,q)}{\delta\phi(p_1)} = \lambda \int dr \delta(-p-q+r+p_1)\, \phi(r) \tag{11.23}$$

$$\Gamma^{(4)}(p,q,p_1,p_2) = \frac{\delta^2\Gamma^{(2)}(p,q)}{\delta\phi(p_1)\,\delta\phi(p_2)} = \lambda\delta(-p-q+p_1+p_2) \tag{11.24}$$

With the definition of the propagator as the inverse of the 2-point vertex function, one calculates functional derivatives against the propagator, Eq. (11.25).

$$\frac{\delta G(p,q)}{\delta\phi(p_1)} = \frac{\delta}{\delta\phi(p_1)}\left(\frac{1}{\Gamma^{(2)}(p,q)}\right) = \int d\xi d\zeta \frac{-1}{\Gamma^{(2)}(p,\xi)} \frac{\delta\Gamma^{(2)}(\xi,\zeta)}{\delta\phi(p_1)} \frac{1}{\Gamma_k^{(2)}(\zeta,q)} = -\int d\xi d\zeta G(p,\xi)\, \Gamma^{(3)}(\xi,\zeta,p_1)\, G(\zeta,q)$$
$$= -\lambda \int d\xi d\zeta dr G(p,\xi)\, G(\zeta,q)\, \delta(-\xi-\zeta+r+p_1)\, \phi(r) = -\lambda \int d\xi dr G(p,\xi)\, G(-\xi+r+p_1,q)\, \phi(r) \tag{11.25}$$

The second functional derivative, Eq. (11.26).

$$\begin{aligned}\frac{\delta^2 G(p,q)}{\delta\phi(p_1)\,\delta\phi(p_2)} = \frac{-\delta}{\delta\phi(p_2)} \int d\xi d\zeta G(p,\xi)\, \Gamma_k^{(3)}(\xi,\zeta,p_1)\, G(\zeta,q) = \\ \underbrace{-\int d\xi d\zeta G(p,\xi)\, \Gamma^{(4)}(\xi,\zeta,p_1,p_2)\, G(\zeta,q)}_{V_{2,1}} \\ \underbrace{-\int d\xi d\zeta \frac{\delta G(p,\xi)}{\delta\phi(p_2)} \Gamma^{(3)}(\xi,\zeta,p_1)\, G(\zeta,q)}_{V_{2,2}} \underbrace{-\int d\xi d\zeta G(p,\xi)\, \Gamma^{(3)}(\xi,\zeta,p_1) \frac{\delta G(\zeta,q)}{\delta\phi(p_2)}}_{V_{2,3}}\end{aligned} \tag{11.26}$$

The first term resolves in Eq. (11.26) above resolves into Eq. (11.27).

$$V_{2,1} = \lambda \int d\xi d\zeta G\left(p,\xi\right) G\left(\zeta,q\right) \delta\left(-\xi-\zeta+p_1+p_2\right) = \int d\xi G\left(p,\xi\right) G\left(-\xi+p_1+p_2,q\right) \tag{11.27}$$

The second and third terms resolve in Eq. (11.26) above resolve into Eqs. (11.28, 11.29). Notice that those expressions are equivalent under swapping the primed and unprimed terms and p_1 and p_2 simultaneously, so we can only consider one of them and add a pre-factor of two in the end.

$$V_{2,2} = \int d\xi d\zeta d\xi' d\zeta' G\left(p,\xi'\right) \Gamma^{(3)}\left(\xi',\zeta',p_2\right) G\left(\zeta',\xi\right) \Gamma^{(3)}\left(\xi,\zeta,p_1\right) G\left(\zeta,q\right) \tag{11.28}$$

$$V_{2,3} = \int d\xi d\zeta d\xi' d\zeta' G\left(p,\xi\right) \Gamma^{(3)}\left(\xi,\zeta,p_1\right) G\left(\zeta,\xi'\right) \Gamma^{(3)}\left(\xi',\zeta',p_2\right) G\left(\zeta',q\right) \tag{11.29}$$

11.4 Next Order Correction to the Propagator

We are now prepared to calculate the perturbative corrections to the propagator. Notice that the stationary solution satisfies Eq. (11.30), such that the first order contributions on the fluctuation field vanish, and we start counting from the second order to the taylor expansion.

$$j\left(x\right) = \left(\frac{\partial \Gamma\left[\phi\right]}{\partial \phi\left(x\right)}\right)_{\phi_*} = \left(\frac{\partial \mathcal{S}\left[\phi\right]}{\partial \phi\left(x\right)}\right)_{\phi_*} = 0 \tag{11.30}$$

The next-order correction to the correlation function is therefore given by integrating the quadratic, Eq. (11.31).

$$\Gamma_1\left[\phi\right] = -log \int D\chi exp\left(-\frac{1}{2}\chi \mathcal{S}_*^{(2)} \chi\right) \Longrightarrow \Gamma_1\left[\phi\right] = TrLog\left(S^{(2)}\left[\phi_*\right]\right) = TrLog\left(\frac{\partial^2 \mathcal{S}}{\partial \phi\left(x\right) \partial \phi\left(y\right)}\right)_{\phi_*} \tag{11.31}$$

First, let us calculate the two-point vertex, Eq. (11.32).

$$\begin{aligned}
V_2 &= \frac{\delta^2 log\left(\Gamma^{(2)}\left(p,q\right)\right)}{\delta\phi\left(p_1\right)\delta\phi\left(p_2\right)} = \frac{\delta}{\delta\phi\left(p_2\right)} \int d\xi \frac{1}{\Gamma_k^{(2)}\left(p,\xi\right)} \frac{\delta\Gamma^{(2)}\left(\xi,q\right)}{\delta\phi\left(p_1\right)} \\
&= \int d\xi \frac{1}{\Gamma^{(2)}\left(p,\xi\right)} \frac{\delta^2\Gamma^{(2)}\left(\xi,q\right)}{\delta\phi\left(p_1\right)\delta\phi\left(p_2\right)} + \int d\xi \frac{\delta}{\delta\phi\left(p_2\right)} \left(\frac{1}{\Gamma^{(2)}\left(p,\xi\right)}\right) \frac{\delta\Gamma^{(2)}\left(\xi,q\right)}{\delta\phi\left(p_1\right)} \\
&= \int d\xi G\left(p,\xi\right) \Gamma^{(4)}\left(\xi,q,p_1,p_2\right) - \int d\xi d\xi' d\zeta' G\left(p,\xi'\right) \Gamma^{(3)}\left(\xi',\zeta',p_2\right) G\left(\zeta',\xi\right) \Gamma^{(3)}\left(\xi,q,p_1\right) \\
&= \lambda \int d\xi G\left(p,\xi\right) \delta\left(-\xi-q+p_1+p_2\right) - \int d\xi d\xi' d\zeta' dr dr' G\left(p,\xi'\right) G\left(\zeta',\xi\right) \phi\left(r\right) \phi\left(r'\right) \delta_{-\xi'-\zeta'+r'+p_2} \delta_{-\xi-q+r+p_1} \\
&= \lambda G\left(p,-q+p_1+p_2\right) - \lambda^2 \underbrace{\int d\xi dr dr' G\left(p,\xi\right) G\left(-\xi+r'+p_2,-q+r+p_1\right) \phi\left(r\right) \phi\left(r'\right)}_{V_{2,2}}
\end{aligned} \tag{11.32}$$

In the second term, we use translational-invariance to resolve the integral over ξ, namely, $G\left(p,\xi\right) = \delta\left(p-\xi\right) G\left(p\right)$, such that the integral resolves to Eq. (11.33).

$$V_{2,2} = \int drdr'\delta\left(p-\xi\right)G\left(p\right)G\left(-p+r'+p_2,-q+r+p_1\right)\phi\left(r\right)\phi\left(r'\right) \tag{11.33}$$

To compute the actual two-point vertex, first note that the fields are dropped to zero, so the second term in Eq. (11.32) vanishes. The result is given by Eq. (11.34), where the trace symmetrizes the propagator.

$$Tr\left(\frac{\delta^2 log\left(\Gamma^{(2)}\left(p,q\right)\right)}{\delta\phi\left(p_1\right)\delta\phi\left(p_2\right)}\right) = \lambda\int dpdqG\left(p,-q+p_1+p_2\right) = \lambda\int dp\frac{1}{\left(p-p_1-p_2\right)^2+m^2} \tag{11.34}$$

Up to translational invariance on the integration measure, the integral resolves, in $d=4$ to Eq. (11.35). This integral is ultraviolet divergent. Formally, we can regulate those divergencies by introducing a cut-off Λ to the integration space.

$$\int^{p^2=\Lambda^2}\frac{d^4p}{\left(2\pi\right)^4}\frac{\lambda}{p^2+m^2} = \int_0^\Lambda\frac{dp}{8\pi^2}\frac{\lambda p^3}{p^2+m^2} = \lambda\int_0^\Lambda\frac{dp}{8\pi^2}\left\{p-\frac{pm^2}{p^2+m^2}\right\} = \frac{\lambda}{8\pi^2}\left\{\frac{\Lambda^2}{2}-m^2 log\left(\frac{\Lambda^2+m^2}{m^2}\right)\right\} \tag{11.35}$$

11.5 4-Point Vertex Correction

A similar correction arises to the coupling strength, by evaluating the correction to the four-point vertex function. For the four-point vertex we need to take two additional functional derivatives against Eq. (11.32). Since the fields are eventually evaluated at zero, in the right-most term in Eq. (11.32), the only non-zero terms come from taking functional derivatives against the fields, as any functional derivative against the propagator would leave fields left in the end, which would be taken to zero. There are two possible permutations. For the first term, the first functional derivative must act on the propagator (as the propagator is the only term) and leaves a field left. the second functional derivative must be taken against the leftover field, as otherwise the term would be zero once we take the field value to zero. Taking the additional two functional derivatives therefore leads to Eq. (11.36). There are three available permuations, $p_1,p_2;p_3,p_4$, $p_1,p_3;p_2,p_4$, $p_1,p_4;p_2,p_3$, corresponding respectively to the s, t and u channels. The terms are equivalent up to those permutations, and therefore we set the factor of 3 to the integral.

$$\begin{aligned}\frac{\delta^4 log\left(\Gamma^{(2)}\left(p,q\right)\right)}{\delta\phi\left(p_1\right)\delta\phi\left(p_2\right)\delta\phi\left(p_3\right)\delta\phi\left(p_4\right)} &= 3\lambda^2\int d\xi G\left(p,\xi\right)G\left(-\xi+p_2+p_4,-q+p_1+p_3\right)\\ &= 3\lambda^2 G\left(p\right)G\left(-p+p_2+p_4,-q+p_1+p_3\right)\end{aligned} \tag{11.36}$$

To get the four-point vertex, we integrate over the trace, Eq. (11.37).

$$\begin{aligned}V_4 = Tr\left(\frac{\delta^4 log\left(\Gamma^{(2)}\left(p,q\right)\right)}{\delta\phi\left(p_1\right)\delta\phi\left(p_2\right)\delta\phi\left(p_3\right)\delta\phi\left(p_4\right)}\right) &= 3\lambda^2\int dpdqG\left(p\right)G\left(-p+p_2+p_4,-q+p_1+p_3\right)\\ &= 3\lambda^2\int dp\frac{1}{p^2+m^2}\frac{1}{\left(p+q\right)^2+m^2}\end{aligned} \tag{11.37}$$

Up to translationalinvariance on the integration measure, the integral resolves in $d=4$ to Eq. (11.38),

where we use the identity $\frac{1}{AB} = \int_0^1 dx \frac{1}{(xA+(1-x)B)^2}$ in order to combine the denominators, and define $k = (p - xq)$ and $\Delta^2 = \left(m^2 + x\left(1-x\right)q^2\right)$,

$$
\begin{aligned}
V_4 &= \frac{3\lambda^2}{(2\pi)^4}\int_0^1 dx \int d^4k \frac{1}{\left[x\left(p^2+m^2\right)(1-x)\left((p+q)^2+m^2\right)\right]^2} \\
&= \frac{3\lambda^2}{(2\pi)^4}\int_0^1 dx \int d^4k \frac{1}{\left(k^2+\Delta^2\right)^2} = \frac{3\lambda^2}{16\pi^2}\int_0^1 \left\{ log\left(\frac{\Lambda^2+\Delta^2}{\Delta^2}\right) - \frac{\Lambda^2+\Delta^2}{\Delta^2}\right\} dx
\end{aligned}
\tag{11.38}
$$

In order to handle the divergencies, first we consider $\Lambda >> \Delta$ so we can simplify the ratios. Second we use the same subtraction scheme as before, by absorbing the divergency to the defition of the bare coupling strength. For low-energy $q \approx 0$, we taylor-expand in powers of $\left(\frac{p}{m}\right)^2$, Eq. (11.39).

$$
\begin{aligned}
V_4 &= \frac{-3\lambda^2}{16\pi^2}\int_0^1 log\left(\frac{m^2+x\left(1-x\right)p^2}{\mu^2}\right) dx = \frac{-3\lambda^2}{16\pi^2}\int_0^1 log\left(\left(\frac{m^2}{\mu^2}\right)\left(1+x\left(1-x\right)\frac{p^2}{m^2}\right)\right) dx \\
&\approx \frac{-3\lambda^2}{16\pi^2}\int_0^1 \left\{ log\left(\frac{m^2}{\mu^2}\right) + \left(x\left(1-x\right)\left(\frac{p}{m}\right)^2 - x^2\left(1-x\right)^2\left(\frac{p}{m}\right)^4\right)\right\} dx \\
&= \frac{-3\lambda^2}{16\pi^2}\left\{ log\left(\frac{m^2}{\mu^2}\right) + \frac{1}{6}\left(\frac{p}{m}\right)^2 - \frac{5}{48}\left(\frac{p}{m}\right)^4 + \mathcal{O}\left(\left(\frac{p}{m}\right)^6\right)\right\} \approx \frac{-3\lambda^2}{16\pi^2} log\left(\frac{m^2}{\mu^2}\right)
\end{aligned}
\tag{11.39}
$$

The beta function shows how the running of the coupling with respect to the scale, and it is given by Eq. (11.40), which is the required result.

$$
\beta\left(\lambda\right) = \mu\frac{\partial\lambda}{\partial\mu} = \frac{3\lambda^2}{8\pi^2} \tag{11.40}
$$

Subsequent orders of the perturbation are similarly calculated, with help of (e.g.) Eq. (11.41).

$$
-\frac{\partial}{\partial\phi}e^{-\Gamma[\phi]} = \frac{\partial\Gamma\left[\phi\right]}{\partial\phi}e^{-\Gamma[\phi]} = -\int D\chi \frac{\partial}{\partial\phi}e^{-\mathcal{S}_*^{(2)}\frac{\chi\chi}{2}} = \frac{1}{2}\int D\chi\left(\mathcal{S}_*^{(3)}\chi\chi\right)e^{-\mathcal{S}_*^{(2)}\frac{\chi^2}{2}} \Longmapsto \frac{\partial\Gamma\left[\phi\right]}{\partial\phi} = \frac{1}{2}\mathcal{S}_*^{(3)}\left\langle\chi\chi\right\rangle \tag{11.41}
$$

11.6 Wetterich Vertex Expansion Algebra

The next logical step would be to proceed in a similar way as Eqs. (11.32, 11.36), in order to derive the formula for the exact propagator (or more specifically to derive the differential equation for the RG flow of the exact propagator, in the case of the Wetterich equation). Starting from Eq. (11.26), we proceed in a similar fashion as we did for Eq. (11.32), to obtain Eq. (11.42).

$$
\begin{aligned}
\partial_k \Gamma_k^{(2)} &= \frac{\partial^2 \left(\partial_k \Gamma_k [\phi]\right)}{\partial\phi\left(p_1\right)\partial\phi\left(p_2\right)} = \frac{\partial^2}{\partial\phi\left(p_1\right)\partial\phi\left(p_2\right)} \int \left(\partial_k R_k\left(q\right)\right) G_k\left(q\right) d^d q \\
&= -\frac{\partial}{\partial\phi\left(p_1\right)} \int \left(\partial_k R_k\left(q\right)\right) G_k\left(q\right) \Gamma_k^{(3)}\left(q,p\right) G_k\left(p\right) dqdp \\
&= \int \left(\partial_k R_k\left(q\right)\right) G_k\left(q\right) \Gamma_k^{(3)}\left(q,p\right) G_k\left(p\right) \Gamma_k^{(3)}\left(q,r\right) G_k\left(r\right) dqdpdr \\
&- \int \left(\partial_k R_k\left(q\right)\right) G_k\left(q\right) \Gamma_k^{(4)}\left(q,p,p_1\right) G_k\left(p\right) dqdp
\end{aligned} \tag{11.42}
$$

This equation is so far exact. One of the known tricks to solve this integral is to propose an Ansatz for Γ_k. For a scalar ϕ^4 theory, since we already considered regularized and renormalized solutions above by means of the perturbation theory, we can make "guesses" that the terms should take the form $\Gamma_k^{(2)} = p^2 + m_k^2$, $\Gamma_k^{(4)} = \lambda_k$, such that $G_k\left(p\right)^{-1} = \left(p^2 + m_k^2 + R_k\right)$. The term $\Gamma_k^{(3)} = 0$ vanishes because of the imposed$\mathbb{Z}^2$ symmetry, so that for this particular theory the first integral in Eq. (11.42) vanishes; and we are left with Eqs. (11.43, 11.44).

$$
\partial_k \Gamma_k^{(2)} = \partial_k p^2 + \partial_k m_k = -\int \frac{\lambda_k \left(\partial_k R_k\left(q\right)\right)}{\left(q^2 + m_k^2 + R_k\right)^2} d^d q \tag{11.43}
$$

$$
\partial_k m_k^2 = -\int \frac{\lambda_k \left(\partial_k R_k\left(q\right)\right)}{\left(q^2 + m_k^2 + R_k\right)^2} d^d q \tag{11.44}
$$

As mentioned previously, the choice of regulator affects the renormalization flow, but it should not affect the actual measurable physical quantities, so lon gas the conditions with respect to the method formulation are satisfied. A common choice is the Litim regulator, $R_k\left(q\right) = \left(q^2 - k^2\right)\Theta\left(q^2 - k^2\right)$. With this choice, Eq. (11.44) is resolved into Eq. (11.45), and in the last step we assume $k >> m_k$.

$$
\partial_k m_k^2 = -\int \frac{\lambda_k \left(\partial_k R_k\left(q\right)\right)}{\left(q^2 + m_k^2 + R_k\right)^2} d^4 q = \frac{2\lambda_k k}{\left(k^2 + m_k^2\right)^2} \int^k d^4 q = \frac{2\lambda_k k}{\left(k^2 + m_k^2\right)^2} \frac{k^4}{32\pi^2} = \frac{\lambda_k}{16\pi^2} \frac{k^5}{\left(k^2 + m_k^2\right)^2} \approx \frac{\lambda_k k}{16\pi^2} \tag{11.45}
$$

The RG flow corresponding to $\Gamma_k^{(4)} = \lambda_k$ can furthermore be calculated. Starting from Eq. (11.42), we need to take two additional functional derivatives. A similar analysis as carried out in Eq. (11.36) shows that only three terms, identical up to index permutation, survive, corresponding to Eq. (11.46).

$$
\begin{aligned}
\partial_k \Gamma_k^{(4)} &= \partial_k \lambda_k = 3 \int \left(\partial_k R_k\left(q\right)\right) G_k\left(q\right) \Gamma_k^{(4)}\left(q,p\right) G_k\left(p\right) \Gamma_k^{(4)}\left(q,r\right) G_k\left(r\right) d^d q d^d p d^d r \\
&= 3 \int \frac{\lambda_k^2 \left(\partial_k R_k\left(q\right)\right)}{\left(Z_k q^2 + m_k^2 + R_k\right)^3} d^4 = \frac{\lambda_k^2}{16\pi^2} \frac{k^5}{\left(k^2 + m_k^2\right)^3} \approx \frac{1}{k} \frac{3\lambda_k^2}{16\pi^2}
\end{aligned} \tag{11.46}
$$

Differently from the result obtained through steps of regularization and renormalization in perturbation theory, notice that in this case the renormalization scale has a mass dimension $[k] = \left[m^1\right]$. we define the renormalization time $t = log\left(k/\Lambda\right)$, and Eqs. (11.45, 11.46) reduce to Eq. (11.47), such that the choice $\frac{\mu}{m^2} \to \frac{1}{k^2}$ makes the equivalence of both methods explicit.

$$\frac{1}{k^2}\partial_t m_t^2 = \frac{\lambda_t}{16\pi^2}\ ; \qquad \partial_t \lambda_t = \frac{3\lambda_t^2}{16\pi^2} \tag{11.47}$$

11.7 Derivation of the LPA-Wetterich for a ϕ^4 Scalar Theory

One considers the Ansatz specified by Eq. (11.48) below.

$$\Gamma_k[\phi] = \int d^d x \left\{ \frac{1}{2}\partial_\mu \phi \partial^\mu \phi + U_k(\rho) \right\} \tag{11.48}$$

One requires the $\mathbb{Z}^2$ symmetry explicitly, and one accordingly defines $\rho = \frac{1}{2}\phi^2$. As a preparation, one computes the identity Eq. (11.49), which will be implemented later.

$$\frac{\partial}{\partial\phi} = \frac{\partial\rho}{\partial\phi}\frac{\partial}{\partial\rho} = \phi\frac{\partial}{\partial\rho} \Longrightarrow \frac{\partial^2}{\partial\phi^2} = \frac{\partial}{\partial\phi}\left(\phi\frac{\partial}{\partial\rho}\right) = \frac{\partial}{\partial\rho} + \phi\frac{\partial\rho}{\partial\phi}\frac{\partial^2}{\partial\rho^2} = \frac{\partial}{\partial\rho} + 2\rho\frac{\partial^2}{\partial\rho^2} \tag{11.49}$$

The effective action is evaluated through Eq. (11.50). In the last step one replaces $\Gamma_k \to U_k(\rho)$ on the following basis: (i) one would like to derive specifically the RG-flow of the potential; and (ii) such that by setting the momenta $p \to 0$, $U_k(\rho)$ is recovered from Γ_k, as the potential is the momentum-independent part of the running inverse propagator. [ag]

$$\frac{\partial^2\Gamma_k}{\partial\phi^2} = \frac{\partial\Gamma_k}{\partial\rho} + 2\rho\frac{\partial^2\Gamma_k}{\partial\rho^2} = \Gamma_k' + 2\rho\Gamma_k'' = U_k'(\rho) + 2\rho U_k''(\rho) \tag{11.50}$$

One consider an optimized regulator of the form Eq. (11.51).

$$R_k(z) = \left(k^2 - z^2\right)\Theta\left(k^2 - z^2\right) \tag{11.51}$$

By replacing those quantities directly onto the the Wetterich Equation, Eq. (5.7), one obtains Eq. (11.52), whereA_d is a normalization constant corresponding to the integral over a d-dimensional ball of radius k.[ah] Furthermore, one notices that the introduction of the regulator explicitly makes the integrand independent of p.

$$\partial_k\Gamma_k = \partial_k U_k(\rho) = \int \frac{d^d p}{(2\pi)^d}\frac{\partial_k R_k}{p^2 + U'(\rho) + \rho_* U''(\rho) + R_k} = \int_0^{k^2}\frac{d^d p}{(2\pi)^d}\frac{2k}{k^2 + U'(\rho) + 2\rho U''(\rho)} = \frac{A_d k^{d+1}}{k^2 + U'(\rho) + 2\rho U''(\rho)} \tag{11.52}$$

Similar in procedure to the method of vertices, one makes the equation dimensionless in order to extract

[ag] Strictly speaking, those should be evaluated as functionals, not functions; the results are fundamentally the same, up to formalism:

$$\frac{\partial\Gamma_k[\phi]}{\partial\phi_i(x)\,\partial\phi_j(x)} = \left(-\nabla + U_k^{(2)}(\rho)\right)\delta^d(x-y)\,\delta_{ij} = \left(\left(-\nabla + U'(\rho)\right) + \phi_i(x)\,\phi_j(x')\,U''(\rho)\right)\delta^d(x-x')\,\delta_{ij}$$

$$\frac{\partial}{\partial\phi(x)} = \int\frac{d^d p}{(2\pi)^d}e^{-ipx}\frac{\partial}{\partial\phi(p)} \Longmapsto \frac{\partial\Gamma_k[\phi]}{\partial\phi(p)\,\partial\phi(p)} = \left(p^2 + U'(\rho)\right) + \tilde{\phi}^2(p)\,U''(\rho)$$

Where one denotes a convolution, as already described in Appendix Manipulation of Functional Objects:

$$\tilde{\phi}^2(p) = \int\phi^2(x)\,e^{ipx}d^d x = \int\phi(x)\,\phi(x')\,e^{ipx+ik(x-x')}d^d x d^d x' d^d k = \int\phi(k)\,\phi(p-k)\,d^d k$$

[ah] in 4 dimensions, for example, one obtains $A_4 = \frac{1}{32\pi^2}$

relvant information from the system. The relevant dimensionlessquantities are stated in Eq. (11.53) below, where one emphasizes that even the derivative operator itself has a dimension.

$$m = k\tilde{m} \qquad \rho = k^{d-2}\tilde{\rho} \qquad g_4 = k^{-d+4}\tilde{g}_4 \qquad g_{2n} = k^{d-n(d-2)}\tilde{g}_{2n} \qquad \partial_\rho = (\partial_\rho\tilde{\rho})\,\partial_{\tilde{\rho}} = k^{2-d}\partial_{\tilde{\rho}} \tag{11.53}$$

Additionally, the conventional transformation to RG-time gives Eq. (11.54).

$$\partial_k = (\partial_k t)\,\partial_t = k^{-1}\partial_t \tag{11.54}$$

The left-handed side of Eq. (11.52) becomes Eq. (11.55).

$$\begin{aligned}\partial_k U_k(\rho) &= \partial_k\left(k^d\tilde{U}(\tilde{\rho})\right) = k^{d-1}d\cdot\tilde{U}(\tilde{\rho}) + \partial_k\tilde{U}(\tilde{\rho}) \\ &= k^{d-1}d\cdot\tilde{U}(\tilde{\rho}) + k^d(\partial_k\tilde{\rho})\,\partial_{\tilde{\rho}}\tilde{U}_k(\tilde{\rho}) + k^d\partial_k\tilde{U}_k(\tilde{\rho}) = k^{d-1}\left(d\cdot\tilde{U}_t(\tilde{\rho}) - (d-2)\,\tilde{\rho}\tilde{U}_t'(\tilde{\rho}) + \partial_t\tilde{U}_t(\tilde{\rho})\right)\end{aligned} \tag{11.55}$$

The denominator in the right-handed side of Eq. (11.52) becomes Eq. (11.56), where one replaces $(\ldots)' = \partial_\rho(\ldots)$ in order make the step more explicit. Furthermore, one notices that since k is not a function of $\tilde{\rho}$, this implies that k then commutes with the partial derivatives.

$$k^2 + \partial_\rho U(\rho) + 2\rho\partial_\rho^2 U(\rho) = k^2 + k^{2-d}\partial_{\tilde{\rho}}k^d\tilde{U}(\tilde{\rho}) + 2k^{d-2}\tilde{\rho}k^{2-d}\partial_{\tilde{\rho}}k^{2-d}\partial_{\tilde{\rho}}k^d\tilde{U}(\tilde{\rho}) = k^2\left(1 + \partial_{\tilde{\rho}}\tilde{U}(\tilde{\rho}) + 2\tilde{\rho}\partial_{\tilde{\rho}}^2\tilde{U}(\tilde{\rho})\right) \tag{11.56}$$

Combining everything all-together, the dependancy on k disappears, and one obtains the dimensionless Eq. (11.57).

$$\partial_t\tilde{U}(\tilde{\rho}) = -d\cdot\tilde{U}(\tilde{\rho}) + (d-2)\,\tilde{\rho}\tilde{U}_t'(\tilde{\rho}) + A_d\cdot\frac{1}{1+\tilde{U}'(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}''(\tilde{\rho})} \tag{11.57}$$

11.8 A Remark on Dimensional Analysis

One previously relied on Eq. (11.58) in order to write the Wetterich Equation in adimensionless form. In this Appendix, dimensional analysis is discussed more explicitly. One distinguishes dimensionful and dimensionless variables by denoting the tilde. For example m_k is the dimensionful "mass" at a scale k and $\tilde{m}_k$ the corresonding dimensionless "mass". By definition, the momentum has dimension $[p] = 1$, i.e., $p \sim k^1$.

$$m = k\tilde{m} \qquad \rho = k^{d-2}\tilde{\rho} \qquad g_4 = k^{-d+4}\tilde{g}_4 \qquad g_{2n} = k^{d-n(d-2)}\tilde{g}_{2n} \qquad \partial_\rho = (\partial_\rho\tilde{\rho})\,\partial_{\tilde{\rho}} = k^{2-d}\partial_{\tilde{\rho}} \tag{11.58}$$

The fourier transform is dimensionless, $[p\cdot x] = 0$, such that $[x] = -1$ and $[\partial_\mu] = 1$. The action needs to be dimensionless due to the definition of the partition function, such that the lagrangian always has dimension d. This leads to Eq. (11.59).

$$[\partial_\mu\phi\partial^\mu\phi] = d \Longrightarrow [\phi] = \frac{d-2}{2} \tag{11.59}$$

By extension, one obtains Eq. (11.60), which are the results stated previously, Eq. (11.58).

$$\left[\phi^2 m^2\right] = d \Longrightarrow \left[m^2\right] = 2 \; ; \qquad \left[\phi^{2n} g_{2n}\right] = d \Longrightarrow \left[g_{2n}\right] = -n\left(d-2\right) + d \tag{11.60}$$

Furthermore, all tilde-quantities are dimensionless by construction or definition, $[\tilde{a}] = 0$. When scaling relations are found, for example, Eq. (11.61), in tihs case $\tilde{g_{2n}}$ itself continues being dimensionless, that is to say $[\tilde{g_{2n}}] = 0$.

$$\tilde{g_{2n}} \sim k^{(n(d-2)-2)} \tag{11.61}$$

Rather, one should interpret it as Eq. (11.62), such that the dimensionful quantity is given by Eq. (11.63). In this particular case, one concludes that g_{2n} has dimension k^d, and g_{2n} does not scale with dimension. The dimensionless coupling runs with k, but the scaleful coupling is fixed.

$$\tilde{g_{2n}}\left(k\right) = \tilde{g_{2n}}\left(k_0\right) k^{(n(d-2)-d)} \tag{11.62}$$

$$g_{2n} = k^{-n(d-2)+d} \tilde{g_{2n}} = \tilde{g_{2n}}\left(k_0\right) \tag{11.63}$$

One considers conversely the case of Eq. (11.64). In this case, in $d = 2$, one finds Eq. (11.65).

$$\tilde{\lambda} \sim \frac{1}{log\left(\Lambda/k\right)} \tag{11.64}$$

$$\lambda = k^{-n(d-2)+d} \tilde{\lambda} = \frac{k^2 \tilde{\lambda}\left(k_0\right)}{log\left(\Lambda/k\right)} \tag{11.65}$$

The dimensionality still doesn't change: $[\lambda] = 2$. On the other hand, one finds that λ has an anomalous scaling with respect to k, Eq. (11.66). The conclusion is then that λ has a dimension k^2, but it scales as Eq. (11.66) with respect to the dimension. This is understood as an anomalous scaling of the coupling. In literature, this phenomena is normally referred to as a "running coupling constant". The effect is physically observable, in that the coupling strength changes respect to changes in the scale at which a particular system is being analysed. This phenomena is illustratively characterized and justified in literature, that the interaction term runs because it is being "screened-out by virtual interactions".

$$\lambda \sim \frac{k^2}{log\left(\Lambda/k\right)} \tag{11.66}$$

11.9 Anomalous Dimension Evaluation

One begins by taking functional derivatives against the running inverse propagator, Eqs. (11.67, 11.68).

$$\frac{\delta\Gamma_k^{(2)}(p,q)}{\delta\phi(p_1)} = -\int G(p,\zeta)\,\Gamma_k^{(3)}(\xi,\zeta,p_1)\,G(\xi,q)\,d\xi d\zeta \tag{11.67}$$

$$\begin{aligned}\frac{\delta^2\Gamma_k^{(2)}(p,q)}{\delta\phi(p_1)\,\delta\phi(p_2)} &= -\int G(p,\zeta)\,\Gamma_k^{(4)}(\xi,\zeta,p_1,p_2)\,G(\xi,q)\,d\xi d\zeta \\ &\underbrace{+2\int G(p,\zeta')\,\Gamma_k^{(3)}(\xi',\zeta',p_2)\,G(\xi',\zeta)\,\Gamma_k^{(3)}(\xi,\zeta,p_1)\,G(\xi,q)\,d\xi d\zeta d\xi' d\zeta'}_{V_{2,2}}\end{aligned} \tag{11.68}$$

Only the term in the second line, $V_{2,2}$, is non-zero. It evaluates by means of Eq. (11.69), where additionally one exploits the translational symmetry in $G(p,q)$ and the minima condition, $\phi(r) = \phi_* \delta(r,0)$.

$$\begin{aligned}V_{2,2} &= 2\lambda^2 \int \phi(r)\,\phi(r')\,G(p,\zeta')\,G(\xi',\zeta)\,G(\xi,q)\,\delta(\xi',\zeta',p_2,r')\,\delta(\xi,\zeta,p_1,r)\,d\xi d\zeta d\xi' d\zeta' dr dr' \\ &= 2\lambda^2 \int \phi(r)\,\phi(r')\,G(p)\,G(q)\,G(-p+p_2+r',-q+p_1+r)\,dr dr' \\ &= \lambda^2 \rho_* (G(p))^2\,G(p-p_1)\end{aligned} \tag{11.69}$$

One considers the Ansatz and regulator, Eq. ().

$$\Gamma_k^{(2)}[\phi] = \frac{Z_\phi}{2} p^2\phi^2 + U_k^{(2)}[\phi]\ ; \qquad R_k = Z_\phi\left(k^2-p^2\right)\Theta\left(k^2-p^2\right) \tag{11.70}$$

One notices that the anomalous dimension is found via

$$\frac{k}{Z_\phi}\partial_t Z_\phi = \frac{d^2}{dp^2}\frac{\delta^2\Gamma_k^{(2)}(p,q)}{\delta\phi(p_1)\,\delta\phi(p_2)} ==$$

11.10 Results for $O(N)$ Models

The derivation of the Wetterich Equation for the $O(1)$ model can be easilyextended to $O(N)$ models. Specifically, for the $O(N)$ model, one obtains Eq. (11.71).

$$\frac{\delta U_k}{\delta\phi^a\delta\phi^b} = U_k'\delta_{ab} + U_k''\phi^a\phi^b \tag{11.71}$$

By choosing a rotated, diagonalized basis with $\phi^a = \begin{pmatrix} 0 & \dots & \sqrt{2\rho} \end{pmatrix}$, one obtains $(N-1)$ transverse directions with $\lambda_\sigma = U'$ and one massive direction with $\lambda_\pi = U' + 2\rho U''$. Since the potential is now diagonalized, the wetterich equation is derived by simply summing each linearly-inedependent potential for each of the nodes, that is to say, Eq. (11.72).

$$\partial_t\tilde{U}(\tilde{\rho}) = -d\cdot\tilde{U}(\tilde{\rho}) + (d-2)\,\tilde{\rho}\tilde{U}_t'(\tilde{\rho}) + A_d\cdot\left\{\frac{(N-1)}{1+\tilde{U}'(\tilde{\rho})} + \frac{1}{1+\tilde{U}'(\tilde{\rho})+2\tilde{\rho}\tilde{U}''(\tilde{\rho})}\right\} \tag{11.72}$$

To second order truncation, one obtains the flows, Eqs. (11.73, 11.74).

$$\partial_t \tilde{m}^2 = -2\tilde{m}^2 - \frac{A_d}{3} \cdot \frac{(N+2)\,\tilde{\lambda}}{\left(1+\tilde{m}^2\right)^2} \tag{11.73}$$

$$\partial_t \tilde{\lambda} = (-4+d)\,\tilde{\lambda} + \frac{2A_d}{3} \cdot \frac{(N+8)\,\tilde{\lambda}^2}{\left(1+\tilde{m}^2\right)^3} \tag{11.74}$$

The fixed point is at Eq. (11.75).

$$\tilde{m}^2 = \frac{(N+2)\,(d-4)}{8\,(N+5)-(N+2)\,d}\;; \qquad \tilde{\lambda} = \frac{6\tilde{m}^2}{A_d}\frac{\left(1+\tilde{m}^2\right)^2}{(N+2)} = -\frac{96}{A_d}\frac{(d-4)\,(N+8)^3}{\left(8\,(N+5)-(N+2)\,d\right)^4} \tag{11.75}$$

The stability matrix, up to 2-level truncation, becomes Eq. (11.76).

$$D_* = \begin{pmatrix} \frac{\beta_{\tilde{m}^2}}{\partial \tilde{m}^2} & \frac{\beta_{\tilde{\lambda}}}{\partial \tilde{m}^2} \\ \frac{\beta_{\tilde{m}^2}}{\partial \tilde{\lambda}} & \frac{\beta_{\tilde{\lambda}}}{\partial \tilde{\lambda}} \end{pmatrix} = \begin{pmatrix} -2 + \frac{(N+2)(4-d)}{N+8} & -\frac{A_d}{3}\,(N+2)\left[\frac{8(N+5)-(N+2)d}{4(N+8)}\right]^2 \\ -\frac{2\times 36}{A_d}\cdot(N+8)\left[\frac{(d-4)}{8(N+5)-(N+2)d}\right]^2 & -(d-4) \end{pmatrix} \tag{11.76}$$

The quadratic reads Eq. (11.77).

$$\lambda^2 + \lambda\left[(d-4) - \frac{2\,(N-4)-d\,(N+2)}{N+8}\right] - (d-4)\,\frac{2\,(N-4)-d\,(N+2)}{N+8} - \frac{3}{2}\frac{(N+2)}{(N+8)}\,(d-4)^2 = 0 \tag{11.77}$$

As a sanity check, one can replace $N=1$ in Eqs. (11.72 — 11.77) above, and the correspondence to the $O\,(1)$ model becomes explicit.

Part II

Conformal Field Theories

12 Introduction

A Conformal Field Theory (CFT) is defined as a Quantum Field Theory that is invariant under Conformal Transformations. A Conformal Transformation is a transformation that leaves the metric invariant, up to a scaling factor $\Lambda\,(x)$ (this condition is equivalent to the preservation of angles). CFTs received great interest in recent years, as they describe the physics of fixed points and second order phase transitions. CFTs also encounter applications in areas of QFT, for example, to describe the dynamics of strings in string theory, as well as the $\mathcal{N}=4$ supersymmetric Yang-Millis theory. In the following sections, conformal invariance and scale invariance are treated as almost equivalent terms. This is justified for a few reasons:[70, 82]

- conformal symmetry can be seen as a generalization of scale symmetry;
- it is largely believed that theories that are simultaneously unitary and scale invariant, are automatically conformal invariant (this has been shown for $d=2$, and a perturbative argument exists for $d=4$); and

- the generators of the conformal group for $d \geq 3$ are isomorphic to $SO(d+2)$ (or $SO(d+1,1)$ in Minkowsky metric), and, specifically, they are the generators of the Poincaré group, plus scale invariance, plus the special conformal transformation (SCT). This gives insight that these three classes of transformaitons are deeply connected.

Although most interacting theories are not conformal invariant, the interest in CFTs is motivated by the fact that theories become conformal invariant in the vicinities of fixed points in the RG flow, because, at those points, the correlation length diverges, $\xi \to \infty$, such that the theory loses its scale. One can therefore construct interacting QFTs as "perturbations" to CFTs in the vicinity of fixed points of the RG flow. CFTs are also of interest, because they give the maximum number of symmetries allowed, given d.[70, 73, 79, 81]

Conformal invariance is an anomalous symmetry in QFT. A scale anomaly means that a theory, scale-invariant in the classical regime, may acquire dependencies in the quantum level, due to quantum fluctuations. The origin of anomalies comes therefrom, that whilst in the classical theory we one requires only that the action be invariant under the symmetry, in the quantum theory, on the other hand, the path integral formalism requires that the partition function, that is, the action and the integration measure of the partition function simultaneously, be invariant under the symmetry; which may induce those anomalies. Different from gauge anomalies, the scale anomaly doesn't make the theory inconsistent. Unitarity, as well as correlation functions and the hilbert space, are still preserved and well defined; and energy is still conserved. The scale anomaly leads to a non-zero expectation value for the trace of the energy-stress tensor $\langle T^{\mu}_{\mu} \rangle$ after quantization — whereas this quantity is zero in the classical theory. The trace of the energy-stress tensory acquires a dependency on the RG flow, once the theory is quantized; such that the scale-invariance is only recovered at fixed points. In $d = 2$ the effect is even more significant, since $\langle T^{\mu}_{\mu} \rangle$ becomes proportional to the ricci tensor and doesn't vanish, even when the RG flow is at a fixed point. This means that the curvature induced by the metric acts as a scale. The way anomalies can be dealt with are threefold:[71, 72, 74]

- one may formulate supersymmetric theories that have a zero-beta-function by construction, that is, they are scale invariant by construction; or
- one considers exclusively fixed points of the theory; or
- one gives up on the lagrangian formalism, and describes the theory exclusively from correlation functions and the S-matrix, such that those objects are constructed to be confromal-invariant. In that context, come the Ward Identities to use, as those identities can be used to verify, among others, that correlation functions will remain invariant under generators of conformal transformations.

Particular emphasis in this thesis is given to CFTs in $d = 2$. In two dimensions, the theory can be constructed in a complex plane, where the usual coordinate system is replaced by a holomorphic and anti-holomorphic basis. The apparatus of complex analysis can then be implemented, and one obtains a significantly stronger restrictions to the theory. A $d = 2$ CFT contains infinitely many symmetry generators. This infinite-dimensional conformal algebra restricts the CFT theories available. This fact gives hope of a complete classification of critical phenomena in two dimensions.[70, 72, 74, 75]

The following sections will be organized as follows:

- The main ideas of quantum field theory, and specifically the noether theorem, are revisited.
- Conformal invariance is formally based on its action on the metric, and from there the algebra of infinitesimal conformal transformations in d dimensions is derived, as well as the noether current induced by the conformal transformation. The requirement that $\langle T^{\mu}_{\mu} \rangle = 0$ for classical fields is shown.

- The theory is specialized to two dimensions, in hand of complex analysis. The main results are derived, including, for example, the so-called Witt Algebra, describing the generators of conformal symmetries in two dimensions. The theory is subsequently quantized, and the natural emergence of the Virasoro Algebra from the Witt Algebra is shown, by means of introduction of a central charge to the theory.

- The creaction of physical states from operators is described (the so-called "state-operator equivalence"), as well as the construction of conformal minimal models, by means of null vectors in the verma module. It is shown that those models can be solved analytically — from conformal symmetry alone — by means of deriving differential equations for the correlation functions, with the help of the Ward identities. The fusion rules are shown and used to truncate minimal models to a finite number of primary fields, that are sufficient completely describe the theory.

13 Formal Developments in Field Theories

This section follows closely from [23, 29, 30].

13.1 Field Theory Formalism

Let a scalar field be described by Eq. (13.1).

$$\mathcal{S}\left[\phi\right] = \int \mathcal{L}\left[\phi;\partial_\mu\phi\right] dx = -\frac{1}{2}\int \left(\partial_\mu\phi\partial^\mu\phi + m^2\phi^2\right) dx \tag{13.1}$$

The least action principle implies Eq. (13.2). The second integral is a total derivative and therefore vanishes at appropriate boundary conditions. The first term is the Euler-Lagrange Equation.

$$\frac{\delta S\left[\phi\right]}{\delta\phi\left(y\right)} = \int \left(\frac{\delta\mathcal{L}\left[\phi;\partial_\mu\phi\right]\left(x\right)}{\delta\phi\left(y\right)} - \partial_\mu\frac{\delta\mathcal{L}\left[\phi;\partial_\mu\phi\right]\left(x\right)}{\delta\left(\partial_\mu\phi\left(y\right)\right)}\right)\frac{\delta\phi\left(y\right)}{\delta\phi\left(x\right)}dx + \int \partial_\mu\left(\frac{\delta\mathcal{L}\left[\phi;\partial_\mu\phi\right]\left(x\right)}{\delta\left(\partial_\mu\phi\left(y\right)\right)}\frac{\delta\phi\left(y\right)}{\delta\phi\left(x\right)}\right)dx = 0 \tag{13.2}$$

By employing Eq. (13.2) to Eq. (13.1), one obtains standard solutions given by Eq. (13.3)

$$\phi\left(x\right) = \int \left(a\left(\vec{p}\right)e^{ipx} + a^*\left(\vec{p}\right)e^{-ipx}\right) d\tilde{p} \tag{13.3}$$

Here, the following quantities are defined, Eq. (13.4).

$$d\tilde{p} = \frac{d^dp}{\left(2\pi\right)^d 2\omega}\ ; \qquad \omega = \sqrt{m^2+\vec{p}^2}\ ; \qquad p \equiv p^\mu \tag{13.4}$$

One moves to the hamiltonian formalism by defining a conjugate field, Eq. (13.5).

$$\pi\left(x\right) = \frac{\delta\mathcal{L}\left[\phi;\partial_\mu\phi\right]\left(y\right)}{\delta\dot{\phi}\left(x\right)} \tag{13.5}$$

The Hamiltonian reads from Eq. (13.6).

$$\mathcal{H}\left[\phi;\pi\right]\left(x\right) = \pi\left(x\right)\dot{\phi}\left(x\right) - \mathcal{L}\left(x\right) \tag{13.6}$$

The commutation relations for the field are found via the poisson bracket, Eq. (13.7).

$$\{F(y), G(z)\} = \int dx \left(\frac{\delta F(y)}{\delta \phi(x)} \frac{\delta G(z)}{\delta \pi(x)} - \frac{\delta G(z)}{\delta \phi(x)} \frac{\delta F(y)}{\delta \pi(x)} \right) \tag{13.7}$$

One obtains exclusively the non-trivial relation, Eq. (13.8).

$$\{\phi(x), \pi(y)\}_{\pm} = \delta(x-y) \tag{13.8}$$

By taking the fourier transforms with respect to ϕ and π, one finds Eq. (13.9).

$$a(\vec{p}) = \int e^{-ipx} \left(i\omega\phi(x) + \pi(x) \right) dx \tag{13.9}$$

The non-trivial commutation relation is Eq. (13.10).

$$\{a(\vec{p}), a^*(\vec{q})\}_{\pm} = (2i\omega)\, \delta(p-q) \tag{13.10}$$

With those relations, the Hamiltonian Eq. (13.6) can be re-written in momentum space via Eq. (13.11).

$$\mathcal{H} = \frac{1}{2} \int \omega \left(a(\vec{p})\, a^*(\vec{p}) + a^*(\vec{p})\, a(\vec{p}) \right) d\tilde{p} + \text{const.} \tag{13.11}$$

To extend this formalism to a QFT, one promote the fields to operators, Eq. (13.12).

$$\phi \to \hat{\phi}\ ; \qquad \pi \to \hat{\pi}\ ; \qquad a \to \hat{a}\ ; \qquad a^* \to \hat{a}^\dagger \tag{13.12}$$

From the commutation relations, one naturally interprets $\hat{a}^\dagger$and $\hat{a}$ as the creation and elimination operators, such that states are prepared by acting with the creation operator on the vacuum, Eq. (13.13).

$$a^\dagger(\vec{p})\,|0\rangle = |\vec{p}\rangle\ ; \qquad a(\vec{p})\,|0\rangle = 0 \tag{13.13}$$

The correlation function is defined by taking the time-ordered operator of two fields, Eq. (13.14).[ai]

$$G(x-y) = \langle 0|\mathcal{T}\phi(x)\,\phi(y)\,|0\rangle = \int \frac{d^3\vec{p}}{(2\pi)^3\, 2\omega} e^{ip|x-y|} \tag{13.14}$$

13.2 Noether Theorem

13.2.1 Formulation

One considers an infinitesimal symmetry transformation, Eq. (13.15)

[ai] In more details,

$$\begin{aligned}\langle 0|\mathcal{T}\phi(x)\,\phi(y)\,|0\rangle &= \int \langle \left(a_{\vec{p}} e^{ipx} + a^*_{\vec{p}} e^{-ipx} \right) \left(a_{\vec{q}} e^{iqx} + a^*_{\vec{q}} e^{-iqx} \right) \rangle d\tilde{p} d\tilde{q} \\ &= \int \left(\underbrace{\langle a_{\vec{p}} a_{\vec{q}} \rangle e^{ipx+iqy} + \langle a^*_{\vec{p}} a^*_{\vec{q}} \rangle e^{-ipx-iqy}}_{=0} + \langle a_{\vec{p}} a^*_{\vec{q}} \rangle e^{ipx-iqy} + \langle a^*_{\vec{p}} a_{\vec{q}} \rangle e^{-ipx+iqy} \right) d\tilde{p} d\tilde{q} = \int \frac{d^3\vec{p}}{(2\pi)^3\, 2\omega} e^{ip|x-y|}\end{aligned}$$

$$\mathcal{L} \to \mathcal{L} + \delta\mathcal{L} \; ; \qquad \phi \to \phi + \delta\phi \tag{13.15}$$

The action is left invariant, Eq. (13.16).

$$\delta S\left[\phi\right] = \underbrace{\int \left(\frac{\delta\mathcal{L}\left[\phi;\partial_\mu\phi\right](x)}{\delta\phi\left(x\right)} - \partial_\mu \frac{\delta\mathcal{L}\left[\phi;\partial_\mu\phi\right](x)}{\delta\left(\partial_\mu\phi\left(x\right)\right)} \right) \delta\phi\left(x\right) dx}_{=0 \text{ due to Euler-Lagrange Equation.}} + \int \partial_\mu \left(\frac{\delta\mathcal{L}\left[\phi;\partial_\mu\phi\right](x)}{\delta\left(\partial_\mu\phi\left(x\right)\right)} \delta\phi\left(x\right) \right) dx = 0 \tag{13.16}$$

With the definitions, Eq. (13.17), if the infinitesimal symmetry transformation leaves the action invariant, one finds Eq. (13.18). This is Noether's theorem.

$$\delta\mathcal{L} = \partial_\mu K^\mu \; ; \qquad \mathcal{J}^\mu = j^\mu - K^\mu \tag{13.17}$$

$$\partial_\mu \mathcal{J}^\mu = \partial_\mu \left(\frac{\delta\mathcal{L}\left[\phi;\partial_\mu\phi\right](x)}{\delta\left(\partial_\mu\phi\left(x\right)\right)} \delta\phi\left(x\right) - K^\mu \right) = 0 \; ; \qquad \mathcal{Q} = \int d^{d-1}x \mathcal{J}^0\left(x\right) \tag{13.18}$$

The existence of a symmetry leaving the action invariant, implies the continuity equation for a conserved noether current, and a corresponding noether charge $\mathcal{Q}$. The noether charge is the corresponding symmetry generator for the respective symmetry. This is shown easily from Eq. (13.19).

$$i\left[\mathcal{Q}, \phi\left(x'\right)\right] = \int d^{d-1}x \left[\frac{\delta\mathcal{L}\left[\phi;\partial_\mu\phi\right](x)}{\delta\left(\partial_\mu\phi\right)(x)} \delta\phi, \phi\left(x'\right) \right] = \int d^{d-1}x \left[\pi\left(x\right)\delta\phi, \phi\left(x'\right)\right] = \delta\phi \tag{13.19}$$

One may therefore define the following transformation, Eq. (13.20).

$$U_{\mathcal{Q}} = e^{i\alpha\mathcal{Q}} \; ; \qquad U_{\mathcal{Q}}^{-1}\left(\alpha\right)\phi\left(x\right)U_{\mathcal{Q}}\left(\alpha\right) = \phi'\left(x'\right) \tag{13.20}$$

13.2.2 Example: Spacial Translation

An infinitesimal spacial translation is specified by Eq. (13.21).

$$x^\mu \to x^\mu + \delta x^\mu \; ; \quad \phi\left(x^\mu + \delta x^\mu\right) = \phi\left(x\right) + \delta x^\mu \partial_\mu \phi\left(x\right) \; ; \quad \mathcal{L}\left(x^\mu + \delta x^\mu\right) = \mathcal{L}\left(x\right) + \delta x^\mu \partial_\mu \mathcal{L}\left(x\right) \tag{13.21}$$

Correspondingly, one obtains the noether charge, Eq. (13.22), where the energy-momentum tensor $T^{\mu\nu}$ is defined, as well as the noether charge $\mathcal{Q}^\mu$, corresponding to the conserved energy and momenta.

$$\partial_\mu \mathcal{J}^\mu = dx_\nu \partial_\mu T^{\mu\nu} = 0 \; ; \qquad \mathcal{Q}^\mu = \int d^{d-1}x T^{\mu 0} \tag{13.22}$$

13.2.3 Example: Relativistic Transformation

Relativistic transformations can be implemented in a similar way, Eq. (13.23).

$$x^\mu \to x^\mu + \delta\omega^\mu_\nu x^\nu \ ; \quad \phi\left(x^\mu + \delta x^\mu\right) = \phi\left(x\right) + \delta\omega^\mu_\nu x^\nu \partial_\mu \phi\left(x\right) \ ; \quad \mathcal{L}\left(x^\mu + \delta x^\mu\right) = \mathcal{L}\left(x\right) + \delta\omega^\mu_\nu x^\nu \partial_\mu \mathcal{L}\left(x\right) \tag{13.23}$$

One obtains the corresponding conserved current and charge, Eq. (13.24). $\mathcal{Q}^{\mu\nu}$ are therefore the 10 symmetry generators of the Lorenz group, corresponding to translations, rotations, and boosts.

$$\mathcal{J}^\mu = \frac{\delta\mathcal{L}\left[\phi;\partial_\mu\phi\right]\left(x\right)}{\delta\left(\partial_\mu\phi\left(x\right)\right)}\delta\phi\left(x\right) - K^\mu = \delta\omega_{\rho\nu}x^\nu\left(\partial^\mu\phi\partial^\rho\phi - g^{\rho\mu}\mathcal{L}\right) = \delta\omega_{\rho\nu}x^\nu T^{\rho\mu} \ ; \quad \mathcal{Q}^{\mu\nu} = \int d^{d-1}x\left(x^\mu T^{0\nu} - x^\nu T^{0\mu}\right) \tag{13.24}$$

13.2.4 Example: Scale Transformation

For a scale symmetry, one defines the transformations, Eqs. (13.25, 13.26).

$$x \to \lambda x \ ; \qquad \phi \to \lambda^{-\left(\frac{d-2}{2}\right)}\phi \ ; \qquad \mathcal{L} \to \lambda^d\mathcal{L} \tag{13.25}$$

$$\lambda = \left(1+\varepsilon\right) \ ; \qquad \delta\phi = = \varepsilon\left(-\left(\frac{d-2}{2}\right) + x^\mu\partial_\mu\right)\phi \ ; \qquad \delta\mathcal{L} = \partial_\mu\left(\varepsilon x^\mu\mathcal{L}\right) \tag{13.26}$$

The current becomes Eq. (13.27).

$$\mathcal{J}^\mu = \varepsilon x_\nu T^{\mu\nu} + \varepsilon\left(\frac{2-d}{2}\right)\phi\partial^\mu\phi \tag{13.27}$$

To verify charge conservation, one finds Eq. (13.28).[aj]

$$\partial_\mu\mathcal{J}^\mu = m^2\phi^2 \tag{13.28}$$

This result shows that the dilation current is not generally conserved, unless $m = 0$. This finding makes sense, since, in Part I, it has already been shown that a ϕ^4 theory has a non-zero beta-function, so that it is, in general, not scale invariant. Near a fixed point, if such a point exists, the beta function vanishes, and, in this case, the theory becomes scale invariant, such that the dilation current, Eq. (13.28), vanishes.

13.2.5 Generic Symmetry Transformation

Finally, one considers a generic symmetry transformation, Eq. (13.29).

$$x^\mu \to x^\mu + \varepsilon^\mu\left(x\right) \ ; \qquad \delta\phi = \varepsilon^\mu\partial_\mu\phi \tag{13.29}$$

One obtains the following variation to the action, Eq. (13.30).

$$\delta\mathcal{S} = \int\left(\delta\left(dx\right)\mathcal{L} + dx\delta\mathcal{L}\right) = \int dx\left(\left(\partial_\mu\varepsilon^\mu\right)\mathcal{L} + \varepsilon^\mu\partial_\mu\mathcal{L}\right) = \int dx\partial_\mu\left(\varepsilon^\mu\mathcal{L}\right) \tag{13.30}$$

[aj] In more details,

In addition, the field variation is given by Eq. (13.31).

$$\delta\mathcal{L} = \frac{\delta\mathcal{L}}{\delta\left(\partial_\mu\phi\right)}\partial_\mu\left(\delta\phi\right) + \frac{\delta\mathcal{L}}{\delta\phi}\delta\phi = \partial_\mu\left(\varepsilon^\rho\partial^\mu\phi\partial_\rho\phi\right) \tag{13.31}$$

It follows that, generically, Eq. (13.32) holds. $T^{\mu\nu}$ can therefore be seen as the response of the action with respect to an infinitesimal transformation given by Eq. (13.29).

$$\mathcal{J}^\mu = \varepsilon_\nu\left(x\right)T^{\mu\nu} \tag{13.32}$$

Eq. (13.33) is therefore required, in order to protect current conservation for a generic symmetry transformation of the type specified by Eq. (13.29).[ak] This expression is used in the subsequent section, to show that conformal transformations require the tracelessness of the stress-energy tensor, $\left\langle T^\mu_\mu\right\rangle = 0$.

$$\partial_\mu j^\mu = \left(\partial_\mu T^{\mu\nu}\right)\varepsilon_\nu + T^{\mu\nu}\left(\partial_\mu\varepsilon_\nu\right) = \frac{1}{2}T^{\mu\nu}\left(\partial_\mu\varepsilon_\nu + \partial_\nu\varepsilon_\mu\right) = 0 \tag{13.33}$$

14 Conformal Invariance in d Dimensions

In this section, conformal symmetries are introduced. This section follows closely from [70, 72, 85, 86].

A conformal symmetry, is a symmetry that conserves angles, but not necessarily distances — it can be seen as a "generalization" of the scale invariance. If a theory is symmetric under conformal transformations, 2- and 3-point correlation functions in any d dimensions can be completely specified by symmetry considerations alone. The conformal treansformation is formally implemented by Eq. (14.1).

$$g'_{\mu\nu} = \frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu}g_{\alpha\beta} = \Lambda^2\left(x\right)g_{\mu\nu} \tag{14.1}$$

The angle-preservation property can be checked explicitly, Eq. (14.2).

$$cos\left(\theta\right) = \frac{g_{\mu\nu}v^\mu v^\nu}{\left(g_{\mu\nu}v^\mu v^\nu\right)^{\frac{1}{2}}\left(g_{\mu\nu}u^\mu u^\nu\right)^{\frac{1}{2}}} = \frac{g'_{\mu\nu}v^\mu v^\nu}{\left(g'_{\mu\nu}v^\mu v^\nu\right)^{\frac{1}{2}}\left(g'_{\mu\nu}u^\mu u^\nu\right)^{\frac{1}{2}}} = cos\left(\theta'\right) \tag{14.2}$$

The goal is to derive the algebra of infinitesimal conformal transformations. To accomplish this goal, one first considers the infinitesimal transformation, Eq. (14.3).

$$x'^\mu = x^\mu + \varepsilon^\mu\left(x\right) \tag{14.3}$$

By replacing Eq. (14.3) onto Eq. (14.1), one obtains Eq. (14.4); where the rightmost equation is obtained by traing the leftmost equation with respect to the metric, and d is the dimension of the metric.

$$\partial_\mu\varepsilon_\nu + \partial_\nu\varepsilon_\mu = f\left(x\right)g_{\mu\nu}\ ; \qquad\qquad 2\partial_\rho\varepsilon^\rho = d\cdot f\left(x\right) \tag{14.4}$$

From the previous section it has been shown that a transformation of the type of Eq. (14.3) can lead

$$\partial_\mu\mathcal{J}^\mu = \left(\partial_\mu x_\nu\right)T^{\mu\nu} + x_\nu\left(\partial_\mu T^{\mu\nu}\right) + \left(\frac{2-d}{2}\right)\partial_\mu\phi\partial^\mu\phi + \left(\frac{2-d}{2}\right)\phi\partial^2\phi = T^\mu_\mu + \left(\frac{2-d}{2}\right)\partial_\mu\phi\partial^\mu\phi + \left(\frac{2-d}{2}\right)\phi\partial^2\phi = m^2\phi^2$$

where one writes $\partial_\mu T^{\mu\nu} = 0$ due to the translational-invariance symmetry, and $\phi\partial^2\phi = m^2\phi^2$ due to the equations of motion.

[ak] Here, one uses $\partial_\mu T^{\mu\nu} = 0$ to eliminate the first term, and the symmetry $T^{\mu\nu} = T^{\nu\mu}$ to reduce the second expression.

to a conservation law, only if Eq. (13.33) is satisfied. By combining Eq. (13.33) with Eq. (14.4) above, one therefore concludes that conformal transformations require the tracelesness of the energy-momentum tensor. This condition immediately leads, furthermore, to the requirement of masslessness. Here, it is worth emphasizing that although tracelessness is a requirement to ensure the most generic conformal symmetry, in principle, there are conformal transformations that could be enforced even in the absence of tracelessness, namely, when $\partial_\mu \varepsilon_\nu = 0$, i.e., when ε_ν is independent of the four-position. Translations are an example of this.

Eq. (14.5) is obtained from Eq. (14.4), by contracting with $g_{\mu\nu}$, multiplying by ∂_ρ, and the indicies.

$$2\partial_\mu \partial_\nu \varepsilon_\rho = g_{\rho\nu}\partial_\mu f(x) + g_{\mu\rho}\partial_\nu f(x) - g_{\mu\nu}\partial_\rho f(x) \tag{14.5}$$

Contracting Eq. (14.5) with $g_{\mu\nu}$ gives Eq. (14.6).

$$(d-2)\,\partial_\mu \partial_\nu f(x) = \partial_\nu \left(2\partial^2 \varepsilon_\rho\right) = d\partial_\mu \partial_\nu f = 0 \tag{14.6}$$

Therefore, one finds that $f(x)$ is a first order polynomial in x^μ, Eq. (14.7)

$$f(x) = A + B_\mu x^\mu \tag{14.7}$$

Plugging Eq. (14.7) back to Eq. (14.5), one finds Eq. (14.8).

$$\varepsilon_\mu = a_\mu + b_{\mu\nu}x^\nu + c_{\mu\nu\rho}x^\nu x^\rho \tag{14.8}$$

By plugging Eq. (14.8) onto Eq. (14.4), one finds Eq. (14.9). The coefficients $b_{\mu\nu}$ are therefore proportional to the metric minus an anti-symmetric part.

$$\partial_\mu \varepsilon_\nu + \partial_\nu \varepsilon_\mu = (b_{\mu\nu} + b_{\nu\mu}) + \mathcal{O}(x^\mu) = \left(A + B_\theta x^\theta\right) g_{\mu\nu} \Longrightarrow b_{\mu\nu} + b_{\nu\mu} = A \cdot g_{\mu\nu} \tag{14.9}$$

Translations are represented by a_μ. Rotations and lorenz boosts are represented by $b_{\mu\nu}x^\nu$. The special conformal transformation arises from third contribution, $c_{\mu\nu\rho}x^\nu x^\rho$. By plugging Eq. (14.8) onto Eq. (14.5), one obtains Eq. (14.10). This is the special conformal transformation, and it corresponds to an inversion followed by a translation followed by an inversion.

$$c_{\rho\mu\nu}x^\mu x^\nu = 2x_\rho (B \cdot x) - B_\rho x^2 \tag{14.10}$$

Using Eq. (14.8), the field transforms as $\delta\phi = \varepsilon^\mu \partial_\mu \phi$, and the induced noether current is Eq. (14.11).

$$\mathcal{J}^\mu = \varepsilon_\nu T^{\mu\nu} = \left(2x_\nu (B \cdot x) - B_\nu x^2\right) T^{\mu\nu} = B_\rho \left(2x^\rho x_\nu T^{\mu\nu} - x^2 T^{\mu\rho}\right) \tag{14.11}$$

15 Conformal Theories in 2 Dimensions

15.1 Introduction

Conformal field theories are of great interest in statistical and quantum field theory, because they describe physical theories without scales, where the structure of the model is forced by symmetries alone. Conformal field theories describe the emergence of universality classes and fixed points in the RG flow. In two dimen-

sions, the conformal condition, Eq. (14.1), simplifies drastically, due to properties of holomorphic functions. Conformal theories in two dimensions dispose over infinitely many symmetry generators, because of the factorization into left- and right-handed chiralities. The high level of symmetry in those theories, allows for exact-solvability of many models, as well as the description of non-trivial physics. At higher dimensions, the richness of the theory vanishes, because no decoupling into chiralities occurs.[69, 74—77]

In two dimensions, one obtains an infinite-dimensional algebra corresponding to the generators of symmetry transformations, and, accordingly, an infinite number of conserved quantities due to noether's theorem. Such models can then be solved exactly, by symmetry considerations alone. This occurs, because the infinite-dimensional symmetry algebra leads to very restrictive requirements. Here, exact solvability is meant to be understood, that analytical expressions for the correlation functions corresponding to the model can be obtained, and critical exponents for the model can be obtained explicitly. As already discussed in previous sections, scale-invariance requires that all excitations be massless. This is not problematic, because:[70, 73, 74, 77]

1. many high energy theories, and theories near criticality, are correctly described by massless excitations;
2. starting from a CFT, deformations can be applied to the CFT, in order to create massive theories; and
3. critical points are accordingly described by CFTs, and the critical exponents are universal quantities. QFTs at criticality thus tend to CFTs.

15.2 Formulation of the Theory

15.2.1 Holomorphic Basis

One considers an euclidean space with metric given by Eq. (15.1). The formulation can be easly changed to Minkowsky by simpky considering a Wick rotation of the euclidean metric in the complex plane.[70]

$$g_{\mu\nu} = diag\begin{pmatrix} 1 & 1 \end{pmatrix} \tag{15.1}$$

In $d = 2$, one decomposes the basis onto a holomorphic and anti-holomorphic parts, Eqs. (15.2, 15.3).

$$z = x^0 + ix^1 \ ; \qquad \partial = \partial^0 + i\partial^1 \tag{15.2}$$

$$\overline{z} = x^0 - ix^1 \ ; \qquad \overline{\partial} = \partial^0 - i\partial^1 \tag{15.3}$$

The jacobian of the transformation is given by Eq. (15.4).

$$J^\mu_\nu = \frac{1}{2}\begin{pmatrix} \frac{\partial x^0}{\partial z} & \frac{\partial x^0}{\partial \overline{z}} \\ \frac{\partial x^1}{\partial z} & \frac{\partial x^1}{\partial \overline{z}} \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix} \tag{15.4}$$

Correspondingly, the metric and its inverse are given by Eq. (15.5) below. The metric becomes off-diagonal and multiplied by a factor of two. The raising and lowering of indicies changes the variable from holomorphic to anti-holomorphic (or vice-versa), in addition to introducing additional multiplicative factors.

$$g'_{\mu\nu} = J^\alpha_\mu J^\beta_\nu g'_{\alpha\beta} = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \ ; \qquad g'^{\mu\nu} = 2\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{15.5}$$

15.2.2 Holomorphic Noether Theorem

The conserved current condition, is modified to Eq. (15.6).

$$\partial^\mu j_\mu = \partial j_z + \overline{\partial} j_{\overline{z}} = g_{\mu\nu}\partial^\mu j^\nu = \frac{1}{2}\left(\partial j^{\overline{z}} + \overline{\partial} j^z\right) \tag{15.6}$$

This condition shows that, let $f(z)$ be a holomorphic function (or, equivalently, let $\overline{f}(\overline{z})$ an anti-holomorphic function), then by choosing any such function, and writing any holomorphic (or anti-holomorphic) current, Eq. (15.7), then the condition the condition $\partial j_z = 0$ is immediately satisfied, since $\partial j_z = \overline{\partial j_{\overline{z}}} = 0$. This leads to infinitely many conserved currents and symmetry generators.[70, 75]

$$j_z = \begin{pmatrix} f(z) & 0 \end{pmatrix} \; ; \qquad \overline{j}_{\overline{z}} = \begin{pmatrix} 0 & f(\overline{z}) \end{pmatrix} \tag{15.7}$$

This property is seen more explicitly, by combining the theorems of Green and Cauchy. First, consider an infinitesimal motion dr along a tangent direction t^μ, and consider a normal vector to this motion, n_μ, Eq. (15.8), where $\hat{z}$ is the unit vector in the z-direction.

$$n_\mu = \varepsilon_{\mu\nu\rho}\hat{z}^\nu dr^\rho \; ; \qquad t^\mu = \frac{dx^\mu}{ds} \tag{15.8}$$

Due to Stokes' theorem, one re-writes the divergence of some function over a closed surface, as a line integral over the contour of such surface. On the other hand, Cauchy's theorem states that the line integral of a (anti-)holomorphic function over a simply-connected region, is path independent and therefore vanishes. Since, in the construction above, one considers a (anti-)holomorphic j_μ defined in a simply connected region, those theorems automatically lead to a conserved current, Eq. (15.9). This is the first strong property of CFTs in $d = 2$, which is permitted by complex analysis.[28, 74, 75]

$$\int_\Sigma \partial_\mu j^\mu dz d\overline{z} = \oint_\gamma dn_\mu j^\mu = \oint_\gamma \left(j^{\overline{z}} dz - j^z d\overline{z}\right) = 2\oint_\gamma \left(j_z dz - j_{\overline{z}} d\overline{z}\right) = 0 \tag{15.9}$$

The conserved charge associated with symmetries is furthermore promoted to a much stronger condition in two-dimensions, due to the new properties introduced by complex analysis. First, any (anti-)holomorphic function can be expressed in terms of a series of Laureant, Eq. (15.10). By using this property, one decomposes those infinitely-many conserved currents and charges onto a representation on a complete basis, where an infinitesimal transformation along some "eigen-direction" n is described Eq. (15.11).[28, 70, 85]

$$f(z) = \sum_{n=-\infty}^{+\infty} \alpha_n z^{-n-1} \; ; \qquad \alpha_n = \oint f(z)\, z^{n+1} dz \tag{15.10}$$

$$z \to z + \varepsilon_n(z) \; ; \qquad \varepsilon_n(z) = z^{n+1} \tag{15.11}$$

Furthermore, again due to Cauchy, the analytic continuation allows time-slice integrals to be replaced by contours. The charge conservation statement is therefore promoted to a much stronger contition, Eq. (15.12).

$$Q = \oint_{\gamma} j_z dz \tag{15.12}$$

Poisson brackets translate naturally to the language of complex analysis, Eq. (15.13).[70]

$$\{\phi(z), \partial\phi(w)\} \sim \frac{1}{z-w} \; ; \qquad \oint_{\gamma} \frac{f(z)}{z-w} dz = f(w) \tag{15.13}$$

Other correlation functions can then be calculated, by integrating or differentiating against Eq. (15.13) on both sides, for example, Eq. (15.14).

$$\{\partial\phi(z), \partial\phi(w)\} \sim \frac{1}{(z-w)^2} \; ; \qquad \{\phi(z), \phi(w)\} \sim log\,|z-w|^2 \tag{15.14}$$

Although those are, so far, not operators, as the theory has not yet been quantized, the Laureant Series will later be promoted into a so-called "Operator Power Expansion" (OPE), where the coefficients $\hat{\alpha}_n$ are interpreted as operators. Although operator-valued distributions evaluated at the same point are ill-defined and lead to divergencies, the OPE allows these divergencies to be absorbed by the power expansion, whilst leaving the operators $\hat{\alpha}_n$ well-behaved. One already observes this property here: the laureant series for $\partial\phi(z)$ is given by Eq. (15.15), where the divergence $\partial\phi \sim \frac{1}{z-w}$ is absorbed by the $n = 0$ term. α_n thus becomes regular, and the divergency is mediated instead by the power series.[78, 80, 85, 86]

$$\partial\phi(z-w) = \sum_{n=-\infty}^{+\infty} \alpha_n \cdot (z-w)^{-n-1} \tag{15.15}$$

15.2.3 Witt Algebra

The square of Eq. (15.15) is given by Eq. (15.16).

$$(\partial\phi(z-w))^2 = \sum_{j=-\infty}^{+\infty} \sum_{k=-\infty}^{+\infty} \alpha_j \alpha_k \cdot (z-w)^{-j-k-2} = \sum_{n-k=-\infty}^{+\infty} \sum_{k=-\infty}^{+\infty} \alpha_{n-k} \alpha_k \cdot (z-w)^{-n-2} = \sum_{n=-\infty}^{+\infty} L_n \cdot (z-w)^{-n-2} \tag{15.16}$$

In the last line inequality, one defines Eq. (15.17) and replaces the sum over $n-k \to n$, since it spans an infinite domain. The n-pole-structure of these formulas is made explicity by the laureant series.

$$L_n = \sum_{k=-\infty}^{+\infty} \alpha_{n-k} \alpha_k \tag{15.17}$$

An alternative way of showing this formula explicitly is by looking directly at the coefficients, Eq. (15.18).

$$L_n = \oint (\partial\phi(z))^2 z^{n+1} dz = \sum_{j,k=-\infty}^{+\infty} \alpha_j \alpha_k \oint z^{n-j-k-1} dz = \sum_{j,k=-\infty}^{+\infty} \alpha_j \alpha_k \delta(n-j-k) = \sum_{j=-\infty}^{+\infty} \alpha_j \alpha_{n-j} \tag{15.18}$$

Furthermore, by integrating Eq. (15.15), one finds Eq. (15.19), where q comes from the constant of

integration, and $p = a_0$ comes from the $n = 0$ term. The emergence of the logarithmic term in $\phi(z-w)$ critically means that ϕ is not a local operator, and, therefore, it cannot be a fundamental field of the theory, nor is it a well-defined operator in the hilbert space. The emergence of the logarithmic term is directly connected to the Mermin-Wagner theorem: the logarithmic divergence in the infrared means that no order parameter can exist, and no spontaneous symmetry breaking of a continuous symmetry occurs.[70, 75]

$$\phi(z-w) = q + p \cdot log(z-w) - \sum_{n=-\infty}^{+\infty} \frac{\alpha_n}{n} (z-w)^{-n} \tag{15.19}$$

The poisson brackets of those terms appear naturally, Eqs. (15.20, 15.21), as well as the commutation relation for $\{L_m, L_n\}$, which is called the Witt Algebra, Eq. (15.22).[al]

$$\{\alpha_m, \alpha_n\} = m\delta(m+n) \tag{15.20}$$

$$\{L_m, \alpha_n\} = -n\alpha_{m+n} \tag{15.21}$$

$$\{L_m, L_n\} = (m-n) L_{m+n} \tag{15.22}$$

This mathematical formulation connects directly to physics. The Witt Algebra describes the classical symmetry generators representing infinitesimal conformal transformations on classical fields. Furthermore, the generators $\{L_{-1}, L_0, L_1\}$ form a closed sub-algebra by itself, which is called the $SL(2,\mathbb{C})$ group, and corresponds to the global conformal group. Those three generators correspond to global transformations, namely the translations, dilations, and the special conformal transformation. The other generators L_m correspond to local transformations that exist only in $d = 2$, and that are only defined locally. The existence of those infinitely-many symmetry generators means, furthermore, that the theory is constraint. The generators corresponding to anti-holomorphic transformations can be defined in a similar way, and it is easy to see that the holomorphic and anti-holomorphic parts commute.[72, 74, 75, 77]

15.2.4 Equations of Motion and Conserved Charges

The lagrangian needs to be massless to describe a conformal-invariant theory. In terms of the holomorphic and anti-holomorphic basis, the lagrangian and equations of motion take the form, Eq. (15.23).

$$\mathcal{L} = \partial\phi\bar{\partial}\phi \; ; \qquad \partial\bar{\partial}\phi = 0 \tag{15.23}$$

[al] Derived more explicitly,

$$\{\alpha_m, \alpha_n\} = \oint \{\partial\phi(z), \partial\phi(w)\} z^{m+1} w^{n+1} dz dw = \oint \frac{z^m w^n}{(z-w)^2} dz dw = m \oint z^{m+n-1} dz = m\delta(m+n)$$

$$\{L_m, \alpha_n\} = \sum_{j=-\infty}^{+\infty} \{\alpha_j \alpha_{m-j}, \alpha_n\} = \sum_{j=-\infty}^{+\infty} (\alpha_j \{\alpha_{m-j}, \alpha_n\} + \{\alpha_j, \alpha_n\} \alpha_{m-j}) = -n\alpha_{m+n}$$

$$\{L_m, L_n\} = \sum_{j=-\infty}^{+\infty} \{L_m, \alpha_j \alpha_{n-j}\} = \sum_{j=-\infty}^{+\infty} (\alpha_j \{L_m, \alpha_{n-j}\} + \{L_m, \alpha_j\} \alpha_{n-j}) = \sum_{j=-\infty}^{+\infty} ((-n+j) \cdot \alpha_j \alpha_{m+n-j} - j \cdot \alpha_{m+j} \alpha_{n-j})$$

$$= \sum_{j=-\infty}^{+\infty} ((-n+j) \cdot \alpha_j \alpha_{m+n-j} + (-j+m) \cdot \alpha_j \alpha_{m+n-j}) = (m-n) \sum_{j=-\infty}^{+\infty} \alpha_j \alpha_{m+n-j} = (m-n) L_{m+n}$$

One easily verifies the correspondence, by replacing the equations of motion, Eq. (15.23), onto the green function, Eq. (15.13). One obtains Eq. (15.24).

$$\partial\overline{\partial}G\left(z-w\right)\sim\partial\overline{\partial}log\left|z-w\right|^{2}=\partial\overline{\partial}log\left(\left(z-w\right)\left(\overline{z}-\overline{w}\right)\right)=\frac{1}{z-w}+\frac{1}{\overline{z}-\overline{w}}=\delta\left(z-w\right) \tag{15.24}$$

Suppose a transformation, Eq. (15.11). The variation on the lagrangian is Eq. (15.25).

$$\delta\mathcal{L}=\frac{\delta\mathcal{L}}{\delta\left(\partial\phi\right)}\partial\left(\delta\phi\right)+\frac{\delta\mathcal{L}}{\delta\left(\overline{\partial}\phi\right)}\overline{\partial}\left(\delta\phi\right)+\frac{\delta\mathcal{L}}{\delta\phi}\delta\phi=\partial\left(\varepsilon_{n}\left(z\right)\partial\phi\overline{\partial}\phi\right)+\overline{\partial}\left(\varepsilon_{n}\left(z\right)\left(\partial\phi\right)^{2}\right) \tag{15.25}$$

One obtains the noether current, Eq. (15.26).

$$j_{z}=\varepsilon_{n}\left(z\right)\left(\partial\phi\right)^{2}=\varepsilon_{n}\left(z\right)T_{zz}\ ;\qquad j_{\overline{z}}=\varepsilon_{n}\left(z\right)\left(\partial\phi\overline{\partial}\phi\right)=\varepsilon_{n}\left(z\right)T_{z\overline{z}} \tag{15.26}$$

Here, one defines the energy-momentum tensor with components, Eq. (15.27), where one sets $T_{z\overline{z}}=0$, because conformal theories require the tracelessness of the energy-momentum tensor.

$$T_{zz}=\left(\partial\phi\right)^{2}\ ;\qquad T_{z\overline{z}}=T_{\overline{z}z}=\left(\partial\phi\overline{\partial}\phi\right)=0\ ;\qquad T_{\overline{z}\overline{z}}=0 \tag{15.27}$$

The conserved charges are thus given by Eq. (15.28), which correspond to Eq. (15.18).

$$L_{m}=\oint\varepsilon_{m}\left(z\right)T_{zz}dz=\oint z^{m+1}\left(\partial\phi\right)^{2}\left(z\right) \tag{15.28}$$

Indeed, those conserved charges induce infinitesimal symmetry transformations to the field, Eq. (15.29). The Witt Algebra can therefore be independently derived from the Jacobi Identity.[am]

$$\left\{L_{m},\phi\right\}=\frac{1}{2}\oint_{\gamma}\varepsilon_{m}\left(z\right)\left\{\left(\partial\phi\right)^{2},\phi\right\}dz=\oint_{\gamma}\frac{\varepsilon_{m}\left(z\right)\partial\phi\left(z\right)}{z-w}dz=\varepsilon_{m}\left(w\right)\partial\phi\left(w\right) \tag{15.29}$$

15.3 Local and Global Symmetry Generators

The generators $\{L_{-1},L_{0},L_{1}\}$ form a closed sub-algebra by itself, which is called the $SL\left(2,\mathbb{C}\right)$ group, and corresponds to global transformations of translations, dilations, and the special conformal transformation. On the other hand, the generators L_{m} correspond to local transformations that only exist in $d=2$, and that only form algebras, but not groups; since a group requires the mappings to be invertible everywhere, and this requirement is not satisfied for general L_{m}. Those statements are shown here explicitly:[72, 74, 75, 77]

- First, one requires that the conformal transformation is well-defined at infinity. By considering a

[am] Specifically,

$$\left\{L_{m},\left\{L_{n},\phi\right\}\right\}+\left\{\phi,\left\{L_{m},L_{n}\right\}\right\}+\left\{L_{n},\left\{\phi,L_{m}\right\}\right\}=\left\{L_{m},\left\{L_{n},\phi\right\}\right\}-\left\{\left\{L_{m},L_{n}\right\},\phi\right\}-\left\{L_{n},\left\{L_{m},\phi\right\}\right\}=0$$
$$\left\{\left\{L_{m},L_{n}\right\},\phi\right\}=\left\{L_{m},\left\{L_{n},\phi\right\}\right\}-\left\{L_{n},\left\{L_{m},\phi\right\}\right\}=\left[-\left(n+1\right)+\left(m+1\right)\right]z^{n+m+1}\partial\phi+\left(1-1\right)z^{n+m+2}\partial^{2}\phi$$

Which implies

$$\left\{L_{m},L_{n}\right\}=\left(m-n\right)L_{m+n}$$

change of coordinates , Eq. (15.30), this requirement is satisfied.

$$w = \frac{1}{z} \; ; \qquad \partial_z = -w^2 \partial_w \tag{15.30}$$

- Next, one considers an infinitesimal conformal transformation given by Eq. (15.31).

$$v\left(z\right)\partial_z = -w^2 v\left(\frac{1}{w}\right)\partial_w \tag{15.31}$$

- By construction, $v\left(z\right)$ is holomorphic, such that Eq. (15.31) can be written as Eq. (15.32).

$$w^2 v\left(\frac{1}{w}\right) = \sum_n \alpha_n w^{2-n} \tag{15.32}$$

- Regularity of this expression requires $n \leq 2$, and, furthermore, $n \geq 0$ is required in order for $v\left(z\right)$ itself to be regular. Therefore Eq. (15.33) is the only transformation for which the expression is also regular at zero. Since $L_m = z^{m+1}\partial_z$, one easily associates the coefficients $\alpha_0, \alpha_1, \alpha_2$ with the generators L_{-1}, L_0, L_1 respectively, which conlcudes the proof.

$$v\left(z\right) = \alpha_0 + \alpha_1 z + \alpha_2 z^2 \tag{15.33}$$

16 Quantization

16.1 Introduction

The next step is quantization. Formally, quantization consists of introducing a central charge to the Witt Algebra; that is, Witt Algebra is extended by a central element that commutes with all elements of the algebra. It is thereby promoted to a Virasoro Algebra. The Virasoro Algebra is therefore called the central extension of the Witt Algebra, and ordinary representations of the Virasoro Algebra lead to projective representations of the Witt Algebra. This can be understood as follows: some symmetries belonging to a group, $g, h \in G$, act on a hilbert space by means of a unitary matrix $U_g = U_g^\dagger$. Naively, one would expect that the simultaneous action by two symmetries is equal to the composite action, Eq. (16.1).[72, 74]

$$U_g U_h |\psi\rangle = U_{gh}|\psi\rangle \tag{16.1}$$

In reality, the condition is more laxed, and these operations can differ by a phase factor $c\left(g,h\right) \in U\left(1\right)$, Eq. (16.2). U_g is therefore called a projective representation of g.

$$U_g U_h |\psi\rangle = c\left(g,h\right) U_{gh}|\psi\rangle \tag{16.2}$$

The central extension of G, $\tilde{G} \cong G \times U\left(1\right)$, is such that $\tilde{G}$ acts linearly on the hilbert space. In other words, projective representations of G correspond to linear representations of the central extension $\tilde{G}$. Anomalies are a failure of the classical symmetry to lift to a true linear representation of the Hilbert space after quantization. This can be seen in analogy with the way spins are defined in Quantum Electrodynamics: classical angular momenta has $SO\left(3\right)$ symmetry; and quantization consists of implementing the central extension of $SO\left(3\right)$, $SU\left(2\right)/\mathbf{1}_{2\times 2} \simeq SO\left(3\right)$. This discussion is easily extended to lie algebras. The introduction of the Virasoro Algebra leads thereto, that the structure of the conformal symmetry is not preserved after quantization — conformal symmetry is, therefore, an anomalous symmetry.[70, 75]

16.2 Formulation

Quantization is implemented by requiring normal ordering, Eq. (16.3).[70, 72]

$$\alpha_m \alpha_n := \begin{cases} \alpha_m \alpha_n & n \geq m \\ \alpha_n \alpha_m & n < m \end{cases} ; \qquad [\alpha_m, \alpha_n] = m\delta\left(m+n\right) \tag{16.3}$$

The commutation relations are modified by the quantization, such that a central charge c is introduced, Eqs. (16.4, 16.4). The central charge vanishes for $m = \{-1, 0, +1\}$, that is, for Möbius transformations.[86],[an]

$$[L_m, \alpha_n] = -n\alpha_{m+n} \tag{16.4}$$

$$[L_m, L_n] = (m-n)\, L_{m+n} + \frac{c}{12} m \left(m^2 - 1\right) \delta_{m+n} \tag{16.5}$$

The central charge has significant implications to the theory:[70, 71, 75, 76]

1. CFTs are described by local operators whose derivatives are also local. Those operators are called fields. one distinguishes those fields onto primary, semi-primary, and secondary fields. Primary fields are those that transform covariantly under any conformal transformation L_m, both local and global; semi-primary fields are those that transform covariantly under global transformations exclusively, L_m , $m = \{-1, 0, +1\}$. Finally, secondary fields are all the others fields, that don't transform so trivially, and, instead, dispose of complicated transformation laws.

2. Primary fields are of particular interest, because, due to their transformation laws, they correspond to the conventoinal fields of QFT. Primary fields, furthermore, fix the structure of the theory completely. They are the building-blocks of CFTs. Each primary field has an associated (holomorphic and anti-holomorphic) conformal weight associated to it. Conformal weights quantifies the covariant transformation of primary fields under rotations and scaling, i.e., the conformal weight is associated with spin and scaling eigenvalues. The main object of interest in CFTs, is the spectrum of conformal weights that specify the primary fields of the theory. Primary fields, furthermore, play a role in uniquely specifying the correlation functions, scaling laws, and critical exponents of a given CFT.

17 Construction of a Hilbert Space

17.1 Primary and Descendant Fields

A lagrangian formulation is not even guaranteed in a CFT. Theory is formulated in terms of correlation functions and the S-matrix, by fixing those objects with respect to conformal symmetries. The high degree of symmetry, especially in $d = 2$, heavily constrains the possible correlation functions that can be defined for such a theory. One assumes, therefore, exclusively the existence of a Hilbert space that local fields $\phi\left(z, \overline{z}\right)$ can be associated with, as well as other conventional assumptions, such as the existence of an invariant vacuum, causality, etc. Here, those local fields $\phi\left(z, \overline{z}\right)$ can in principle be any operator, and one must constrain the set of operators that are actually allowed to belong to the Hilbert space.[70, 73, 74, 85, 86]

[an] *See* Appendix 26.1 for detailed developments of this section .

Since, in $d = 2$, conformal transformations factorize into holomorphic and anti-holomorphic parts, an infinite-dimensional algebra is obtained, and, therefore, operators are expected to be highly constrained. Primary fields are thus defined, such that they transform contravariantly subject to a conformal transformation, and they are (accordingly) elimianted by all positive Virasoro generators, such that they define the highest-weight states of the theory. Primary fields are, therefore, the irreductible building-blocks that define conformal families. All descendant states of a family can be found by acting with symmetry generators on this highest-weight state, which in turn is specified by a primary field. Conformal families are then be perfectly defined by the primary states plus conformal symmetries alone.[70, 74, 75, 76]

Primary fields transform tensor-like subject to a conformal operation. Primary fields are defined by Eq. (17.1), where the conformal transformation is given by holomorphic and anti-holomorphic functions $f(z)$ and $\overline{f}(\overline{z})$, and one defines h and $\overline{h}$ as the holomorphic and anti-holomorphic conformal weights.[75]

$$\phi'\left(z', \overline{z'}\right) = \left(\frac{\partial f(z')}{\partial z}\right)^{-h} \left(\frac{\partial \overline{f}\left(\overline{z'}\right)}{\partial \overline{z}}\right)^{-h} \phi(z, \overline{z}) \tag{17.1}$$

The conformal weights are the eigenvalues quantifying how a state transforms under conformal symmetries. Consider, for example, dilations and rotations described by Eqs. (17.2,17.3).

$$z \to \lambda z \ ; \qquad \overline{z} \to \lambda \overline{z} \tag{17.2}$$

$$\delta z = i\theta z \ ; \qquad \delta \overline{z} = i\theta \overline{z} \tag{17.3}$$

The primary field, Eq. (17.1), transforms as Eq. (17.4). Conformal weights thus associate naturally to the scaling dimension and the spin, by identifying $\left(h \pm \overline{h}\right)$ correspondingly.

$$\phi'\left(z', \overline{z'}\right) = \lambda^{-\left(h+\overline{h}\right)} \phi(z, \overline{z}) \ ; \qquad \phi'\left(z', \overline{z'}\right) = \theta^{-\left(h-\overline{h}\right)} \phi(z, \overline{z}) \tag{17.4}$$

17.2 State-Operator Equivalence (Insertions in Vacuum)

One considers the infinitesimal conformal transformations, Eq. (17.5).

$$f(z) = z + \varepsilon(z) \ ; \qquad \overline{f}(\overline{z}) = \overline{z} + \varepsilon(\overline{z}) \tag{17.5}$$

The primary field, Eq. (17.1), will transform as Eq. (17.6), where the symbol "$ah.$" denotes the anti-holomorphic part, which is identical up to the overline symbol.[70, 75]

$$\phi'\left(z', \overline{z'}\right) = \left(\frac{\partial f(z')}{\partial z}\right)^{-h} \left(\frac{\partial \overline{f}\left(\overline{z'}\right)}{\partial \overline{z}}\right)^{-h} \phi(z, \overline{z}) = \left(1 + h\partial_z \varepsilon(z) + \varepsilon(z)\partial_z + ah.\right)\phi(z, \overline{z}) \tag{17.6}$$

In order to create states in the Hilbert space, one specifies, first, a change of coordinates, Eq. (17.7), where x^0 is the time dimension. By realizing this transformation, one observes that time-translations correspond to dilations in this radial frame, whereas space-translations correspond to rotations.[75]

$$w = e^z = exp\left(x^0 + ix^1\right) \tag{17.7}$$

In QFT, "in"-states are created by inserting operations on the vacuum at $t \to -\infty$. In the radial quantization, this is equivalent to setting $w \to 0$. Furthermore, the time-ordering of correlation functions is replaced by a radial ordering. A primary field is inserted to the vacuum by means of Eq. (17.8). Similarly, "out"-states are created by taking the complex conjugate, where the factor $w^{-2\overline{h}}\overline{w}^{-2h}$ comes from definition of primary fields, Eq. (17.1).[29, 70, 72]

$$|\phi\rangle = \lim_{w,\overline{w}\to 0} \phi\left(w,\overline{w}\right)|0\rangle \; ; \qquad \langle\phi| = \lim_{w,\overline{w}\to\infty} \langle 0|\phi^\dagger\left(\frac{1}{\overline{w}},\frac{1}{w}\right) w^{-2\overline{h}}\overline{w}^{-2h} \tag{17.8}$$

The state is guaranteed to have a finite norm, since the correlation function $\left\langle\phi^\dagger\phi\right\rangle$ will depend polynomially on $w^{-2\overline{h}}\overline{w}^{-2h}$, such that the terms inside the limit in Eq. (17.9) become independent of $w, \overline{w}$, and are thus guaranteed to converge in the limit $w, \overline{w} \to \infty$.[70, 74]

$$\langle\phi|\phi\rangle = \lim_{w,\overline{w}\to\infty} w^{-2\overline{h}}\overline{w}^{-2h}\langle 0|\phi^\dagger\left(\frac{1}{\overline{w}},\frac{1}{w}\right)\phi\left(w,\overline{w}\right)|0\rangle = \lim_{w,\overline{w}\to\infty} w^{-2\overline{h}}\overline{w}^{-2h}\left\langle\phi^\dagger\left(\frac{1}{\overline{w}},\frac{1}{w}\right)\phi\left(w,\overline{w}\right)\right\rangle \tag{17.9}$$

17.3 Heighest-Weight Representation

The generators satisfy Eq. (17.10).[ao]

$$L_n^\dagger = L_{-n} \tag{17.10}$$

The action of symmetry generators on the state is given by Eq. (17.11).

$$\left[L_m\left(z\right),\phi\left(w\right)\right] = \left(1 + h\left(m+1\right)z^m + z^{m+1}\partial_z + ah.\right)\phi\left(z,\overline{z}\right) \tag{17.11}$$

For $m > 0$, when $z = 0$, the expression is equal to zero, therefore, $L_{m>0}|\phi\rangle = 0$. Meanwhile, for $m = 0$, the second term is eliminated, but the first term becomes $L_0|\phi\rangle = h|\phi\rangle$, or, as it is conventionally written, $L_0|h\rangle = h|h\rangle$. Finally, for the case of $m < 0$, one considers the action of L_{-m}, $m > 0$ on $|h\rangle$, Eq. (17.12).

$$L_{-m}|h\rangle = L_{-m}L_0|h\rangle + \left[L_0, L_{-m}\right]|h\rangle = \left(h+m\right)L_{-m}|h\rangle \tag{17.12}$$

Similarly, given two $m > n > 0$, obtains Eq. (17.13). This shows that L_n, $n > 0$ acts as a lowering operator that either lowers the conformal weight of a state, or eliminates the ground state. If one had, instead, $n > m > 0$, then, in the last inequality, $(n-m) > 0$ would lead to $L_{n-m}|h\rangle = 0$, and the state would be completely eliminated.

$$L_nL_{-m}|h\rangle = L_{-m}L_n|h\rangle + \left[L_n, L_{-m}\right]|h\rangle = \left(m-n\right)L_{n-m}|h\rangle \tag{17.13}$$

The equivalent analysis can be carried with the anti-holomorphic symmetry generators. The aforeout-

[ao] This property is obtain from the following algebraic manipulation,

$$T^\dagger\left(z\right) = \sum L_n^\dagger\overline{z}^{-n-2} = \overline{z}^{-4}T\left(\frac{1}{\overline{z}}\right) = \overline{z}^{-4}\sum L_n\overline{z}^{n+2} = \sum L_n\overline{z}^{n-2} = \sum L_{-n}\overline{z}^{-n-2}$$

carried analysis allows for the assertion of the following statements:[74, 75]

1. The operator L_0 specifies the highest-weight state of the Hilbert space. This corresponds to the ground state. The action of operators L_{-m}, $m > 0$ correspond to raising operators, whose effect is to raise the conformal weight of the state. Conversely, the action of operators L_m, $m > 0$ correspond to lowering operators, and, specifically L_m $m > 0$ acting on $|h\rangle$ annihilates the ground state.

2. The considerations above are sufficient to show that primary fields, in addition to the symmetry generators L_m, are sufficient to specify the Hilbert space. There is an correspondence between states and primary and descendant operators — the state-operator equivalence. This procedure allows for the specification of the Hilbert space, without the need for defining lagrangians or hamiltonians.

3. The vacuum is defined by composition the holomorphic and anti-holomorphic parts, $|0\rangle \otimes |\bar{0}\rangle$. In the analysis carried-out above, anti-holomorphic portions were largely omitted, as all the considerations are made equivalently. On the one hand, the vacuum must be invariant under the global transformations, $L_m|0\rangle = 0$, $m = -1, 0, 1$, and it must also be invariant under L_m, $m > 1$, since L_m act as the lowering operators. On the other hand, the vacuum is not guaranteed to be invariant under local transformations: $L_m|0\rangle \neq 0$ for $m < -1$. This is seen, because the action of L_m, $m < -1$ on the vacuum corresponds — due to the state-operator equivalence — to the insertion of an operator on the vacuum, which in principle is not zero. This is further justified by the following consideration: by considering Eq. (17.14), for all $m \geq -1$, the contour has no singularities and therefore vanishes completely. On the other hand, for $m < -1$, the expression leads to the formation of poles.

$$L_m|0\rangle = \left(\oint \frac{dz}{2\pi i} z^{m+1} T(z)\right)|0\rangle \tag{17.14}$$

18 Operator Power Expansions (OPEs)

The algebraic steps in this section are largely subordinated to Appendix Algebra in Section 18. In order to verify whether the stress-energy operator is a primary field or not, one begins by performing the operator power expansions of the few first operators. One obtains Eqs. (18.1, 18.2, 18.3).

$$\partial\phi(z)\,\partial\phi(w) \sim -\frac{1}{(z-w)^2} + \dots \tag{18.1}$$

$$T(z)\,\partial\phi(w) \sim -\frac{\partial\phi(z)}{(w-z)^2} + \dots \tag{18.2}$$

$$T(z)\,T(w) \sim \frac{c/2}{(z-w)^4} + \frac{hT(z)}{(z-w)^2} + \frac{\partial_z T(z)}{z-w} \tag{18.3}$$

From this analysis, one concludes that $T(z)$ cannot be a primary field, unless in the case where $c = 0$. This follows from the fact that primary fields are expected to transform as Eq. (18.4).

$$T(z)\,\mathcal{O}(w) \sim \frac{h\mathcal{O}(w)}{(z-w)^2} + \frac{\partial\mathcal{O}(w)}{z-w} \tag{18.4}$$

An alternative way of seeing this, is to consier the action of a symmetry generator on $T(z)$, Eq. (18.5). The first two terms correspond to the action of a generator on a primary field. The third, anomalous term,

shows that $T(z)$ can only be a primary field when $c = 0$. This simple derivation also shows the effect of the conformal anomaly on the stress-energy tensor.[70, 75]

$$[L_m, T(z)] = 2(m+1) z^m T(z) + z^{m+1} \partial_z T(z) + \frac{c}{12} m \left(m^2 - 1\right) z^{m-2} \tag{18.5}$$

One notices that $T(z)$ is still invariant under the global transformations $L_{\pm 1}$ and L_0, since, in those cases, the term proportional to the central charge vanishes. $T(z)$ is thus called a quasi-primary field, since it is anomalous under local transformations, but not under global transformations.[70]

19 Orthogonal States and Gram Matrix

As a next step, one would like to fix the value of the central charge c, and given c, find out the maximum number of descendant fields allowed by the theory. First, one notices that states with different conformal weights $m \neq n$ have zero inner product, and are therefore orthogonal, Eq. (19.1).

$$\langle h | L_{-n}^{\dagger} L_{-m} | h \rangle = (m+n) \langle h | L_{m+n} | h \rangle = 0 \tag{19.1}$$

Furthermore, unitarity requires $\langle \ldots \rangle \geq 0$. By explicitly solving Eq. (19.2) below, where $n > 0$ is assumed, one finds the inner product of states with conformal weight $(n + h)$.

$$\langle h | L_{-n}^{\dagger} L_{-n} | h \rangle = \left(2nh + \frac{c}{12} n \left(n^2 - 1\right)\right) \langle h | h \rangle \tag{19.2}$$

The action of a generator L_{-n} onto a state increases the confromal weight by n. As n increases, the allowed value for c becomes more constained (if unitarity is a requirement of the theory). at $n \to \infty$, the second term in Eq. (19.2) above, dominates, and, ultimately, one requires $c \geq 0$ for unitarity. Calculating for the first few n, one obtain Eq. (19.3). The additional constraint that becomes explicit, is that unitarity further requires $h \geq 0$.

$$\langle L_1 L_{-1} \rangle = 2h \geq 0 \; ; \qquad \langle L_2 L_{-2} \rangle = 4h + \frac{c}{2} \geq 0 \; ; \qquad \langle L_3 L_{-3} \rangle = 6h + 2c \geq 0 \; ; \qquad \ldots \tag{19.3}$$

States with different conformal weight are orthogonal, therefore, no non-zero correlation functions can be built from them. As a next step, one considers the overlap of states at a fixed level. For example, consider for simplicity the level $n = 2$ with conformal weight $(h + 2)$. The Gram Matrix is built by considering all the possible overlap of states at this fixed level. For $n = 2$, the available overlaps are given by Eq. (19.4) and their corresponding complex conjugates (which have the same inner product.[69, 79, 87]

$$\langle L_2 L_{-2} \rangle = 4h + \frac{c}{2} \; ; \qquad \left\langle L_1^2 L_{-1}^2 \right\rangle = 4h(2h+1) \; ; \qquad \left\langle L_1^2 L_{-2} \right\rangle = 6h \tag{19.4}$$

The requirement that inner products $\langle \ldots \rangle \geq 0$, can be lifted to a stronger requirement, that the determinant of the Gram Matrix (the Kac Determinant) is ≥ 0. One obtains Eq. (19.5).[70, 74, 87]

$$4h(2h+1)\left(4h + \frac{c}{2}\right) - 36h^2 = 4h\left[(2h+1)\left(4h + \frac{c}{2}\right) - 9h\right] \geq 0 \tag{19.5}$$

20 Global Conformal Symmetries and n-Point Correlation Functions

Let $\phi_1(z_1)$ be a field of conformal weight h_1 defined at some point z_1. The two-point correlation function, Eq. (20.1), can be highly constrained by global conformal symmetries alone.[70, 86]

$$G_{12}(z_1, z_2) = \langle \phi_1(z_1)\, \phi_2(z_2) \rangle \tag{20.1}$$

Symmetry unter translations and rotations imply Eqs. (20.2, 20.3). Symmetry under scale invariance means that $\phi \to \lambda^{-1}\phi$ as $z \to \lambda z$, and, therefore, G must have a weight of -2 on the field.

$$G_{12}(z_1, z_2) \to G_{12}(z_1 - z_2) \tag{20.2}$$

$$G_{12}(z_1 - z_2) \to G_{12}\left((z_1 - z_2)^2\right) \tag{20.3}$$

Different conformal weights are orthogonal, therefore, their correlation function vanishes. From symmetry considerations alone, the 2-point correlation function, up to C, is fixed to Eq. (20.4).

$$\langle \phi_1(z_1)\, \phi_2(z_2) \rangle = \frac{\delta_{12} \cdot C}{(z_1 - z_2)^{2h_1} \left(\overline{z_1} - \overline{z_2}\right)^{2\overline{h_1}}} \tag{20.4}$$

The 3-point correlation function is also fixed by symmetries, up to a constant C_{123}, Eq. (20.5).

$$\langle \phi_1(z_1)\, \phi_2(z_2)\, \phi_3(z_3) \rangle = \frac{C_{123}}{(z_1 - z_2)^{h_1+h_2-h_3} (z_1 - z_3)^{h_1+h_3-h_2} (z_2 - z_3)^{h_2+h_3-h_1} \times ah.} \tag{20.5}$$

Here, ϕ_i denotes a field with holomorphic conformal weight h_i and anti-holomorphic $\overline{h_i}$. The expression "$ah.$" in Eq. (20.5), denotes the anti-holomorphic portion, which is identical to the holomorphic part, up to the overline. Invariance under the special conformal transformation (SCT) in Eq. (20.5) fixes the exponents in $(z_i - z_j)^{h_i+h_j-h_k}$ in the denominator. More generally, one would require exponents of the form $(ah_i + bh_j - ch_k)$, with $(a + b + c) = 3$. The SCT restricts the choice to $a = b = c = 1$. Local symmetries, on the other hand, do not fix correlation functions in a simmilar manner. Suppose a local symmetry enforced by a holomorphic function $f(z)$. Fields transform under thys symmetry as Eq. (20.6).

$$\phi_1'(z_1') = U_f \phi_1(z_1)\, U_f^{-1} \tag{20.6}$$

Consequently, one expects Eq. (20.7) to be satisfied.

$$\langle h | \phi_1'(z_1')\, \phi_2'(z_2') \ldots \phi_n'(z_n') | h \rangle = \langle h | U_f \phi_1(z_1)\, U_f^{-1} U_f \phi_2(z_2) \ldots U_f \phi_n(z_n)\, U_f^{-1} | h \rangle \tag{20.7}$$

Specifically, this requires the vacuum to be invariant under the transformation, $U_f^{-1}|h\rangle = |h\rangle$. This is generally not true, as the local transformations L_{-n}, $n > 0$ change the conformal weight of $|h\rangle$, except for the cases of global transformations L_0 and $L_{\pm 1}$. In other words, U_f^{-1} can (infinitesimally) be written as a linear combination of linearly independent local and global conformal transformations. Local transformations spontaneously break the vacuum by giving it a conformal weight different from h, and, as a consequence, they cannot be used to implement further constraints to the correlation function directly. On the other hand, those local symmetry generators are still relevant to constrain the theory in $d = 2$, and, accordingly,

they allow for exact-solvability. This is justified by the fact that locally-defined fields are successible to local symmetries, even if those symmetries are not defined globally.[70, 74, 75]

21 Complete-Definiteness of 2-Dimensional CFTs

Given conformal weights, and given C_{ijk} for the 3-point correlation function; then a $d = 2$ CDF is completely well-defined. Namely, consider a 4-point correlation function. This correlation function can be reduced into a sum of 3-point correlation functions, Eq. (21.1). A similar analysis would allow us (inductively) to calculate the $(n+1)$-point correlation function, given an n-point correlation function. 2- and 3-point correlation functions are thus sufficient to completely specify the theory.[78, 80]

$$G\left(z_1, \ldots, z_4\right) = \sum_j \langle h|\phi_1\left(z_1\right)\phi_2\left(z_2\right)|\phi_j\rangle\langle\phi_j|\phi_3\left(z_3\right)\phi_4\left(z_4\right)|h\rangle = \sum_j \left\langle\phi_1\left(z_1\right)\phi_2\left(z_2\right)\phi_j\left(0\right)\right\rangle\left\langle\phi_j\left(\infty\right)\phi_3\left(z_3\right)\phi_4\left(z_4\right)\right\rangle \tag{21.1}$$

A similar analysis can be used to derive fusion rules between two operators. Consider Eq. (21.2). This is the operator power expansion. In further sections, this will be used to create fusion rules, and to expand products of primary fields as a sums of primary fields multiplied by non-zero three-point coefficients C_{12}^k.

$$\phi_1\left(z_1\right)\times\phi_2\left(z_2\right) = \sum_k \langle k|\phi_1\left(z_1\right)\phi_2\left(z_2\right)|0\rangle\langle 0|k\rangle = \sum_k \langle 0|\phi_k\left(\infty\right)\phi_1\left(z_1\right)\phi_2\left(z_2\right)|0\rangle\langle\phi_k\left(0\right)\rangle = \sum_k C_{12}^k\left(z_1 - z_2\right)\phi_k\left(0\right) \tag{21.2}$$

22 Minimal Models

22.1 Verma Modules

A CFT can be fully specified, by fixing the constant C_{ijk} for the 3-point correlation function, and the holomorphic and anti-holomorphic conformal weights h and $\overline{h}$. In order to construct minimal models, therefore, one begins by fixing the spectrum $\left(h, \overline{h}\right)$ and C_{ijk}, by fixing the theory to a finite number of primary fields (the number of descendant fields, on the other hand, will continue to be infinite). This is accomplished by identifying null vectors in the Hilbert space, and quotienting them out. On the one hand, this procedure will reduce number of physical states in the theory. On the other hand, this leads to strong constraints to the 3 point correlation function. The Verma Module is defined as the vector space spammed by Eq. (22.1), where h is the highest-weight state of $\mathcal{V}_h$.[70, 72—75, 87]

$$\mathcal{V}_h = \left\{L_{-n_i}|h\rangle\ ,\ n_i \geq 0\right\} \tag{22.1}$$

A null vector $\chi \in \mathcal{V}_h$ is a vector that is the highest-weight state itself, $L_n|\chi\rangle = 0$, $n > 0$, and that has zero norm, $\langle\chi|\chi\rangle = 0$ — such that χ is orthogonal to all other states in the verma module. Null vectors of the Verma Module therefore violate unitarity, and they can be understood as gauge artifacts that we must quotiented-out of the Verma Module, in order to create irreducible subspacs of $\mathcal{V}_h$, and thus define physical states. This way, one defines Eq. (22.2), where $\mathcal{W}$ is the sub-spam of the Verma Module containing null

vectors and its descendants.[70]

$$\mathcal{H}_{phys} = \mathcal{V}_h/\mathcal{W} \; ; \qquad \mathcal{W} = \{|\chi\rangle \, : \, L_n|\chi\rangle = 0, n > 0\} \tag{22.2}$$

Since states in the Verma module are not necessarily linearly independent, one creates null vectors by considering linear combinations of states. Null states depend only on h and c. For a generic CFT, each h defines a distinct primary field, and, correspondingly, a distinct verma module $\mathcal{V}_h$. This, in turn, leads to infinitely-many primaries. By restricting the values of h and c to those that create verma modules with null vectors, one obtains minimal model containing a finite number of primary fields. By extension, the theory will decompose into a finite sum of irreductible verma modules, which introduce strong constraints to the operator power expansion. The correlation function is found by solving a differential equation.[74—77]

The null vectors are used as an artifact to constrain the theory to special values of h. Minimal models are created by constraining the conformal weight to special values, whose corresponding Verma module contains null vectors, such that only a finite number of primaries survive in this theory. It is important to emphasize that this rule is not optional: if a theory with a finite number of primary fields is sought, verma modules containing null vectors must be chosen (and those null vectors must subsequently be quotiented-out). Any other choice would lead to infinitely-many primary fields.[70, 75]

One expects eigenstates of L_m to fall into representations of the virasoro algebra, which are the verma modules. Verma modules are infinite-dimensional representations of the virasoro algebra, and their highest weights are given by primary operators. Sometimes, these representations are reductible onto a subspace of $\mathcal{V}$ that are also themselves representations of the virasoro algebra. These subspaces are generated from null vectors $|\chi\rangle$ which are the highest-weight states of the sub-spaces of the verma module, since they are also eliminated by all L_n, $n > 0$. The operator-field correspondence implies that the those null vectors $|\chi\rangle$ also correspond to primary operators themselves, although they descend from $|h\rangle$.[70, 73, 74, 86, 87]

22.2 Gram Matrix

The determinant of the Gram Matrix is called the Kac Determinant. Null vectors emerge when the determinant of the Kac Matrix is zero. One considers the example for $n = 2$ by solving the cubic polynomial in h, Eq. (22.3), the conformal weights that lead to null vectors are found.[70, 73, 87]

$$4h\left((2h+1)\left(4h+\frac{c}{2}\right)-9h\right)=0 \Longrightarrow \begin{cases} h_{1,1} & = 0 \\ h_{1,2} & = \frac{1}{16}\left((c-5)-\sqrt{(1-c)\,(25-c)}\right) \\ h_{2,1} & = \frac{1}{16}\left((c-5)+\sqrt{(1-c)\,(25-c)}\right) \end{cases} \tag{22.3}$$

For exmaple, the null vector composed by the linear combination, Eq. (22.4), must be simultaneously eliminated by L_1 and by L_2 (for all L_n, $n > 2$, it is automatically eliminated due to the laddering).

$$|\chi\rangle = \left(L_{-2} + \xi L_{-1}^2\right)|h\rangle \tag{22.4}$$

Accoridngly, one obtains Eqs. (22.5, 22.6).

$$L_1|\chi\rangle = \left(L_1 L_{-2} + \xi L_1 L_{-1}^2\right)|h\rangle = \left([L_1, L_{-2}] + \xi\left[L_1, L_{-1}^2\right]\right)|h\rangle = (3 + 2\xi + 4h\xi)|h\rangle = 0 \tag{22.5}$$

$$L_2|\chi\rangle = \left(L_2 L_{-2} + \xi L_2 L_{-1}^2\right)|h\rangle = \left([L_2, L_{-2}] + \xi\left[L_2, L_{-1}^2\right]\right)|h\rangle = \left(\frac{c}{2} + 4h + 6h\xi\right)|h\rangle = 0 \tag{22.6}$$

This leads to a system of equations that are to be solved for ξ and h, Eq. (22.7), and which corresponds to the same quadratic, Eq. (22.4). In other words, the condition of zero- Kac-determinant is equivalent to the requirement that $|\chi\rangle$ to be a null vector at two-level. This methodology demonstrates a systematic way of constructing and constraining singular vectors.

$$\xi = -\frac{3}{2}\frac{1}{1+2h}\ ; \qquad (1+2h)\left(\frac{c}{2} + 4h\right) - 9h = 0 \tag{22.7}$$

22.3 Kac Classification

Kac has classified all possible singular vectors at a given level, by demonstrating that that the determinant of the Gram matrix at the N-th level factorizes as Eq. (22.8), where $P(\cdot)$ is the partition number.[87]

$$det\mathcal{M} \sim \prod (h - h_{r,s})^{P(N-rs)}\ ; \quad h_{r,s} = \frac{1}{4}\left((r\alpha_+ + s\alpha_-) - (\alpha_+ + \alpha_-)^2\right)\ ; \quad \alpha_\pm = \frac{\sqrt{1-c} \pm \sqrt{25-c}}{\sqrt{24}} \tag{22.8}$$

One further constraints the possible set of solutions ,by requiring that the ratio between α_+ and α_- be rational, that is, by requiring that $p\alpha_- + p'\alpha_+ = 0$ for two coprimes p, p'. Finally, by noticing $\alpha_+\alpha_- = -1$, one re-writes the conformal weights for the null vectors, as well as the central charge, as Eq. (22.9) below.

$$h_{r,s} = \frac{(pr - p's)^2 - (p - p')^2}{4pp'}\ ; \qquad c = 1 - 6\frac{(p-p')^2}{pp'} \tag{22.9}$$

23 Ward Identities

Most of the algebra in this section is subordinated to Appendix Derivations regarding Section 23. One considers the action of $T(z)$ onto a primary field $\phi_i(z_i)$. For convenience the anti-holomorphic parts are omitted for now, as they can be treated independently. Due to translational invariance, one obtains Eq. (23.1) below, where, due to the operator-state equivalence, one considers $\phi_i(z_i)$ as eigenstates. In the last expression, "$\sim$" denotes that only non-regular terms are kept track of. Only irregular terms are kept track of, because the contour integral will eventually lead regular terms to disappear.

$$T(z)\phi_i(z_i) \sim \left(\frac{h_i}{(z-z_i)^2} + \frac{\partial_{z_i}}{(z-z_i)}\right)\phi(0) \tag{23.1}$$

The first Ward identity is given by Eq. (23.2), where, once again, regular terms are ignored, since the expectation value operator corresponds to the calculation of a countour integral, so that regular terms vanish.

$$\langle T(z)\,\phi_1(z_1)\ldots\phi_n(z_n)\rangle \sim \sum_{i=1}^{n}\left(\frac{h_i}{(z-z_i)^2}+\frac{\partial_{z_i}}{(z-z_i)}\right)\langle\phi_1(z_1)\ldots\phi_n(z_n)\rangle \tag{23.2}$$

Correlation functions with further descendant fields can be calculated similarly, by expressing the field as $L_{-m}\phi_i$, and expanding L_{-m} as a Laureant coefficient, Eq. (23.3).

$$L_{-m}\phi(w) = \oint \frac{dz}{2\pi i}(z-w)^{m-1}\,T(z-w)\,\phi(0) \tag{23.3}$$

One ultimately obtains Eq. (23.4).

$$\langle\phi_1(z_1)\ldots\phi_n(z_n)\,L_{-m}\phi(w)\rangle \sim \sum_{i=1}^{n}\left(\frac{h_i}{(w-z_i)^{-m+2}}+\frac{\partial_{z_i}}{(w-z_i)^{-m+1}}\right) \tag{23.4}$$

24 Fusion Rules and Truncation

Exceptionally when considering an expectation value containing a null vector $\chi(z)$, because null vectors have zero norm (or, correspondingly, because they are orthogonal to all other vectors in the verma module), the inner product will be zero. Expectation values of correlation functions with $\chi(z)$ and its descendants are therefore zero. At two-level, singular vectors are defined when $h = h_{1,2} = h_{2,1}$. For example, let $\phi_{(1,2)}$ and $\phi_{(2,1)}$ be primary fields with respectiv conformal wieghts $h_{1,2}$ and $h_{2,1}$. Any correlation function containing χ equals zero, since χ is a null vector. The identity, Eq. (24.1), thus holds, where the the operator $\mathcal{L}_z = \left(L_{-2}+\xi L_{-1}^2\right)_z$ acts exclusively on $\phi_{(1,2)}(z)$, since this is the only z-dependant term.[69, 73, 70, 75, 87]

$$\langle\chi(z)\,\phi_1(z_1)\ldots\phi_n(z_n)\rangle = \left(L_{-2}+\xi L_{-1}^2\right)\left\langle\phi_{(1,2)}(z)\,\phi_1(z_1)\ldots\phi_n(z_n)\right\rangle = 0 \tag{24.1}$$

Combining this with the Ward identity, Eq. (23.4), one re-writes Eq. (24.1) as Eq. (24.2).

$$\left(\xi\partial_z^2+\sum_{i=1}^{n}\left(\frac{h_i}{(w-z_i)^{-m+2}}+\frac{\partial_{z_i}}{(w-z_i)^{-m+1}}\right)\right)\left\langle\phi_{(1,2)}(z)\,\phi_1(z_1)\ldots\phi_n(z_n)\right\rangle = 0 \tag{24.2}$$

In Appendix Ward Identities on a 3-Point Correlation Function, Eq. (24.2) is specialized to the case of a 3-point correlation function. One obtains the differential equation for the 3-point function, Eq. (24.3).

$$\left(\xi\frac{\partial^2}{\partial z^2}+\frac{1}{z}\frac{\partial}{\partial z}+\frac{h_\alpha}{z^2}\right)G(z,0,\infty) = 0 \tag{24.3}$$

Correspondingly, the solution is given by Eq. (24.4). One finds that the correlation function must obey some power law, where, in principle, one of the solutions may need to be discarted, in order to enforce some physicality condition (although not necessarily). This result is somewhat expected: since CFTs describe physical behaviour near criticality, power laws should be recovered by the CFT approach.

$$G(z,0,\infty) = C_+ z^{h_{r,s+1}-h_{1,2}-h_{r,s}} + C_- z^{h_{r,s-1}-h_{1,2}-h_{r,s}} \tag{24.4}$$

When then expands field products in terms of OPEs, Eq. (24.5).

$$\phi_{(1,2)}(z)\times\phi_{(r,s)}(0)=\sum_k C_{12}^k(z_1-z_2)\,\phi_k(0)=C_+\phi_{(r,s+1)}(0)\,z^{h_{r,s+1}-h_{1,2}-h_{r,s}}+C_-\phi_{(r,s-1)}(0)\,z^{h_{r,s-1}-h_{1,2}-h_{r,s}}+\text{desc.} \tag{24.5}$$

This is often written in a condensed form, Eq. (24.6)

$$\phi_{(1,2)}\times\phi_{(r,s)}\sim\phi_{(r,s+1)}+\phi_{(r,s-1)}\ ; \qquad \phi_{(2,1)}\times\phi_{(r,s)}\sim\phi_{(r+1,s)}+\phi_{(r-1,s)} \tag{24.6}$$

Consistency requires Eq. (24.7) to be satisfied, that is, the fusion gets truncated from below at $r,s\geq 1$. Furthemore, due to Kac formula, $rs\leq N$ where N is the level of the null vector χ, such that the fusion rules are also truncated from above. The model thus disposes over a finite number of primary fields.

$$\phi_{(1,2)}\times\phi_{(2,1)}=\phi_{(2,1)}\times\phi_{(1,2)}\Longrightarrow\phi_{(2,0)}=\phi_{(0,2)}=0 \tag{24.7}$$

25 Concluding Remarks

One summarizes in some detail the theoretical developments of the foregoing sections:[70, 72—77, 82, 85, 86]

- **Motivation**: CFTs describe fixed points in statistical models. QFTs are expected to arise by deforming CFTs. QFTs then tend to CFTs at continuous phase transitions. CFTs in two dimensions are of particular interest, because properties of holomorphic functions generate additional, local symmetries, that lead to exact-solvability of a large class of models. Exact solvabilty means that, in principle, n-point correlation functions can be calculated exactly. The goal of the analysis carried out hereinfore consists thereon, that the required mathematical tools to tackle those models was developed.

- **Conformal Invariance**: Conformal transformations are transformations of the same, that leave angles preserved, although not necessarily distances. In the first step of the foregoing analysis, the algebra that those transformations must satisfy was derived, and, correspondingly, the available (global) conformal transformations in generic d dimensions were classified. Furthermore, the requirement of tracelessness in the energy-momentum tensor was explicitly shown (which corresponds to the requirement of massless excitations for generic conformal transformations).

- **Holomorphicity**: One specializes in CFTs in $d=2$. In this case, the symmetry generators of conformal transformations satisfy the so-called Witt Algebra, and there exists infinitely many symmetry generators. Quantization naturally leads to the emergence of the central charge and of the Virasoro Algebra, as an extension of the Witt Algebra. Local and global transformations distinguish themselves from each other, specifically as a result of the central charge.

- **State-Operator Equivalence**: The construction is explicitly turned into a field theory, by choosing suitable states in a Hilbert Space, and by showing how those states can be explicitly inserted on vacua. A heighest-weight state is defined, as well as the action of symmetry generators on the Hilbert Space. From that, the conformal weight can be used to classify ancestry of states, symmetry generators can be used to generate descendant-states, and OPEs can be used to calculate expectation values of operator products. Furthermore, one shows that conditions, such as unitarity, can be used to constrain the valid ranges of conformal weights and central charges describing a given theory.

- **Minimal Models**: The final step is to choose sub-classes of physical models that might be of particular interest. One natural constraint that might be added, consists thereof, that the model must dispose over a finite number of degrees of freedom (or, in CFT language, a finite number of primary fields). This is accomplished by choosing values of h, c, such that the corresponding Verma module contains null vectors, and is accordingly reducible. In those cases: (1) the Ward Identities can be used in order to derive an exactly-solvable differential equation for the three point correlation function; (2) fusion rules for combining primary fields via operator product expansions can be derived; and (3) fusion rules lead to truncations, such that the number of primary fields that describe the theory are finite.

In Appendix The Ising CFT, a simple numerical computation of the Ising CFT is carried out as an example.

26 Appendix

26.1 Algebra in Section 16

From section 16, the first commutator solves as Eq. (26.1).

$$\begin{aligned}
[L_m, \alpha_n] &= \sum_{j \geq \frac{m}{2}} [\alpha_{m-j}\alpha_j, \alpha_n] + \sum_{j < \frac{m}{2}} [\alpha_j \alpha_{m-j}, \alpha_n] \\
&= \sum_{j \geq \frac{m}{2}} \left(\alpha_{m-j} [\alpha_j, \alpha_n] + [\alpha_{m-j}, \alpha_n] \alpha_j\right) + \sum_{j < \frac{m}{2}} \left(\alpha_j [\alpha_{m-j}, \alpha_n] + [\alpha_j, \alpha_n] \alpha_{m-j}\right) \\
&= \sum_{j \geq \frac{m}{2}} \left(\alpha_{m-j} [\alpha_j, \alpha_n] + [\alpha_{m-j}, \alpha_n] \alpha_j\right) + \sum_{j < \frac{m}{2}} \left(\alpha_j [\alpha_{m-j}, \alpha_n] + [\alpha_j, \alpha_n] \alpha_{m-j}\right) \\
&= \sum_{j \geq \frac{m}{2}} j \left(\alpha_{m-j}\delta_{j+n} + \delta_{m+n-j}\alpha_j\right) + \sum_{j < \frac{m}{2}} j \left(\alpha_j \delta_{m+n-j} + \delta_{j+n}\alpha_{m-j}\right) = -n\alpha_{m+n}
\end{aligned} \tag{26.1}$$

For $m \neq -n$, the second commutator solves as Eq. (26.2).

$$\begin{aligned}
[L_m, L_n] &= \sum_{j \geq \frac{m}{2}} [L_m, \alpha_{n-j}\alpha_j] + \sum_{j < \frac{m}{2}} [L_m, \alpha_j \alpha_{n-j}] \\
&= \sum_{j \geq \frac{m}{2}} \left(\alpha_{n-j} [L_m, \alpha_j] + [L_m, \alpha_{n-j}] \alpha_j\right) + \sum_{j < \frac{m}{2}} \left(\alpha_j [L_m, \alpha_{n-j}] + [L_m, \alpha_j] \alpha_{n-j}\right) \\
&= \sum_{j \geq \frac{m}{2}} \left((-j)\, \alpha_{n-j}\alpha_{m+j} + (-n+j)\, \alpha_{m+n-j}\alpha_j\right) + \sum_{j < \frac{m}{2}} \left((-n+j)\, \alpha_j \alpha_{m+n-j} + (-j)\, \alpha_{m+j}\alpha_{n-j}\right) \\
&= \sum_{j \geq \frac{m}{2}} \left((-j+m) + (-n+j)\right) \alpha_{m+n-j}\alpha_j + \sum_{j < \frac{m}{2}} \left((-n+j) + (-j+m)\right) \alpha_j \alpha_{m+n-j} \\
&= (m-n) \left(\sum_{j \geq \frac{m}{2}} \alpha_{m+n-j}\alpha_j + \sum_{j < \frac{m}{2}} \alpha_j \alpha_{m+n-j} \right) = (m-n) \sum_j :\alpha_{m+n-j}\alpha_j: = (m-n)\, L_{m+n}
\end{aligned} \tag{26.2}$$

For $m = -n$, the last line in Eq. (26.2) develops as Eq. (26.3).

$$(m-n)\left(\sum_{j\geq\frac{m}{2}}\alpha_{m+n-j}\alpha_j+\sum_{j<\frac{m}{2}}\alpha_j\alpha_{m+n-j}\right)=2mL_0+\frac{1}{12}m\left(m^2-1\right)\delta_{m+n} \tag{26.3}$$

The Virasoro Algebra is thus given by Eq. (26.4), for a central charge c.

$$[L_m,L_n]=(m-n)\,L_{m+n}+\frac{c}{12}m\left(m^2-1\right)\delta_{m+n} \tag{26.4}$$

26.2 Algebra in Section 18

The algebraic steps in Section 18 are solved more precisely in this appendix. One verifies whether the stress-energy operator is a primary field or not, by performing the operator power expansions of the few first operators, where the identity, Eq. (26.5), is extensively implemented.

$$\sum_m m\left(\frac{w}{z}\right)^m=\frac{wz}{(w-z)^2} \tag{26.5}$$

Here, one denotes as "..." the regularized part of the operator. For $\partial\phi(z)\,\partial\phi(w)$, one obtains Eq. (26.6).

$$\begin{aligned}\partial\phi(z)\,\partial\phi(w)&=\sum_{m,n}\alpha_m\alpha_n z^{-m-1}w^{-n-1}=\sum_{m,n}:\alpha_m\alpha_n:z^{-m-1}w^{-n-1}-\sum_{m,n}[\alpha_m,\alpha_n]\,z^{-m-1}w^{-n-1}\\&=-\sum_{m,n}m\delta_{m+n}z^{-m-1}w^{-n-1}+\ldots=-\sum_m m\left(\frac{w}{z}\right)^m\frac{1}{zw}+\ldots=-\frac{1}{(z-w)^2}+\ldots\end{aligned} \tag{26.6}$$

Similarly, for $T(z)\,\partial\phi(w)=(\partial\phi(z))^2\,\partial\phi(w)$, one obtains Eq. (26.7).

$$\begin{aligned}T(z)\,\partial\phi(w)&=\sum_{m,n,k}:\alpha_m\alpha_n:\alpha_k z^{-m-n-2}w^{-k-1}=\sum_{m,n,k}:\alpha_m\alpha_n\alpha_k:z^{-m-n-2}w^{-k-1}-\sum_{m,n,k}[\alpha_m\alpha_n,\alpha_k]\,z^{-m-n-2}w^{-k-1}\\&=-\sum_{m,n,k}\left(\alpha_m[\alpha_n,\alpha_k]+[\alpha_m,\alpha_k]\,\alpha_n\right)z^{-m-n-2}w^{-k-1}+\ldots=-\sum_{m,n,k}n\alpha_m\delta_{n+k}z^{-m-n-2}w^{-k-1}+\ldots\\&=-\sum_m\alpha_m z^{-m-2}w^{-1}\left(\sum_n n\left(\frac{w}{z}\right)^n\right)+\ldots=-\frac{1}{(w-z)^2}\sum_m\alpha_m z^{-m-1}+\ldots=-\frac{\partial\phi(z)}{(w-z)^2}+\ldots\end{aligned} \tag{26.7}$$

The expression for $T(z)\,T(w)$ requires a longer calculation. First expand it explicitly, Eq. (26.8).

$$\begin{aligned}T(z)\,T(w)&=\sum_{m,n}L_mL_nz^{-m-2}w^{-n-2}=\sum_{m,n}:L_mL_n:z^{-m-2}w^{-n-2}-\sum_{m,n}[L_m,L_n]\,z^{-m-2}w^{-n-2}\\&=-\sum_{m,n}(m-n)\,L_{m+n}z^{-m-2}w^{-n-2}-\sum_{m,n}\frac{c}{12}\left(m^3-m\right)\delta_{m+n}z^{-m-2}w^{-n-2}+\ldots\\&=-\sum_{m,n}(m-2n)\,L_mz^{-m-2}w^{-2}\left(\frac{z}{w}\right)^n-\sum_m\frac{c}{12}\left(-m+m^3\right)z^{-2}w^{-2}\left(\frac{w}{z}\right)^m+\ldots\end{aligned} \tag{26.8}$$

Furthermore, notice Eq. (26.9).

$$\partial_z T(z) = \sum (-m-2) L_m z^{-m-3} = -\sum m L_m z^{-m-3} - \frac{2}{z}\sum L_m z^{-m-2} = -\sum m L_m z^{-m-3} - \frac{2}{z} T(z) \tag{26.9}$$

Computing the terms individually, the first term in Eq. (26.8) reduces to Eq. (26.10), using Eq. (26.5).

$$C_1 = \sum_{m,n} m L_m z^{-m-2} w^{-2} \left(\frac{z}{w}\right)^n = \frac{z}{w}\frac{1}{w-z}\sum_m m L_m z^{-m-2} = \frac{z}{w}\frac{1}{z-w}\left(\partial_z T(z) + \frac{2}{z} T(z)\right) \tag{26.10}$$

The second term reduces to Eq. (26.11).

$$C_2 = \sum_{m,n} -2n L_m z^{-m-2} w^{-2} \left(\frac{z}{w}\right)^n = \frac{-2}{(z-w)^2}\sum_m L_m z^{-m-1} w^{-1} = \frac{-2}{(z-w)^2}\frac{z}{w}\sum_m L_m z^{-m-2} = \frac{-2}{(z-w)^2}\frac{z}{w} T(z) \tag{26.11}$$

The third term reduces to Eq. (26.12).

$$C_3 = \sum_{m,n} \frac{-c}{12} m z^{-2} w^{-2} \left(\frac{w}{z}\right)^n = \frac{-c}{12} z^{-2} w^{-2} \sum_{m,n} m \left(\frac{w}{z}\right)^m = \frac{-c}{12}\frac{1}{zw}\frac{1}{(z-w)^2} \tag{26.12}$$

The fourth term reduces to Eq. (26.13).

$$C_4 = \sum_m \frac{c}{12} m^3 z^{-2} w^{-2} \left(\frac{w}{z}\right)^m = \frac{c}{12} z^{-2} w^{-2} \sum_m m^3 \left(\frac{w}{z}\right)^m = \frac{c}{12}\frac{1}{wz}\frac{\left(z^2 + 4wz + w^2\right)}{(z-w)^4} \tag{26.13}$$

In the first term, Eq. (26.10), one writes Eq. (26.14).

$$\frac{z}{w} = 1 - \frac{z-w}{w} \tag{26.14}$$

The only contribution of order $O\left(\frac{1}{z-w}\right)$ that survives is Eq. (26.15).

$$C_1 \sim \frac{1}{z-w}\left(\partial_z T(z) + \frac{2}{z} T(z)\right) \tag{26.15}$$

Similar analysis in the term, Eq. (26.11), leads to Eq. (26.16). In the limit $z \to w$, the terms $\frac{2}{z} T(z)$ in the first and second expression therefore cancel. A similar but slightly trickier analysis, leads to the cancellation of the third term, Eq. (26.12), by expanding the forth term, Eq. (26.13).

$$C_2 \sim \frac{-2}{(z-w)^2} T(z) + \frac{-2}{w}\frac{T(z)}{(z-2)} \tag{26.16}$$

Consequently, and by taking $z \to w$, the product $T(z)T(w)$ is written (up to regular poles) as Eq. (26.17), where one replaces $2 \to h$, the conformal weight, to make the solution more general.

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{hT(z)}{(z-w)^2} + \frac{\partial_z T(z)}{z-w} \tag{26.17}$$

From this analysis, one concludes that $T(z)$ cannot be a primary field, unless in the case where $c = 0$; as primary fields are expected to transform as Eq. (26.18).

$$T(z)\,\mathcal{O}(w) \sim \frac{h\mathcal{O}(w)}{(z-w)^2} + \frac{\partial\mathcal{O}(w)}{z-w} \tag{26.18}$$

An alternative way of seeing this, is to consier the action of a symmetry generator on $T(z)$, Eq. (26.19). The first two terms correspond to the action of a generator on a primary field. The third, anomalous term, shows that $T(z)$ can only be a primary field when $c = 0$. This simple derivation also shows the effect of the conformal anomaly on the stress-energy tensor.

$$\begin{aligned}
[L_m, T(z)] &= \sum_n [L_m, L_n]\, z^{-n-2} = \sum_n \left((m-n)\,L_{m+n} + \frac{c}{12} m\left(m^2-1\right)\delta_{m+n}\right) z^{-n-2} \\
&= \sum_n \left(((2m-n)\,L_n)\, z^{m-n-2} - \frac{c}{12} m\left(m^2-1\right)\delta_{m+n} z^{-n-2}\right) \\
&= 2\,(m+1)\, z^m T(z) + z^{m+1}\partial_z T(z) + \frac{c}{12} m\left(m^2-1\right) z^{m-2}
\end{aligned} \tag{26.19}$$

Notice that $T(z)$ is still invariant under the global transformations $L_{\pm 1}$ and L_0, since, in those cases, the term proportional to the central charge vanishes. $T(z)$ is therefore called a quasi-primary field, since it is anomalous under local transformations but not under global transformations.

26.3 Derivations regarding Section 23

The first Ward identity is derived, in more details, as Eq. (26.20).

$$\begin{aligned}
\langle T(z)\,\phi_1(z_1)\ldots\phi_n(z_n)\rangle &= \oint_\gamma z^{m+1} T(z)\,\phi_1(z_1)\ldots\phi_n(z_n)\,\frac{dz}{2\pi i} \\
&= \sum_{i=1}^n \oint_{\gamma_i} z^{m+1} T(z)\,\phi_1(z_1)\ldots\phi_n(z_n)\,\frac{dz}{2\pi i} \\
&= \sum_{i=1}^n \left(\frac{h_i}{(z-z_i)^2} + \frac{\partial_{z_i}}{(z-z_i)}\right)\langle\phi_1(z_1)\ldots\phi_n(z_n)\rangle
\end{aligned} \tag{26.20}$$

The second Ward identity is derived, in more details, as Eq. (26.21).

$$\begin{aligned}
&\langle\phi_1(z_1)\ldots\phi_n(z_n)\,L_{-m}\phi(w)\rangle = \oint \frac{dz}{2\pi i}(z-w)^{m-1}\sum_{i=1}^n\left(\frac{h_i}{(z-z_i)^2} + \frac{\partial_{z_i}}{(z-z_i)}\right)\langle\phi_1(z_1)\ldots\phi_n(z_n)\,\phi(w)\rangle = \\
&\sum_{i=1}^n\sum_{k=0}^{m-1}\begin{pmatrix} m-1 \\ k\end{pmatrix}\oint\frac{dz}{2\pi i}\left(\frac{(w-z_i)^{m-1-k}\,h_i}{(z-z_i)^{2-k}} + \frac{(w-z_i)^{m-1-k}\,\partial_{z_i}}{(z-z_i)^{1-k}}\right) = \sum_{i=1}^n\left(\frac{h_i}{(w-z_i)^{-m+2}} + \frac{\partial_{z_i}}{(w-z_i)^{-m+1}}\right)
\end{aligned} \tag{26.21}$$

In the second line, one first the binomial theorem to express $(z-w)^{m-1}$ in terms of $(w-z_i)$and $(z-z_i)$; and second the identity, Eq. (26.22). This solves the integral and the summation over k.

$$\oint \frac{dz}{2\pi i}(z-z_i)^m = \delta_{m,0} \tag{26.22}$$

26.4 Ward Identities on a 3-Point Correlation Function

The goal is to solve Eq. (26.23) for the particular case of a 3-Point Correlation Function.

$$\left(\xi\partial_z^2+\sum_{i=1}^{n}\left(\frac{h_i}{(w-z_i)^{-m+2}}+\frac{\partial_{z_i}}{(w-z_i)^{-m+1}}\right)\right)\left\langle\phi_{(1,2)}(z)\,\phi_1(z_1)\ldots\phi_n(z_n)\right\rangle=0 \tag{26.23}$$

The correlation function is invariant under a Möbius transformation, Eq. (26.24).

$$z\to f(z)=\frac{z-z_1}{z-z_2} \tag{26.24}$$

The 3-point correlation function can be simplified to Eq. (26.25).

$$G(z,z_1,z_2)=\left\langle\phi_{(1,2)}(z)\,\phi_{(\alpha)}(z_1)\,\phi_{(\beta)}(z_2)\right\rangle=\left\langle\phi_{(1,2)}(z)\,\phi_{(\alpha)}(0)\,\phi_{(\beta)}(\infty)\right\rangle=Cz^{h_\alpha-h_\beta-h_{1,2}}\ ; \tag{26.25}$$

Plugging this onto Eq. (26.23), one obtains Eq. (26.26).

$$G(z,z_1,z_2)\left(\xi\partial_z^2+\sum_{i=1}^{n}\left(\frac{h_i}{(w-z_i)^{-m+2}}+\frac{\partial_{z_i}}{(w-z_i)^{-m+1}}\right)\right)=\left(\xi\frac{\partial^2}{\partial z^2}+\frac{1}{z}\frac{\partial}{\partial z}+\frac{h_\alpha}{z^2}\right)G(z,0,\infty)=0 \tag{26.26}$$

By employing the Möbius transformation, one obtains a pure polynomial in z, and Eq. (26.26) thus reduces to easily to Eq. (26.27).

$$\left(\xi a^2-(\xi-1)\,a+h_\alpha\right)=0\ ;\qquad\qquad a=h_\alpha-h_\beta-h_{1,2} \tag{26.27}$$

By using the conventions correpsonding to the Kac classification, with some labour, one concludes that the only valid solutions to this equation are given by Eq. (26.28).

$$a_\pm=h_{\beta\pm}-h_{1,2}-h_{r,s}\ ;\qquad\qquad h_{\beta\pm}=h_{r,s\pm1} \tag{26.28}$$

Consequently, the correlation function must obey some power law, Eq. (26.29), where in principle one of the solutions may need to be discarted, in order to enforce some physicality condition.

$$G(z,0,\infty)=C_+z^{h_{r,s+1}-h_{1,2}-h_{r,s}}+C_-z^{h_{r,s-1}-h_{1,2}-h_{r,s}} \tag{26.29}$$

26.5 The Ising CFT

The simplest unitary CFT is given by $p=4$ and $p'=2$. The central charge is $c=\frac{1}{2}$. The theory has three fields and conformal weights, Eqs. (26.30, 26.31, 26.32), with respective fusion rules.[75]

$$\mathbb{1} = \phi_{(1,1)} \ , \ \phi_{(3,3)} \ ; \qquad h_{\mathbb{1}} = 0 \ ; \qquad \epsilon \times \epsilon = \mathbb{1} \tag{26.30}$$

$$\sigma = \phi_{(1,2)} \ , \ \phi_{(2,2)} \ ; \qquad h_{\sigma} = \frac{1}{16} \ ; \qquad \sigma \times \epsilon = \sigma \tag{26.31}$$

$$\epsilon = \phi_{(2,1)} \ , \ \phi_{(1,3)} \ ; \qquad h_{\epsilon} = \frac{1}{2} \ ; \qquad \sigma \times \sigma = \mathbb{1} + \epsilon \tag{26.32}$$

Given the fusion rules, one immediately constructs the two-point functions correpsonding to the theory, with help of the OPEs, Eqs. (26.33, 26.34, 26.35). The third term equals zero due to $\mathbb{Z}_2$ symmetry.

$$\begin{aligned}\langle \sigma_1 (z_1, \overline{z_1}) \, \sigma_2 (z_2, \overline{z_2}) \rangle &= C_{\sigma\sigma\mathbb{1}} \, (z_1 - z_2)^{h_{\mathbb{1}} - 2h_\sigma} \, (\overline{z_1} - \overline{z_2})^{h_{\mathbb{1}} - 2h_\sigma} + C_{\sigma\sigma\epsilon} \, (z_1 - z_2)^{h_\epsilon - 2h_\sigma} \, (\overline{z_1} - \overline{z_2})^{h_\epsilon - 2h_\sigma} \\ &= \frac{C_{\sigma\sigma\mathbb{1}}}{|z_1 - z_2|^{\frac{1}{4}}} + \frac{C_{\sigma\sigma\epsilon}}{|z_1 - z_2|^{-\frac{3}{4}}} = \frac{1}{|z_1 - z_2|^{\frac{1}{4}}} \left(C_{\sigma\sigma\mathbb{1}} + C_{\sigma\sigma\epsilon} \, |z_1 - z_2|^2 \right)\end{aligned} \tag{26.33}$$

$$\langle \epsilon_1 (z_1, \overline{z_1}) \, \epsilon_2 (z_2, \overline{z_2}) \rangle = C_{\epsilon\epsilon\mathbb{1}} \, (z_1 - z_2)^{h_{\mathbb{1}} - 2h_\epsilon} \, (\overline{z_1} - \overline{z_2})^{h_{\mathbb{1}} - 2h_\epsilon} = \frac{C_{\epsilon\epsilon\mathbb{1}}}{|z_1 - z_2|^2} \tag{26.34}$$

$$\langle \epsilon_1 (z_1, \overline{z_1}) \, \sigma_2 (z_2, \overline{z_2}) \rangle = 0 \tag{26.35}$$

Conformal blocks are built from the 4-point correlation function, Eq. (26.36), where k are primaries.

$$\begin{aligned}&\langle \phi_1 (z_1) \, \phi_2 (z_2) \, \phi_3 (z_3) \, \phi_4 (z_4) \rangle = \langle \phi_1 (\infty, \infty) \, \phi_2 (1,1) \, \phi_3 (x,x) \, \phi_4 (0,0) \rangle \\ &= \langle \phi_1 | \phi_2 (1,1) \, \phi_3 (x,x) \, | \phi_4 \rangle = \sum_k \langle \phi_1 | \phi_2 (1,1) \, | k \rangle \langle k | \phi_3 (x,x) \, | \phi_4 \rangle = \sum_k C_{12k} C_{k34} F_{34}^{21} (k|x)\end{aligned} \tag{26.36}$$

Here, from the the first to the second inequality, the Möbius transformation was exploited, Eq. (26.37).

$$z \to f(z) = \frac{(z - z_4)(z_2 - z_1)}{(z - z_1)(z_2 - z_4)} \ ; \qquad x = \frac{(z_3 - z_4)(z_2 - z_1)}{(z_3 - z_1)(z_2 - z_4)} \tag{26.37}$$

The conformal block $F_{34}^{21} (k|x)$ can be affixated by symmetry alone: if one replaces, for example, $z_3 \to z_1$, this corresponds to $x \to (1 - x)$ along with $F_{34}^{21} (k|x) \to F_{14}^{23} (k|1 - x)$; whereas the correlation function, nevertheless, must remain invariant under $z_3 \to z_1$. This procedure is called conformal bootstrapping, whereby one creates a system of equations for the 3-point correlation functions, given the four-point correlation function, expressed in terms of those conformal blocks. In particular, from the symmetry $x \to (1 - x)$ notice that $F_{34}^{21} (k|x)$ is independent of x, when the fusion rules dictate that the four-point correlation function can be constructed from a single primary. Fusion rules furthermore imply Eqs. (26.38 — 26.43).

$$(\epsilon \times \epsilon) \times (\sigma \times \sigma) = \mathbb{1} \cap (\mathbb{1} + \epsilon) = \mathbb{1} \tag{26.38}$$

$$(\sigma \times \epsilon) \times (\sigma \times \epsilon) = \sigma \cap \sigma = \sigma \tag{26.39}$$

$$(\epsilon \times \epsilon) \times (\epsilon \times \epsilon) = \mathbb{1} \tag{26.40}$$

$$(\epsilon \times \epsilon) \times (\sigma \times \epsilon) = \mathbb{1} \cap \sigma = \emptyset \tag{26.41}$$

$$(\sigma \times \epsilon) \times (\sigma \times \sigma) = \sigma \cap (\mathbb{1} + \epsilon) = \emptyset \tag{26.42}$$

$$(\sigma \times \sigma) \times (\sigma \times \sigma) = (\mathbb{1} + \epsilon) \cap (\mathbb{1} + \epsilon) = \mathbb{1} + \epsilon \tag{26.43}$$

Five of the six correlation functions can immediately be solved, Eqs. (26.44 — 26.48).

$$\langle \epsilon_1 (z_1) \epsilon_2 (z_2) \sigma_3 (z_3) \sigma_4 (z_4) \rangle = \frac{C_{\epsilon\epsilon\mathbb{1}} C_{\sigma\sigma\mathbb{1}}}{|z_{12}|^{2(2h_\epsilon - h_\mathbb{1})} |z_{34}|^{2(2h_\sigma - h_\mathbb{1})}} = \frac{C_{\epsilon\epsilon\mathbb{1}} C_{\sigma\sigma\mathbb{1}}}{|z_{12}|^2 |z_{34}|^{\frac{1}{4}}} \tag{26.44}$$

$$\langle \sigma_1 (z_1) \epsilon_2 (z_2) \sigma_3 (z_3) \epsilon_4 (z_4) \rangle = \frac{C^2_{\sigma\sigma\epsilon}}{|z_{12} z_{34}|^{-2(h_\sigma - h_\epsilon - h_\sigma)}} = \frac{C^2_{\sigma\sigma\epsilon}}{|z_{12} z_{34}|} \tag{26.45}$$

$$\langle \epsilon_1 (z_1) \epsilon_2 (z_2) \epsilon_3 (z_3) \epsilon_4 (z_4) \rangle = \frac{C^2_{\epsilon\epsilon\mathbb{1}}}{|z_{12} z_{34}|^{-2(h_\mathbb{1} - 2h_\epsilon)}} = \frac{C^2_{\epsilon\epsilon\mathbb{1}}}{|z_{12} z_{34}|^2} \tag{26.46}$$

$$\langle \epsilon_1 (z_1) \epsilon_2 (z_2) \sigma_3 (z_3) \epsilon_4 (z_4) \rangle = 0 \tag{26.47}$$

$$\langle \sigma_1 (z_1) \epsilon_2 (z_2) \sigma_3 (z_3) \sigma_4 (z_4) \rangle = 0 \tag{26.48}$$

The four-point correlation corresponding to Eq. (26.43) is obtained as Eq. (26.49).

$$\begin{aligned} \mathcal{G} = \langle \sigma_1 (z_1) \sigma_2 (z_2) \sigma_3 (z_3) \sigma_4 (z_4) \rangle &= \frac{C^2_{\sigma\sigma\mathbb{1}} F_\mathbb{1} (x)}{|z_{12} z_{34}|^{-2(h_\mathbb{1} - 2h_\sigma)}} + \frac{C^2_{\sigma\sigma\epsilon} F_\epsilon (x)}{|z_{12} z_{34}|^{-2(h_\epsilon - 2h_\sigma)}} \\ &= \frac{1}{|z_{12} z_{34}|^{\frac{1}{4}}} \left(C^2_{\sigma\sigma\mathbb{1}} F_\mathbb{1} (x) + C^2_{\sigma\sigma\epsilon} F_\epsilon (x) |z_{12} z_{34}|^2 \right) \end{aligned} \tag{26.49}$$

Due to Möbius, one writes $z_{12} z_{34} = (x (1 - x))$. Plugging the ward identities, one obtains Eq. (26.50), such that one obtains two independent solutions for $F_\mathbb{1} (x)$ and $F_\epsilon (x)$.

$$\left(x (x - 1) \frac{\partial^2}{\partial x^2} + \left(\frac{1}{2} - x \right) \frac{\partial}{\partial x} + \frac{1}{16} \right) F (x) = 0 \Longrightarrow \begin{cases} F_\mathbb{1} (x) & = \sqrt{1 + \sqrt{1 + x}} \\ F_\epsilon (x) & = \sqrt{1 + \sqrt{1 - x}} \end{cases} \tag{26.50}$$

By choosing $C_{\sigma\sigma\mathbb{1}} = 1$ (since one fixes at least one of the parameters of the theory freely), and enforcing symmetry under $x \to (1 - x)$ (as discussed above), one finds $C_{\sigma\sigma\epsilon} = \frac{1}{2}$, such that the four point correlation function has the form Eq. (26.51), which has a scaling behaviour $x^{-\frac{1}{4}}$ as $x \to 0$.

$$\langle \sigma_1 (z_1) \sigma_2 (z_2) \sigma_3 (z_3) \sigma_4 (z_4) \rangle = \frac{1}{((x (1 - x)))^{\frac{1}{4}}} \left(\sqrt{1 + \sqrt{1 + x}} + \frac{1}{4} \sqrt{1 + \sqrt{1 - x}} (x (1 - x))^2 \right) \tag{26.51}$$

Part III

FRG Approach to Long-Range Models

27 FRG Framework in Long-Ranged Models

27.1 Introduction

Statistical models with long range interactions are of great interest in the field of statistical physics, due to the diversified properties they exhibit at criticality; such as the exhibition of phase transitions even in one dimension;[107] universality classes indexed by continously-varying exponents; among others.[13–31] The field has furthermore made substantial advances in the experimental side in the last 50–60 years, with expectation that the combined theoretical and experimental advances in the area can eventually be used for the fabrication of materials with exotic properties that could be used in industry for the fabrication of new technologies.[97] The successful use of mathematical tools to describe the universality of critical phenomena has allowed physicists to obtain deep insight in relation to continuous second order phase transitions; and particular interest has grown recently in the introduction of models with non-local couplings, with the goal of describing the competition and hybridization between long- and short-range interactions in those models.[107, 97–113]

Although significant theoretical progress has been made in the last couple years, adequate qualitative as well as quantitative descriptions of long range models are at best incomplete, even for the simplest models.[88, 102, 89–103] Methods of approximatedly mapping long-ranged models to short-range models have shown relative success at mean-field level approximations in particular regimes;[97, 94–119] and most recently, proposed infrared-duality models, where two different models characterized by different UV description (normally described evenso by different operators) capture the same IR physics, have shown promising results in uncovering properties of long-ranged models at previously untouched regimes.[88, 102, 89–94, 91–92]

In this connection, the increasing prevalence of functional renormalization group methods pioneered by Wilson, 1971, 1974, and Wetterich, 1991–1994, among others,[35–45] have shown themselves time and time again as reliable and successful tools in order to study critical phenomena in a systematic and analytical way;[94, 96, 95] with significant count of works both in the areas of statistical and condensed matter physics, as well as quantum field theory, being successful in utilizing this method for accurately and in a standardized way describing, characterizing, and quantifying critical phenomena and continuous second order phase transitions.[42, 44, 37–47] The recent characterization of conformal field theories, particularly in two dimensions, has furthermore presented itself as a powerful tool in this endeavour; as it in principle allows for the complete analytical computation and classification of critical phenomena in two dimensions.[69–76]

27.2 Model formulation

In the contemporary section, one proceds in a nearly-identical manner, as proceeded in the previous section in relation to the short-ranged scalar model. Because of this, most intermediary steps are subortinated to appendicies (which are indicated on the footnotes), and one quickly develops the formalism up to the discussions and results. The long-ranged model is given by the action term, Eqs. (27.1, 10.20), where "..." denotes (potentially) additional self-interaction terms.

$$S[\chi] = \frac{1}{2}\int \frac{(\chi(x)-\chi(y))^2}{|x-y|^{s+d}} d^d x d^d y + \frac{m^2}{2}\int \chi^2(x)\,dx + \frac{\lambda}{4!}\int \chi^4(x)\,dx + \dots \tag{27.1}$$

One finds that the action can be re-written in momentum space as Eq. (27.2).[ap]

$$S[\chi] = \chi^2 \left(p^2 + m^2\right) + \frac{\lambda}{4!}\chi^4 + \dots \tag{27.2}$$

One proposes the LPA' Ansatz and regulator given by Eq. (27.3). Additionally, one makes use of $\mathbb{Z}^2$ symmetry and writes $\rho = \frac{\chi^2}{2}$.

$$\Gamma_k[\chi] = Z_\chi p^s + U_k[\chi] \ ; \qquad R_k = Z_\chi \left(k^s - p^s\right) \Theta\left(k^s - p^s\right) \tag{27.3}$$

The Wetterich equation becomes Eq. (27.4).[aq]

$$\partial_t \tilde{U}_t(\tilde{\rho}) = -d \cdot \tilde{U}_t(\tilde{\rho}) + (d - s + \eta_\chi)\, \tilde{\rho}\tilde{U}_t^{(1)}(\tilde{\rho}) + A_d \frac{s \times C_\chi}{1 + \tilde{U}_k^{(1)} + 2\tilde{\rho}\tilde{U}_k^{(2)}} \tag{27.4}$$

In section 27.5, the author will demonstrate that $C_\chi = 1$ and $\eta_\chi = 0$. For the remaining of the analysis hereinafter, let this be assumed as fact.

27.3 Flow Analysis

The leading order contributions to the couplings in the long-range model are given by Eq. (27.5).[ar]

$$\begin{cases} \partial_t \tilde{m}^2 & \sim -s\tilde{m}^2 \\ \partial_t \tilde{g_4} & \sim (d - 2s)\, \tilde{g_4} \\ \partial_t \tilde{g_{2n}} & \sim ((n-1)\, d - ns)\, \tilde{g_{2n}} \end{cases} \tag{27.5}$$

One observes the following facts:

1. For $s = \frac{d}{2}$, the coefficient $\tilde{g_4}$ becomes marginal and the remaining $\tilde{g_{2n}}$ coefficients become irrelevant. The theory accordingly flows to the gaussian fixed point universality class; and the critical coefficients correspond to those for a short-ranged model in an effective dimensiond $d_{eff} = \frac{2d}{s}$.

2. Given a fixed s, the theory endows of at most $n \leq \frac{d}{d-s}$ relevant and marginal coefficients, or, equivalently, given a fixed n number of relevant and marginal coefficients, the theory will require at least $s \geq \left(\frac{n-1}{n}\right) d$.

3. Given the marginality condition identified in the bullet-point above, in the region specified by $\frac{d}{2} \leq s < \frac{2d}{3}$, the coupling $\tilde{g_4}$ will always be relevant or at least marginal, and $\tilde{g_6}$ will always be irrelevant, so that the fixed point is correctly identified by a perturbative, 2–level truncation to the flow equations, and a stability matrix truncated at the 2×2 level will predict accurate critical coefficients.

4. More generally, given the marginality condition in bullet-point (2) above, and by considering the anomalous dimensions corresponding to the fixed points for $d = 4, 3, 2, 1$, one identifies the Sak points $s^\star_{d=4} = 2$, $s^\star_{d=3} \approx 1.98$, $s^\star_{d=2} = 1.75$ and $s^\star_{d=1} = 1$; such that one avers the following constraints, given a dimensionality:

[ap] *See* Appendix Momentum Transformation of Kinetic Term.
[aq] *See* Appendix Derivation of LPA'-Wetterich in a Long-Ranged Model.
[ar] *See* Appendix section 30.3.

(a) $d = 4$: one obtains at most one relevant and one marginal coupling for all $s < s^\star$;

(b) $d = 3$: one obtains at most one relevant and one marginal coupling for all $s < s^\star$;

(c) $d = 2$: one obtains at most 7 relevant and one marginal coupling for all $s < s^\star$;

(d) $d = 1$: one obtains infinite relevant couplings, and therefore the theory cannot be well-described by truncation methods such as the implemented above;

(e) for all $d > 4$, the theory will have only a single relevant coupling for all $s < s^\star$ and therefore falls to the gaussian fixed point universality class; and

(f) in principle this analysis can be arbitrarily extended to fractional dimensions.

5. In general, the assessment above indicates that for $d = 4$ and $d = 3$, the stability matrix method with truncation at the second level will generally provide accurate results, even beyond the regime $\frac{d}{2} \leq s < \frac{2d}{3}$. On the other hand, for $d = 2$, one expects the two-level truncation to lead to increasingly inaccurate critical coefficients as s approaches the Sak Point $s^\star$.

6. Conversely, even for $d = 2$, the perturbation method at a eight-order truncation is expected to yield generally accurate results. Furthermore, even at $d = 2$, so long as the theory is investigated in the range $\frac{d}{2} \leq s < \frac{2d}{3}$, that is to say, in the range $1 \leq s < \frac{4}{3}$, one expects that the two-level truncation to the stability matrix will continue to generate accurate predictions for the critical exponents.

27.4 Criticality Analysis – Semi-Analytical

The critical points, up to second-level truncation, are given by Eq. (27.6).[as]

$$\tilde{m}^2 = \frac{d - 2s}{8s - d} \; ; \qquad \tilde{\lambda} = \frac{-(d - 2s)}{6s \times Ad} \left(\frac{6s}{8s - d} \right)^3 \tag{27.6}$$

The RG eigenvalues are correspondingly obtained from Eq. (27.7).[at]

$$6\lambda^2 + 2\,(4d - 5s)\,\lambda + (8s - d)\,(d - 2s) = 0 \tag{27.7}$$

One obtains a continuous family of universalities indexed by the coefficient s. Figures 27.1, 27.2 and 27.3 display the critical coefficients as a function of s for dimensions $d = 2, 3, 4$, as well as the relevant eigenvalue y_t and the irrelevant eigenvalue ω. The curve labelled as "adjusted y_t" corresponds to y_t adjusted to the effective dimension $d_{eff} = \frac{2d}{s}$. The plot starts at $s = \frac{d}{2}$ on the basis that Eq. (27.7) is derived specifically from the assumption that $s > \frac{d}{2}$ and is therefore invalid below this regime; namely because the flows in Eq. (27.5) lead to irrelevant $\tilde{g}_4$, such that the stability matrix Eq. (27.7)[au] actually reduces to a single entry, Eq. (27.8); and all the other matrix elements vanish; such that $y_t = s$ becomes fixed.

$$\frac{\partial \beta_{\tilde{m}^2}}{\partial \tilde{m}^2} = -s \tag{27.8}$$

The black "×"s mark the following points:

[as] *See* Appendix Tower Equations for Long Range Model.
[at] *See* Appendix Tower Equations for Long Range Model.
[au] *See* Appendix Tower Equations for Long Range Model.

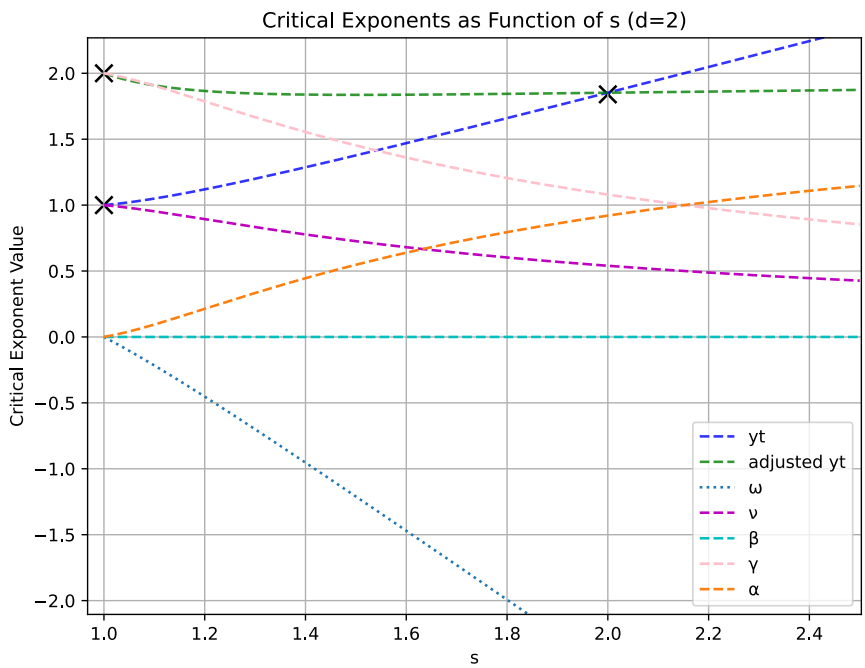

Figure 27.1: Value of main critical coefficients as function of s for a long-ranged model in $d = 2$. Marked with a black "×" are the points of cross-over to the short-ranged model ($s^\star = 2 - \eta$) and to the free-field universality class ($s = 1$).

1. At $s = \frac{d}{2}$ and $s = \frac{d_{eff}}{2}$, the points where the model resolves onto the gaussian fixed point universality class; and

2. At $s = 2$, and $s = 2 - \eta_{SR}$, the the Sak Points, respectively on the assumption of zero and non-zero anomalous dimension; that is, to the LPA and LPA' predictions for the Sak point; corresponding to the point(s) where the model is expected to transition from the long-range to the short-range universality class.

Additionally, figure 27.4 diplays y_t adjusted to the effective dimension for $d = 2, 3, 4$ as a function of s, zoomed to the inteval $[1.84, 2.00]$, and figure 27.5 provides the asymptotic behaviour y_t adjusted to the effective dimension for $d = 2, 3, 4$ as a function of s, zoomed to the inteval $[1.84, 2.00]$.

One makes the following observations:

- For $d = 4$, no long-range universality class emerges; because the Sak Point and the gaussian free field point coincide, such that the long-range regime for $\overline{s} \leq s \leq s^\star$ is completely squeezed and disappears. Namely, in $d = 4$ model falls to the short-range, gaussian free field universality class for all s as is of no interest. As discussed previously, one avers that this behaviour is also reproduced for $d > 4$ on the basis that at $d > 4$ one expects at most one relevant coefficient. In other words, the behaviour of y_t as function of s in figure 27.3 is mostly uninteresting.

- For $d = 3$, the model contains one relevant and one marginal coupling in the range $\frac{3}{2} \leq s < 2$. Since the Sak point occurs at $s^\star \approx 1.98$, the model transitions to the short-range universality class before

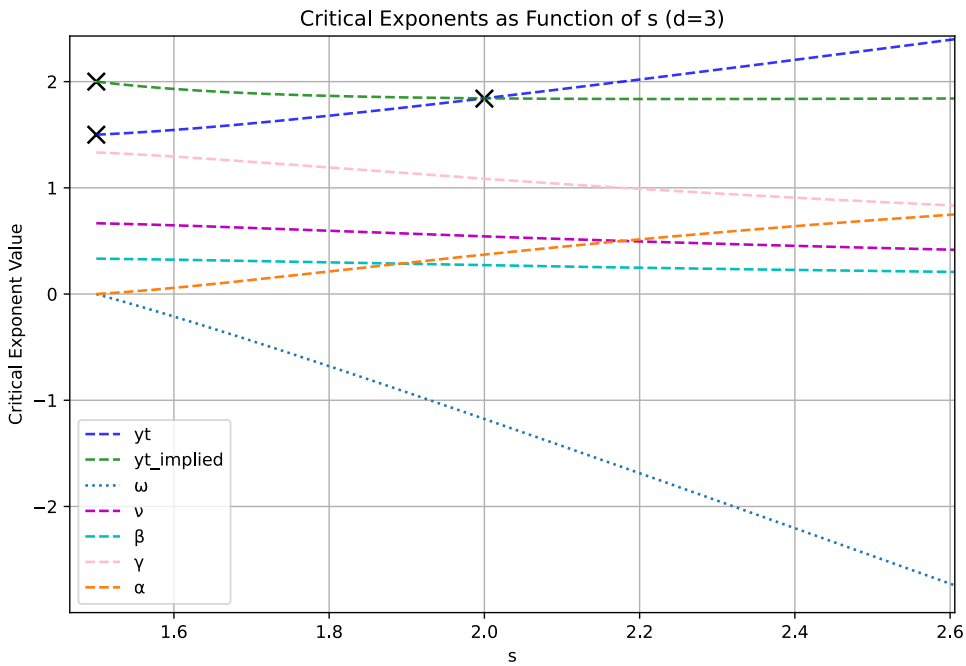

Figure 27.2: Value of main critical coefficients as function of s for a long-ranged model in $d = 3$. Marked with a black "$\times$" are the points of cross-over to the short-ranged model ($s^\star = 2 - \eta$) and to the free-field universality class ($s = 1.5$).

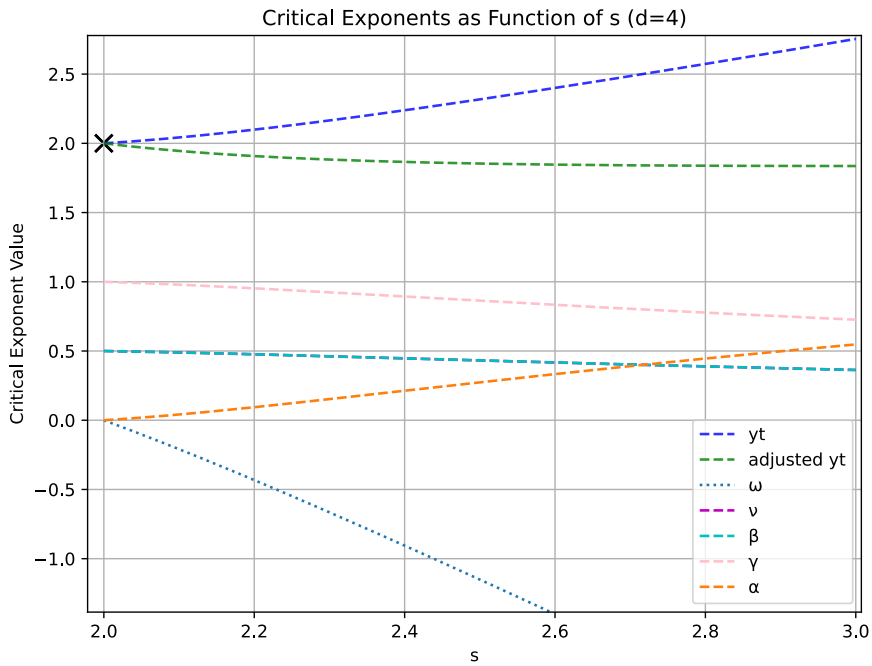

Figure 27.3: Value of main critical coefficients as function of s for a long-ranged model in $d = 4$. Marked with a black "×" are the points of cross-over to the short-ranged model ($s = 2$) and to the free-field universality class ($s = 2$). — In this case, one notices that the two points coincide.

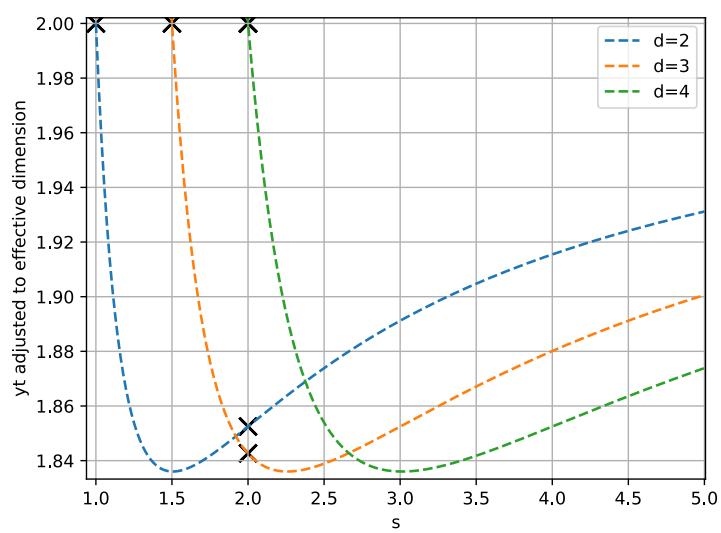

Figure 27.4: y_t adjusted to the effective dimension for $d = 2, 3, 4$ as a function of s, zoomed to the inteval $[1.84, 2.00]$. Marked with a black "×" are the points of cross-over to the short-ranged model ($s^\star = 2 - \eta$), and to the free-field universality class.

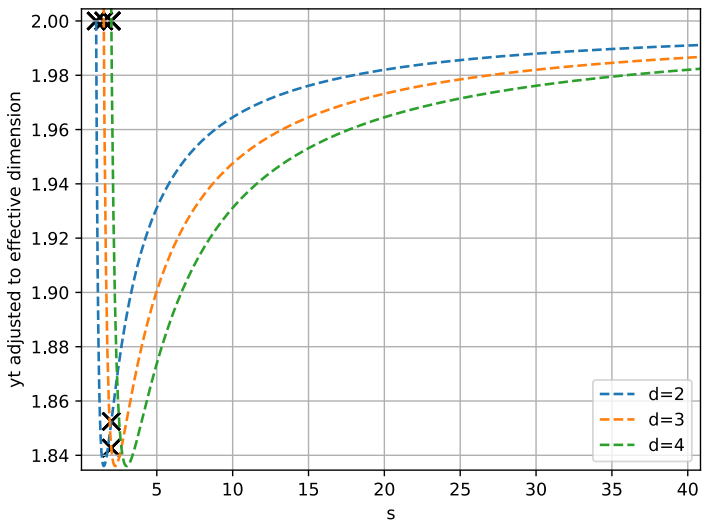

Figure 27.5: Asymptotic behaviour y_t adjusted to the effective dimension for $d = 2, 3, 4$ as a function of s, zoomed to the inteval $[1.84, 2.00]$. Marked with a black "×" are the points of cross-over to the short-ranged model ($s^\star = 2 - \eta$), and to the free-field universality class.

additional relevant couplings emerge, such that the theory the stability matrix truncated at two-level gives reliable predictions to the theory at all interval displayed in figure 27.2.

- For $d = 2$, one obtains a hirearchy of s, $1 < s_1 < \frac{4}{3} < s_2 < \frac{3}{2} < s_3 < \frac{8}{5} < \ldots$, such that within the interval of s_1 the theory is well-described by a 2-level truncation, in the interval of s_2 the theory is well-described by a 3-level truncation, etc., up to $\frac{12}{7} < s < \frac{7}{4}$, where the theory is described by a 7-level truncation. Beyond this point, the theory falls to the short-range universality class. One expects, therefore, that in figure 27.1 the prediction for the eigenvalue y_t and the respective critical coefficients should be fairly precise up to $s = 1.33\ldots$, and perhaps even up to $s = 1.5\ldots$; whilst the results progressively get worse as $s \to 1.75$.

- One emphasizes once more the emergence of a continuous family of universality classes indexed by s for $d = 2$ and $d = 3$ (whereas for $d = 4$ no long-range universality emerges, as discussed already in the first bullet-point above). Furthermore, one emphasizes again that for $s < \frac{d}{2}$, the flows in Eq. (27.5) lead to irrelevant $\tilde{g}_4$, such that the stability matrix Eq. (27.7)[av] actually reduces to a single entry $(D_*)_{11}$, such that $y_t = s$ becomes fixed. Up to an effective dimension $d_{eff.} = \frac{2d}{s}$, one therefore obtains $y_t^{eff.} = 2$, which corresponds to the earlier prediction that the model collapses to the gaussian fixed point/ mean field universality class at an effective dimension $d_{eff.} = \frac{2d}{s}$ for all $s < \frac{d}{2}$.

- Furthermore, also corresponding to expectation, at the point of the cross-over, $s = s^\star$, the long-ranged model predicts critical exponents corresponding to the short-range ising model universality class, up to the effective dimension $d_{eff} = \frac{2d}{s}$. That is to say, at the cross-over, the model falls into the short-range universality class. On the other hand, from figure 27.4, one observes that this, for $s > s^*$, the prediction immediately breaks down because the regime of validity of Eq. (27.7 is only up to the point where

[av] *See* Appendix Tower Equations for Long Range Model.

the long-range kinetic term is a relevant operator. In other words, Sak argues that a transition to the short range universality class occurs at $s^\star$. This piucture is supported by Defenu. Defenu argues that a short-range term in the style of Eq. (27.9) must be added to the LPA(') Ansatz, for the RG flow to correctly project the behaviour above the cross-over point.

$$\Gamma_k^{(2)}[\phi] = Z_{k,2}p^2 + Z_{k,s}p^s + U_k^{(2)}[\phi] \tag{27.9}$$

- Figure 27.5 — and in fact also the quadratic equation corresponding to Eq. (27.7) — predicts that asympthotically $y_t = s$, such that one again the model should fall onto the gaussian free field universality class at the limit of $s \to \infty$. On the other hand, the observation in bullet-point above indicates that this is in fact a false prediction, and the model instead simply falls to the short-range universality class for all $s > s^\star$.

- Even at two-level truncation, the stability matrix correctly predict the point of cross-over to the gaussian free field universality class at $s = \frac{d}{2}$ and to the short range universality class at $s^\star$. Furthermore, the two-level truncation correctly predicts the critical coefficients corresponding to those regimes. That the interval corresponding to the long-range universality class, $\frac{d}{2} < s < s^\star$, can be approximately modeled by an interpolation between those regimes, with y_t falling proportional to some power law.

- In principle, this analysis can be arbitrarily extended to fractional dimensions, although one would not expect meaningful phenomenological changes.

27.5 The Anomalous Dimension in the Long-Ranged Model

28 Critical Coefficients in $O(N)$ Models

29 Connection to Mermin-Wagner

29.1 Introductory Remarks

In essence, the Mermin-Wagner Theorem argues that no spontaneous symmetry breaking of a *continuous* symmetry can occur for infinite-sized $d \leq 2$ systems with short-ranged interactions, on the basis that long-range fluctuations can be created with little energy cost, such that in those systems continuous long-range order cannot exist. On the other hand, one emphasizes that *discrete* symmetries are still allowed to be broken. For example, the scalar $O(1)$ theory disposes over a discrete $\mathbb{Z}_2$ symmetry that in principle can still be broken, even for $d \leq 2$, without violating the mermin-wagner theorem.

It is often claimed in literature that the LPA solution to the Wetterich Equation violates the Mermin-Wagner theorem, because it purportedly predicts a continuous phase transition for $d = 2$ in $O(N)$ models. In this section, the author argues that the self-consistent treatment of the LPA method does, in fact, correctly predicts the Mermin-Wagner theorem; and that conversely, the correct conclusion is that the method fails to predict the discrete $\mathbb{Z}_2$ SSB in $d = 2$ for $O(1)$ models. More precisely, in this section the author aims to prove the following statement:

> *Let a theory with lower critical dimension d be given . Then either (i) a spontaneously broken phase does not exist (i.e., trivial $\tilde{\rho} = 0$); or (ii) if it does exist, then the LPA solution to the Wetterich Equation in this spontaneously-broken phase predicts a non-physical wavefunction, such*

that in practice this spontaneously-broken phase is not reachable with a physical choice of the wavefunction. That is to say, a spontaneously broken phase at d exists only for $\tilde{\rho} \leq 0$.

29.2 Argument

29.2.1 Short Range $O(1)$ Model

Numerical manipulations are subordinated to Appendicies in Concern to Mermin-Wagner Theorem. For the $O(1)$ scalar, short-ranged model, the flow equation for the wavefunction in the spontaneously broken phase is given by Eq. (29.1).

$$\partial_t \tilde{\rho} = (2 - d - \eta)\, \tilde{\rho} + \frac{3A_d}{\left(1 + 2\tilde{\rho}\tilde{U}''(\tilde{\rho})\right)^2} \tag{29.1}$$

For $d > 2 - \eta$, the linear term $(2 - d - \eta)\, \tilde{\rho_*}$ dominates in the IR limit, such that $\tilde{\rho_*} \sim k^{(2-d-\eta)}$. The exponent is negative, such that $\tilde{\rho_*} \to \infty$ in the IR and therefore $\tilde{\rho_*}$ is a physical and relevant parameter. For $d < 2 - \eta$, conversely, the linear term $(2 - d - \eta)\, \tilde{\rho_*}$ scales as $\sim k^{(2-d-\eta)}$, but the exponent is negative, such that this term goes to zero in the IR limit. The dominant contribution is therefore given by Eq. (29.2). Similarly, in the marginal case, $d = 2 - \eta$, the linear term is fully suppressed, thus, once again, one obtains a dominant contribution from Eq. (29.2).

$$\partial_t \tilde{\rho} \sim \frac{3A_d}{\left(1 + 2\tilde{\rho_*}\tilde{U}_t^{(2)}(\tilde{\rho_*})\right)^2} \tag{29.2}$$

In the vicinity of the fixed point, the contribution, Eq. (29.2), is roughly constant, such that the scaling law follows Eq. (29.3), such that in the IR limit, $\tilde{\rho_*} \to -\infty$ as $k \to 0$. But this is an unphysical behaviour, since $\tilde{\rho_*} = \frac{1}{2}\left|\tilde{\phi_*}\right|^2$ is strictly defined in $\mathbb{R}^+$; or alternatively, becuase as $\tilde{\rho_*} \to -\infty$, because $\tilde{U}_t^{(2)}(\tilde{\rho_*})$ is positive, the denominator in Eq. (29.2) diverges for some value of $\tilde{\rho_*}$ such that there exists at least one point where the potential is not continuous and therefore unphysical. Therefore, one demonstrates that in the truncated LPA' scheme, if a SSB phase exists at the lower critical dimension $d \leq 2 - \eta$, this SSB has $\tilde{\rho_*} \to -\infty$, which leads to unphysical behaviour of the wavefunction or alternatively of the potential. By contradiction one thus finds that no SBB phase can exist at the lower critical dimension; whereas for $d > 2 - \eta$ the wave function is well-defined. In the particular truncation where $\eta = 0$, the lower critical dimension corresponds then to $d = 2$. One furthermore observes that in general for $\eta > 0$ the effect of the anomalous dimension is, in effect, to lower the lower critical dimension below its $\eta = 0$ LPA counterpart.

$$\tilde{\rho_*} \sim log\,(k/\Lambda) \tag{29.3}$$

29.2.2 Short Range $O(N)$ Model.

The analysis is identical for $O(N)$ models, since the wavefunction is given by Eq. (29.4) below, such that one again the linear term dominates for $d > 2 - \eta$, such that $\tilde{\rho_*} \to \infty$ in the IR; whereas for $d \leq 2 - \eta$ one obtains the same leading term, Eq. (29.2), plus the constant contribution $A_d\,(N - 1)$; that ultimately the scaling law once again follows Eq. (29.3). In this case one observes again that the non-zero anomalous dimension, in effect, lowers the lower critical dimension.

$$\partial_t \tilde{\rho} = (2 - d - \eta)\, \tilde{\rho} + \frac{3A_d}{\left(1 + 2\tilde{\rho}\tilde{U}''(\tilde{\rho})\right)^2} + A_d\,(N-1) \tag{29.4}$$

29.2.3 Long Range Model

This analysis can be furthermore extended to the long-ranged model. For the long-ranged model, the flow equation for the wavefunction in the spontaneously broken phase is given by Eq. (29.5).

$$\partial_t \tilde{\rho} = (s - d)\, \tilde{\rho} + \frac{3sA_d}{\left(1 + 2\tilde{\rho}\tilde{U}_t^{(2)}(\tilde{\rho})\right)^2} \tag{29.5}$$

By an identical consideration as above, one finds that for $d > s$, the term linear in $\tilde{\rho}$ dominates in the IR limit, such that $\tilde{\rho_*} \to \infty$. On the other hand, for $d \leq s$, the term linear in $\tilde{\rho}$ vanishes, such that the wavefunction scales as Eq. (29.3) and therefore this SSB has $\tilde{\rho_*} \to -\infty$, which leads to the same unphysicality as described above.

On the other hand, for $d = 4, 3, 2, 1$, one identifies the Sak points $s^\star_{d=4} = 2$, $s^\star_{d=3} \approx 1.98$, $s^\star_{d=2} = 1.75$ and $s^\star_{d=1} = 1$; such that the long-range *always* crosses the Sak point and transitions to the short-range universality class before it can possibly reach the threshold s for which no SSB phase is allowed for a given dimensionality d. In other words, this treatment predicts that SSBs are always allowed for long-ranged models. For the particular case of $d = 4$, as already investigated in previous sections, one finds no long-range universality class, as the Sak point and the point of transition to the GFF universality class coincide. In this particular case, one finds that no SSB phase exists to begin with (or, more precisely, the SBB phase is trivial); that is to say, in the SSB phase $\tilde{\rho}_* = 0$ always. Furthermore, for $s < \frac{d}{2}$, one obtains only the trivial SBB phase, that is, $\tilde{\rho_*} = 0$, such that no SSB actually occurs.

29.2.4 Long-Range Short-Range Mixed Model

Finally one aims to investigate the long-range-short-range-mixed-model. Referring back to Appendicies in Concern to Mermin-Wagner Theorem, one expands the RG flow of the potential by taylor expansion near $g \to 0$, such that the theory simplifies to a linear combination of RG flows for the ρ-dependant and σ-dependant potentials, plus linear couplings of the ρ- and σ-potentials mediated by orders $\mathcal{O}\left(g^2\right)$.

One finds that for $g = 0$, the long- and short-range nodes decouple, and one recovers Eq. (29.1) and Eq. (29.5) corresponding to the RG flow of the wavefunctions at the spontaneously-broken phase for the short- and long-range fields respectively. In summary, the analysis reduces to what has already been done in this section.

In effect, for a small $g \neq 0$, to leading order, the additional contribution arises from the perturbation term, Eq. (29.6). Because this term not entail any new dependence on $\tilde{\rho}$ or $\tilde{\sigma}$ in the numerator when functional derivatives with respect to $\tilde{\rho}$ or $\tilde{\sigma}$ are taken, that is, corresponding to the terms $\delta^{(1,0)}_{\rho\sigma}$ and $\delta^{(0,1)}_{\rho\sigma}$; it follows (absent of the need of a detailed derivation) that the introdction of this small perturbation g does not lead to contributions in the leading order in $\tilde{\rho}$ and $\tilde{\sigma}$, respectively to the flows of $\partial_t\tilde{\rho}$ and $\partial_t\tilde{\sigma}$. Accordingly, the perturbation contributes only marignally, and the lower critical dimension is not changed by the introduction of the perturbation.

$$\delta = g^2 A_d \left(\frac{C_\rho}{\left(\Gamma^{(1,0)}_{t,\rho\sigma}\right)^2 \Gamma^{(0,1)}_{t,\rho\sigma}} + \frac{C_\sigma}{\Gamma^{(1,0)}_{t,\rho\sigma}\left(\Gamma^{(0,1)}_{t,\rho\sigma}\right)^2} \right) \tag{29.6}$$

29.3 Discussion

The conclusions and/or commends drawn in this section may be regarded as overly-optimistic. On this basis, the author draws a few remarks, and summarizes some of the conclusions in more precise words:

1. First of all, the key result obtained in this section is summarized as follows: given a theory (i) at a lower critical dimension; (ii) that befinds itself in a spontaneously broken phase; and (iii) for which one considers an LPA(') solution to the Wetterich Equation with minimal truncation; then:

   (a) the wavefunction is either trivial, $\tilde{\rho}_* = 0$, or $\tilde{\rho}_* \to -\infty$ in the IR limit; accordingly

   (b) for some value of $\tilde{\rho}_*$ reached by the RG flow, the potential diverges; and furthermore

   (c) $\tilde{\rho}_*$ is ultimately unphysical.

2. Crucially, one emphasizes that those conclusions follow from the theory-description on the basis of an LPA(') solution with minimal truncation; and one should not classify those conclusions as an exact physical theorem. The key result, therefore, is that the LPA(') solution to the Wetterich Equation with minimal truncation forbids the existence of SSB at the lower critical dimension. This must be distinguished from the statement that the truncated LPA(') method predicts the Mermin Wagner theorem. More precisely, it should instead be claimed that the LPA(') method *is consistent with* the Mermin Wagner theorem (to the extend that it finds no physically-reachable SSB phase); and that the LPA(') method incorrectly fails to predict discrete phase transitions at the lower critical dimension, even though they are theoretically known to exist.

3. The additional result obtained in this section is that LPA(') solution to the Wetterich Equation with minimal truncation does not forbit the existence of SSB for the long-range model for any value of s for which the model falls into the long-range universality class; and furthermore the the introduction of the short-range long-range mixture models does not lower or raise the lower critical dimension.

4. Furthermore, one finds that the introduction of an anomalous dimension $\eta > 0$ effectively lowers the lower critical dimension of short-ranged models. In particular, this has the implication that, for example, although the $d = 2$ short-range ising model cannot be accurately described by the LPA method, because the lower critical dimension $d = 2$ means that all couplings become relevant; the introduction of an anomalous dimension $\eta = \frac{1}{4}$ in the LPA' method has the effect of reducing the lower critical dimension to $d = 1.75$, such that truncation methods become precise once again.

30 Appendicies in Concern to (Simple) Long-Ranged Models

30.1 Momentum Transformation of Kinetic Term

Starting from Eq. (30.1), one writes the fields χ in momentum space and realizes the substitution $(x - y) = r$, whilst solving the exponentials that reduce to delta functions. The final result obtained is Eq. (30.2).

$$K = \frac{1}{2}\int \frac{\left(\chi\left(x\right)-\chi\left(y\right)\right)^2}{\left|x-y\right|^{s+d}} d^d x d^d y = \int \frac{\chi^2\left(x\right)-\chi\left(x\right)\chi\left(y\right)}{\left|x-y\right|^{s+d}} d^d x d^d y = \int \chi\left(p\right)\chi\left(q\right)\frac{e^{i(p+q)x}-e^{ipx+iqy}}{\left|x-y\right|^{s+d}} d^d x d^d y d^d p d^d q \tag{30.1}$$

$$= \int \chi\left(p\right)\chi\left(q\right)e^{i(p+q)y}\left(\frac{e^{i(p+q)r}-e^{ipr}}{\left|r\right|^{s+d}}\right) d^d r d^d y d^d p d^d = \int \chi\left(p\right)\chi\left(-p\right)\left(\int \frac{1-e^{ipr}}{\left|r\right|^{s+d}} d^d r\right) d^d p = \int p^s \chi\left(p\right)\chi\left(-p\right) d^d p \tag{30.2}$$

One may opt to interpret this result as to mean that the long-ranged model is equivalent to a short-ranged model, only that the kinetic term is instead mediated by a "fractional laplacian", Eq. (30.3).

$$K = \frac{1}{2}\int \chi\left(x\right)\left(-\partial^{\frac{s}{2}}\right)^2 \chi\left(x\right) d^d x \tag{30.3}$$

30.2 Derivation of LPA'-Wetterich in a Long-Ranged Model

The derivation is similar to Appendix Derivation of the LPA-Wetterich for a ϕ^4 Scalar Theory. One considers the Ansatz and regulator specified by Eq. (30.4).

$$\Gamma_k\left[\chi\right] = Z_\chi p^s + U_k\left[\chi\right] \ ; \qquad R_k = Z_\chi\left(k^s - p^s\right)\Theta\left(k^s - p^s\right) \tag{30.4}$$

One additionally requires $\mathbb{Z}^2$ symmetry explicitly, and accordingly defines $\rho = \frac{1}{2}\phi^2$. The identity, Eq. (30.5), is once again satisfied.

$$\frac{\partial}{\partial\phi} = \frac{\partial}{\partial\rho} + 2\rho\frac{\partial^2}{\partial\rho^2} \tag{30.5}$$

The definitions, Eq. (30.6), are used.

$$C_\chi = \left(1 - \frac{\eta_\chi}{s+d}\right) \ ; \qquad \eta_\chi = -\frac{1}{Z_\chi}\frac{\partial Z_\chi}{\partial log\left(k\right)} \tag{30.6}$$

The Wetterich Equation is writen as Eq. (30.7).

$$\partial_k\Gamma_k = \int \frac{\partial_k R_k}{\Gamma_k^{(2)} + R_k} dp = \frac{Z_\chi k^{s-1}}{Z_\chi k^s + U_k^{(2)}}\left\{\left(s-\eta_\chi\right)\int d^d p + \eta_\chi k^{-s}\int p^s dp\right\} = k^{d-1}\frac{s \times A_d \times C_\chi}{1 + \tilde{U}_k^{(2)} + 2\tilde{\rho}\tilde{U}_k^{(2)}} \tag{30.7}$$

Once again one defines dimensionless quantities, in order to re-write the left-handed side of the wetterich equation, Eq. (30.8)

$$m = k\tilde{m} \qquad \rho = k^{d-2}\tilde{\rho} \qquad g_4 = k^{-d+4}\tilde{g}_4 \qquad g_{2n} = k^{d-n(d-2)}\tilde{g}_{2n} \qquad \partial_\rho = \left(\partial_\rho\tilde{\rho}\right)\partial_{\tilde{\rho}} = k^{2-d}\partial_{\tilde{\rho}} \tag{30.8}$$

One utilizes furthermore the RG-time, Eq. (30.9).

$$\partial_k = (\partial_k t)\,\partial_t = k^{-1}\partial_t \tag{30.9}$$

One the other hand, left-handed side of Eq. (30.7) becomes Eq. (30.10).

$$\begin{aligned}\partial_k U_k(\rho) &= \partial_k\left(k^d \tilde{U}(\tilde{\rho})\right) = k^{d-1} d\cdot\tilde{U}(\tilde{\rho}) + \partial_k \tilde{U}(\tilde{\rho}) \\ &= k^{d-1} d\cdot\tilde{U}(\tilde{\rho}) + k^d(\partial_k\tilde{\rho})\,\partial_{\tilde{\rho}}\tilde{U}_k(\tilde{\rho}) + k^d\partial_k\tilde{U}_k(\tilde{\rho}) = k^{d-1}\left(d\cdot\tilde{U}_t(\tilde{\rho}) - (d-s)\,\tilde{\rho}\tilde{U}_t'(\tilde{\rho}) + \partial_t\tilde{U}_t(\tilde{\rho})\right)\end{aligned} \tag{30.10}$$

One therefore finds Eq. (30.11). As expected, the formula is fairly similar to the short-ranged model, only that

$$\partial_t\tilde{U}_t(\tilde{\rho}) = -d\cdot\tilde{U}_t(\tilde{\rho}) + (d-s)\,\tilde{\rho}\tilde{U}_t^{(1)}(\tilde{\rho}) + A_d\frac{s\times C_\chi}{1+\tilde{U}_k^{(2)}+2\tilde{\rho}\tilde{U}_k^{(2)}} \tag{30.11}$$

30.3 Tower Equations for Long Range Model

From Eq. (30.11) one derives immediately Eqs. (30.12, 30.13, 30.14).

$$\partial_t\tilde{U}_t^{(1)}(\tilde{\rho}) = -s\tilde{U}_t^{(1)}(\tilde{\rho}) + (d-s)\,\tilde{\rho}\tilde{U}_t^{(2)}(\tilde{\rho}) - A_d s\cdot\frac{3\tilde{U}_t^{(2)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(3)}(\tilde{\rho})}{\left(1+\tilde{U}_t^{(1)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(2)}(\tilde{\rho})\right)^2} \tag{30.12}$$

$$\begin{aligned}\partial_t\tilde{U}_t^{(2)}(\tilde{\rho}) &= (d-2s)\,\tilde{U}_t^{(2)}(\tilde{\rho}) + (d-s)\,\tilde{\rho}\tilde{U}_t^{(3)}(\tilde{\rho}) \\ &\quad - A_d\cdot\frac{5\tilde{U}_t^{(3)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(4)}(\tilde{\rho})}{\left(1+\tilde{U}_t^{(1)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(2)}(\tilde{\rho})\right)^2} + 2A_d s\cdot\frac{\left(3\tilde{U}_t^{(2)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(3)}(\tilde{\rho})\right)^2}{\left(1+\tilde{U}_t^{(1)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(2)}(\tilde{\rho})\right)^3}\end{aligned} \tag{30.13}$$

$$\begin{aligned}\partial_t\tilde{U}_t^{(3)}(\tilde{\rho}) &= (2d-3s)\,\tilde{U}_t^{(3)}(\tilde{\rho}) + (d-s)\,\tilde{\rho}\tilde{U}_t^{(4)}(\tilde{\rho}) - A_d s\cdot\frac{7\tilde{U}_t^{(4)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(5)}(\tilde{\rho})}{\left(1+\tilde{U}_t^{(1)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(2)}(\tilde{\rho})\right)^2} + \\ &\quad + 6A_d s\cdot\frac{\left(5\tilde{U}_t^{(3)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(4)}(\tilde{\rho})\right)\left(3\tilde{U}_t^{(2)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(3)}(\tilde{\rho})\right)}{\left(1+\tilde{U}_t^{(1)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(2)}(\tilde{\rho})\right)^3} - 6A_d s\cdot\frac{\left(3\tilde{U}_t^{(2)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(3)}(\tilde{\rho})\right)^3}{\left(1+\tilde{U}_t^{(1)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(2)}(\tilde{\rho})\right)^4}\end{aligned} \tag{30.14}$$

By similar consideration as done for the short range model, one obtains Eqs. (30.15,30.16, 30.17).

$$\partial_t\tilde{m}^2 = -s\tilde{m}^2 - \frac{A_d s\tilde{g}_4}{(1+\tilde{m}^2)^2} \tag{30.15}$$

$$\partial_t\tilde{g}_4 = (d-2s)\,\tilde{g}_4 + \frac{6A_d s\tilde{g}_4{}^2}{(1+\tilde{m}^2)^3} - \frac{3A_d s\tilde{g}_6}{(1+\tilde{m}^2)^2} \tag{30.16}$$

$$\partial_t\tilde{g}_6 = (2d-3s)\,\tilde{g}_6 - \frac{30A_d s\tilde{g}_4{}^3}{(1+\tilde{m}^2)^4} + \frac{30A_d s\tilde{g}_4\tilde{g}_6}{(1+\tilde{m}^2)^3} \tag{30.17}$$

The computation of the beta functions thus reads Eq. (30.18).

$$D_* = \begin{pmatrix} \frac{\beta_{\tilde{m}^2}}{\partial \tilde{m}^2} & \frac{\beta_{\tilde{\lambda}}}{\partial \tilde{m}^2} \\ \frac{\beta_{\tilde{m}^2}}{\partial \tilde{\lambda}} & \frac{\beta_{\tilde{\lambda}}}{\partial \tilde{\lambda}} \end{pmatrix} = \begin{pmatrix} -\frac{(d+s)}{3} & -A_d s \left(\frac{8s-d}{6s}\right)^2 \\ -\left(\frac{6s(d-2s)}{8s-d}\right)^2 \frac{1}{2A_d s} & -(d-2s) \end{pmatrix} \tag{30.18}$$

The eigenvalues are correspondingly obtained from Eq. (30.19).

$$6\lambda^2 + 2(4d-5s)\lambda + (8s-d)(d-2s) = 0 \tag{30.19}$$

31 Appendicies in Concern to Mermin-Wagner Theorem

31.1 Derivation of Wetterich Equation in Spontaneously-Broken Phase

31.1.1 Short Range Model

One considers the first derivative of the potential, Eq. (31.1).

$$\partial_t \tilde{U}'(\tilde{\rho}) = -2 \cdot \tilde{U}'(\tilde{\rho}) + (d-2+\eta)\tilde{\rho}\tilde{U}_t''(\tilde{\rho}) - A_d \cdot \frac{\left(3\tilde{U}''(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}'''(\tilde{\rho})\right)}{\left(1+\tilde{U}'(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}''(\tilde{\rho})\right)^2} \tag{31.1}$$

In the spontaneously-broken phase, one obtains $\tilde{U}'(\tilde{\rho}) = 0$ and $\rho_* \neq 0$. This condition leads to the identity given by Eq. (31.2).

$$\frac{d}{dt}\tilde{U}'(\tilde{\rho}) = \partial_t \tilde{U}' + \tilde{U}''\partial_t\tilde{\rho} = 0 \tag{31.2}$$

By replacing Eq. (31.2) onto Eq. (31.1) and choosing the spontaneously-broken phase, one obtains Eq. (31.3).

$$\partial_t\tilde{\rho} = (2-d-\eta)\tilde{\rho} + \frac{3A_d}{\left(1+2\tilde{\rho}\tilde{U}''(\tilde{\rho})\right)^2} \tag{31.3}$$

31.1.2 $O(N)$ model

For the $O(N)$ model, one considers instead the potential given by Eq. (31.4).

$$\partial_t\tilde{U}(\tilde{\rho}) = -d \cdot \tilde{U}(\tilde{\rho}) + (d-2)\tilde{\rho}\tilde{U}_t'(\tilde{\rho}) + A_d \cdot \left\{\frac{(N-1)}{1+\tilde{U}'(\tilde{\rho})} + \frac{1}{1+\tilde{U}'(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}''(\tilde{\rho})}\right\} \tag{31.4}$$

The exact algebraic manipulations easily lead to Eq. (31.5).

$$\partial_t\tilde{\rho} = (2-d-\eta)\tilde{\rho} + \frac{3A_d}{\left(1+2\tilde{\rho}\tilde{U}''(\tilde{\rho})\right)^2} + \frac{A_d(N-1)}{\left(1+\tilde{U}'(\tilde{\rho})\right)^2} \tag{31.5}$$

31.1.3 Long Range Model

By similar consideration, for the long range model, one starts from the flow, Eq. (31.6).

$$\partial_t \tilde{U}_t^{(1)}(\tilde{\rho}) = (d-s)\,\tilde{\rho}\tilde{U}_t^{(2)}(\tilde{\rho}) - A_d s \cdot \frac{3\tilde{U}_t^{(2)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(3)}(\tilde{\rho})}{\left(1+\tilde{U}_t^{(1)}(\tilde{\rho}) + 2\tilde{\rho}\tilde{U}_t^{(2)}(\tilde{\rho})\right)^2} \tag{31.6}$$

By doing the same replacements, one obtains Eq. (31.7).

$$\partial_t \tilde{\rho} = (s-d)\,\tilde{\rho} + \frac{3sA_d}{\left(1+2\tilde{\rho}\tilde{U}_t^{(2)}(\tilde{\rho})\right)^2} \tag{31.7}$$

31.1.4 Combined Model (with and without coupling g)

From (...), one finds the potential, Eq. (31.8), corresponding to the short-long-range model.

$$\partial_t \tilde{U}_{t,\rho\sigma}(\tilde{\rho}) = -d\tilde{U}_{t,\rho\sigma}(\tilde{\rho}) + (d-2+\eta)\,\tilde{U}_{t,\rho\sigma}^{(1,0)}(\tilde{\rho}) + (d-s)\,\tilde{U}_{t,\rho\sigma}^{(0,1)}(\tilde{\rho}) - A_d \underbrace{\frac{C_\rho \Gamma_{t,\rho\sigma}^{(0,1)} + C_\sigma \Gamma_{t,\rho\sigma}^{(1,0)}}{\Gamma_{t,\rho\sigma}^{(1,0)}\Gamma_{t,\rho\sigma}^{(0,1)} - g^2}}_{\mathcal{P}} \tag{31.8}$$

By taylor-expanding the renominator in final term in Eq. (31.8) above, one obtains Eq. (31.9).

$$\begin{aligned}\mathcal{P} &= \frac{C_\rho \Gamma_{t,\rho\sigma}^{(0,1)} + C_\sigma \Gamma_{t,\rho\sigma}^{(1,0)}}{\Gamma_{t,\rho\sigma}^{(1,0)}\Gamma_{t,\rho\sigma}^{(0,1)} - g^2} = \frac{C_\rho \Gamma_{t,\rho\sigma}^{(0,1)} + C_\sigma \Gamma_{t,\rho\sigma}^{(1,0)}}{\Gamma_{t,\rho\sigma}^{(1,0)}\Gamma_{t,\rho\sigma}^{(0,1)}} - g^2 \frac{C_\rho \Gamma_{t,\rho\sigma}^{(0,1)} + C_\sigma \Gamma_{t,\rho\sigma}^{(1,0)}}{\left(\Gamma_{t,\rho\sigma}^{(1,0)}\Gamma_{t,\rho\sigma}^{(0,1)}\right)^2} + \mathcal{O}\left(g^2\right) \\ &= \frac{C_\rho}{\Gamma_{t,\rho\sigma}^{(1,0)}} + \frac{C_\sigma}{\Gamma_{t,\rho\sigma}^{(0,1)}} - g^2 \left(\frac{C_\rho}{\left(\Gamma_{t,\rho\sigma}^{(1,0)}\right)^2 \Gamma_{t,\rho\sigma}^{(0,1)}} + \frac{C_\sigma}{\Gamma_{t,\rho\sigma}^{(1,0)} \left(\Gamma_{t,\rho\sigma}^{(0,1)}\right)^2} \right) + \mathcal{O}\left(g^2\right)\end{aligned} \tag{31.9}$$

The potential reduces to Eq. (31.10).

$$\partial_t \tilde{U}_{t,\rho\sigma}(\tilde{\rho}) = -d\tilde{U}_{t,\rho\sigma}(\tilde{\rho}) + (d-2+\eta)\,\tilde{\rho}\tilde{U}_{t,\rho\sigma}^{(1,0)}(\tilde{\rho}) + (d-s)\,\tilde{\sigma}\tilde{U}_{t,\rho\sigma}^{(0,1)}(\tilde{\rho}) - \frac{A_d C_\rho}{\Gamma_{t,\rho\sigma}^{(1,0)}} - \frac{A_d C_\sigma}{\Gamma_{t,\rho\sigma}^{(0,1)}} \tag{31.10}$$

$$+ g^2 A_d \left(\frac{C_\rho}{\left(\Gamma_{t,\rho\sigma}^{(1,0)}\right)^2 \Gamma_{t,\rho\sigma}^{(0,1)}} + \frac{C_\sigma}{\Gamma_{t,\rho\sigma}^{(1,0)} \left(\Gamma_{t,\rho\sigma}^{(0,1)}\right)^2} \right) + \mathcal{O}\left(g^2\right) \tag{31.11}$$

In the case where $g \to 0$, the second line, Eq. (31.11), vanishes; and the flow of the potential reduces to the first line only, Eq. (31.10). In this case, the model reduces to two linearly-independent solutions corresponding to $\tilde{\rho}$ and $\tilde{\sigma}$. By taking a derivative with respect to ρ, one therefore finds in this case Eq. (31.12). This equations correpsonds exactly to Eq. (31.1), such that the analysis reduces to that of a simple short-ranged model.

$$\partial_t \tilde{U}_{t,\rho\sigma}^{(1,0)}(\tilde{\rho}) = (-2+\eta)\,\tilde{U}_{t,\rho\sigma}^{(1,0)}(\tilde{\rho}) + (d-2+\eta)\,\tilde{\rho}\tilde{U}_{t,\rho\sigma}^{(2,0)}(\tilde{\rho}) - \frac{A_d C_\rho \Gamma_{t,\rho\sigma}^{(2,0)}}{\left(\Gamma_{t,\rho\sigma}^{(1,0)}\right)^2} \tag{31.12}$$

By similar consideration, by taking a derivative with respect to σ, one therefore finds in this case Eq. (31.13) which corresponds to Eq. (31.6) in the spontaneouslty-broken phase, such that the analysis reduces

to that of a simple long-ranged model.

$$\partial_t \tilde{U}^{(0,1)}_{t,\rho\sigma}(\tilde{\rho}) = -s\tilde{U}^{(0,1)}_{t,\rho\sigma}(\tilde{\rho}) + (d-s)\,\tilde{\sigma}\tilde{U}^{(0,2)}_{t,\rho\sigma}(\tilde{\rho}) - \frac{A_d C_\sigma \Gamma^{(0,2)}_{t,\rho\sigma}}{\left(\Gamma^{(0,1)}_{t,\rho\sigma}\right)^2} \tag{31.13}$$

In effect, for a small $g \neq 0$, and by choosing a truncation $\Gamma^{(1,1)}_{t,\rho\sigma}$ for simplicity, the perturbation term, Eq. (31.14), does not obtain any new dependence on $\tilde{\rho}$ or $\tilde{\sigma}$ in the numerator, when functional derivatives with respect to $\tilde{\rho}$ or $\tilde{\sigma}$ are taken, that is, corresponding to the terms $\delta^{(1,0)}_{\rho\sigma}$ and $\delta^{(0,1)}_{\rho\sigma}$. In fact, even when no truncation is chosen and $\Gamma^{(1,1)}_{t,\rho\sigma} \neq 0$, this remains true.

$$\delta = g^2 A_d \left(\frac{C_\rho}{\left(\Gamma^{(1,0)}_{t,\rho\sigma}\right)^2 \Gamma^{(0,1)}_{t,\rho\sigma}} + \frac{C_\sigma}{\Gamma^{(1,0)}_{t,\rho\sigma}\left(\Gamma^{(0,1)}_{t,\rho\sigma}\right)^2} \right) \tag{31.14}$$

Given the observation above, absent of the need of a detailed derivation, one finds that the introdction of this small perturbation g does not lead to contributions in the leading order in $\tilde{\rho}$ and $\tilde{\sigma}$, respectively to the flows of $\partial_t\tilde{\rho}$ and $\partial_t\tilde{\sigma}$. Accordingly, the perturbation contributes only marignally, and the lower critical dimension is not changed by the introduction of the perturbation.

Part IV

Conformal Field Theories

Part V

Long Range Minimal Models

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